

AKADEMIKA

**FCL-01: FIBER OPTIC ANALOG
TRANSMITTER KIT**

&

**FCL-02: FIBER OPTIC ANALOG
RECEIVER KIT**

**EXPERIMENTAL & SERVICE
MANUAL**

.... A MESSAGE FROM

Today's system designers are faced with tomorrow's problems. INSTRUMENTATION & CONTROL is one of the important subject need to teach while learning electronics. It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

OUR MOTTO;

- Light years ahead – refers to leadership.

As leaders in our industry in India, we are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that we have a complete, truly innovative & superior training products we are so committed to quality that we back our products with a complete comprehensive warranty.

AKADEMIKA LAB SOLUTIONS

Sr. No. 15/8/1, Unit No.9, Kruti Industrial Estate, Opp.

Karishma Society, Off Karve Road, Kothrud, Pune,

Maharashtra- 411038

INDIA

Contact No.:+91-7447438443

Email: lab@akademika.in

www.akademika.in

SAFETY RULES

Carefully follow the instructions contained in this handbook as they give important points on safety during the installation, use and maintenance. Keep this handbook for any further help.

After unpacking, set all accessories in order so that they are not lost and check the kit integrity. In particular, check that the kit is integral and shows no visible damage.

Before connecting the power supply to the kit, be sure that the jumpers and connecting chords are connected correctly as per experiment.

In FIBER OPTIC COMMUNICATION LAB modules, signals are to be monitored with an oscilloscope as explained in the manual. The scope input channel should be ac coupled, unless otherwise indicated. For observation of signals on oscilloscope either use X10 (Attenuation Probe) or 180 Ω resistance in series with normal oscilloscope probe.

This kit must be employed only for the use for which it has been conceived, i.e. as educational kit and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of any fault or bad operation, turn off the power supply and do not tamper the kit. In case servicing is required, contact the service center for technical assistance.

The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we would charge for repairs, regardless of the warranty period. The warranty does not cover include perishable items like connecting chords, crystals, etc. and other imported items.

Conditions and Limitations

The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product will be Akademika Lab Solutions sole judgment. The defective product will be replaced with a new one or repaired, without charge or with charge.

In the warranty period if the service is needed, the purchaser should get in touch with the service center or the sales outlet. The purchaser should return the product to the service center or the sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of the kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case the warranty does not cover the product on Akademika Lab Solutions judgment, we would repair the product after obtaining prior permission from purchaser who would receive an estimate statement about the repairing charges. In such cases, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but they have no rights to stretch the meaning of original warranty contents or to offer an additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or the sales outlet.

INDEX

SR. NO	CHAPTER	PAGE NO
1.	Technical Specifications	06
2.	Functional Block	09
3.	Introduction	12
4.	Switch Fault	23
5.	Circuit Diagram	24
6.	Experiment No. 1 Initial Fiber End Preparation (Connectorisation)	41
3.	Experiment No. 2 Transmission through a Gap between Fibers	45
4.	Experiment No. 3 Fiber Optic Transmission Sensors	51
5.	Experiment No. 4 Fiber Optic Reflection Sensors	55
6.	Experiment No. 5 Measurement of Numerical Aperture	61
7.	Experiment No. 6 Setting Up Fiber Optic Analog Link	65
8.	Experiment No. 7 Study of Losses in Optical Fiber I) Measurement of Propagation Loss Ii) Measurement of Bending Loss Iii) Measurement of Connector Loss	69
9.	Experiment No. 8 Study Effect of Lateral, Longitudinal & Angular Displacement	75
10.	Experiment No. 9 Study of Pulse Amplitude Modulation (Pam) and Demodulation	83

11.	Experiment No. 10 Study of 4 Channel Time Division Multiplexing and Demultiplexing	89
12.	Experiment No. 11 Study of Amplitude Modulation And Demodulation	97
13.	Experiment No. 12 Measurement of Length of Fiber Cable	105
14.	Experiment No. 13 To Study Effect of Emi Interference on Copper Medium and Optical Fiber Medium	113
15.	Experiment No. 14 Study of Characteristics Of Fiber Optic LED's And Detectors	119
16.	Experiment No. 15 Forming Simple Fiber Optic Voice Link Using MIC & Speaker	125

TECHNICAL SPECIFICATIONS

Transmitter	: 2 Nos. Siemens Fiber Optic LEDs
Transmitter 01	: Peak wavelength of emission 660 nm visible (SFH756V)
Transmitter 02	: Peak wavelength of emission 950 nm Infrared (SFH450V)
Receiver	: 2 Nos. Siemens Photo Detectors
Receiver 01	: Photo Diode with responsivity of 0.3 $\mu\text{A}/\mu\text{W}$ (SFH250V)
Receiver 02	: Photo transistor with responsivity of 80 $\mu\text{A}/\mu\text{W}$ (SFH350V)
Reference pulse generator	
Frequency	: 660 KHz,
Amplitude	: 2Volt
Analog Bandwidth	: 2 MHz
Modulation Techniques	: i) Pulse Amplitude Modulation (PAM with variable clock of 16KHz & 32KHz) ii) Amplitude Modulation. Carrier Input : 1-500 KHz Modulating Input : 0.1-100 KHz Carrier Null : Adjustable Output Amplitude : upto 2Vpp iii) Intensity Modulation
Multiplexing	: 4 Channel Analog Time Division Multiplexing & De-multiplexing
Clock Frequency	: 32 KHz, Duty Cycle 50%
Signal Strength Meter	: 8 LEDs provided to measure optical power strength
Amplitude Demodulation:	Diode Envelope Detector. Detection of the envelope with RC filter
Filters	: 4 Nos. 4 th order Butterworth low pass filters
Voice Communication	: Fiber Optic Voice Link using dynamic MIC and speaker
Audio Preamplifier	
Input impedance	: 600 ohms
Voltage gain	: 1 to 100
Audio Amplifier	
Input Impedance	: 50 K Ω

Output Voltage : Adjustable
Speaker : 8Ω , 0.5 W

Fiber Optic Cable

Type : Plastic Fiber
Step Index, Multimode

Core Refractive Index -n1:1.492

Clad Refractive Index -n2:1.406

Numerical Aperture : 0.50

Acceptance Angle : 60 degrees

Fiber Diameter : 1000 microns

Outer diameter : 2.2 mm

Fiber lengths : 15cms (1No.), 1(2 Nos.), 3 Meter (1 No.) and
20 Meter (1No.)

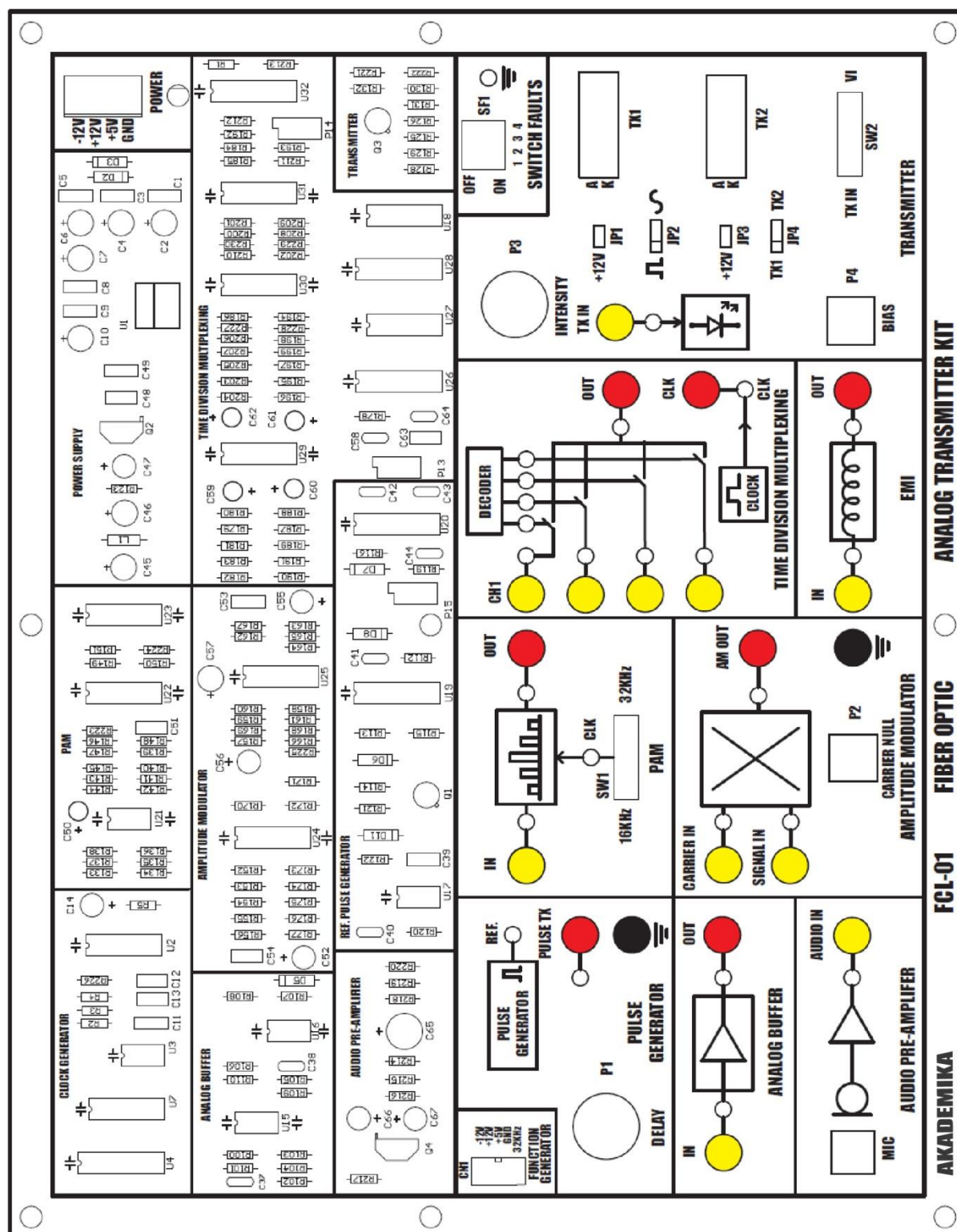
Switch Faults : 4 Nos. on FCL-01 and 4 Nos. on FCL-02 are
provided to study different effect on circuit.

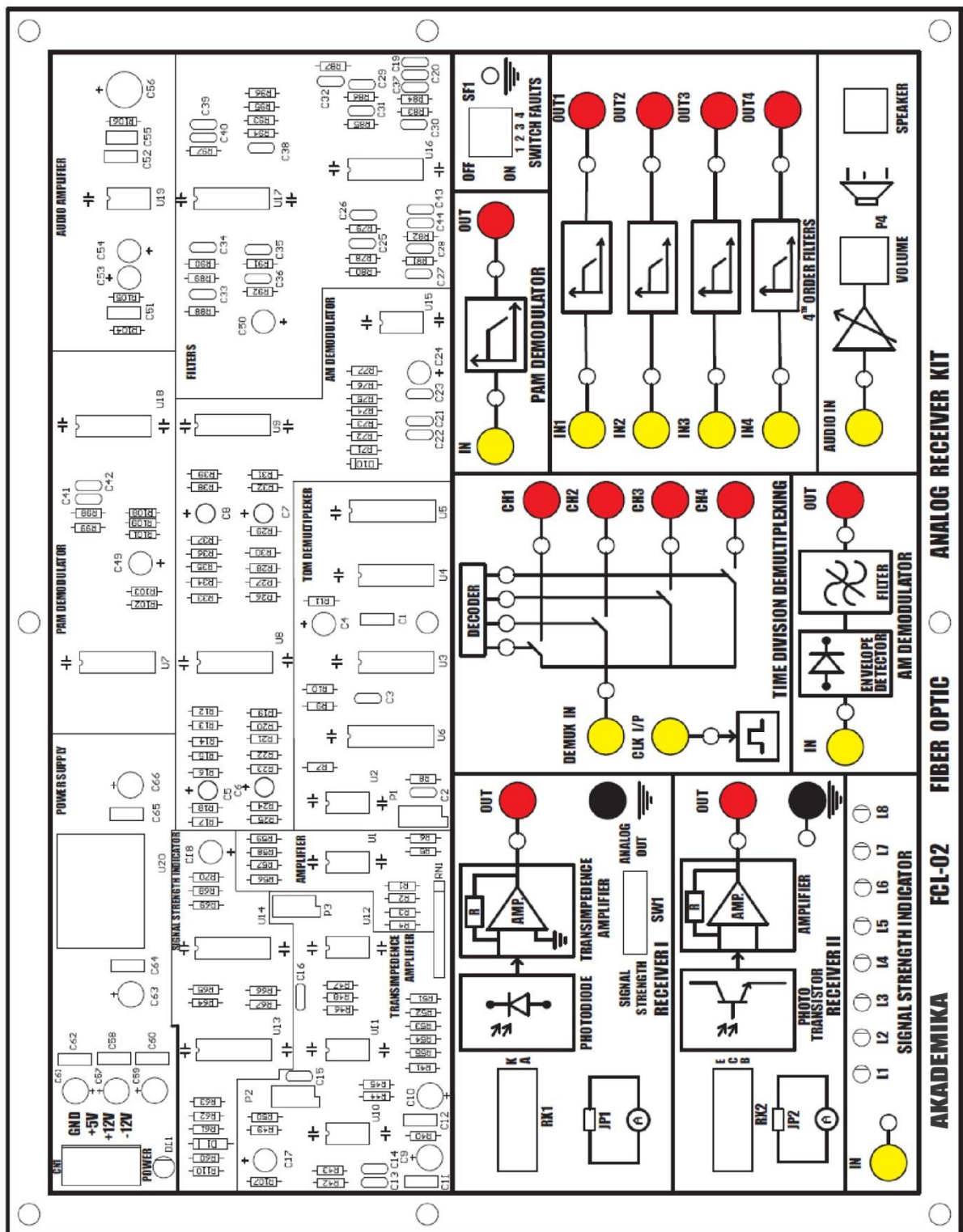
Power Supply : Double connector (GND, +5V, +12V, -12V)

Deliverables

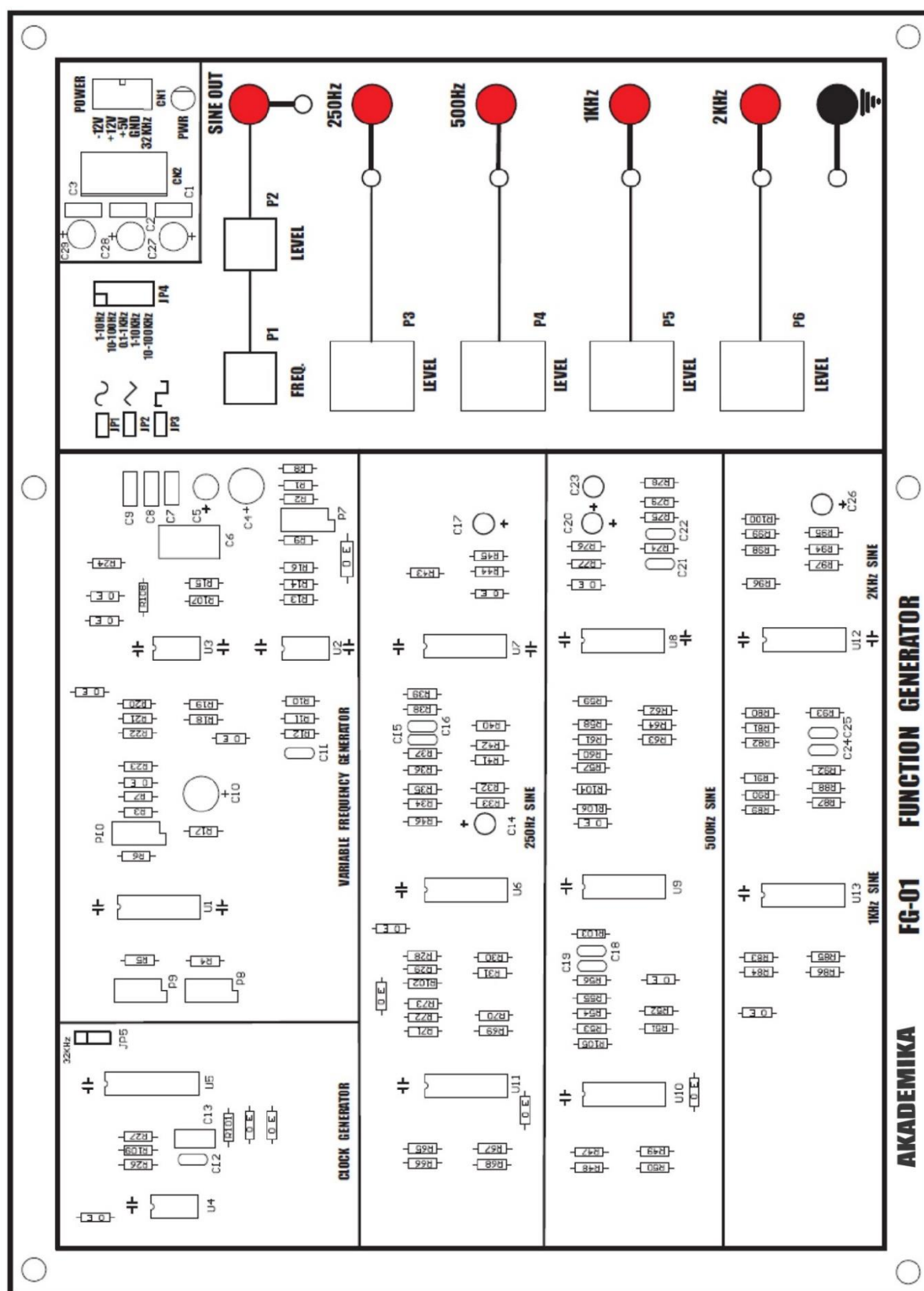
Long Links	02
Mini Links	08
Short Links	06
FCL-01 & 2 EXP & SER Manual	01
Power Supply	01
Power Chord	01
Jumper to crocodile 30cm with Double wire	03
Micro Phone	01
1.5V Battery (AA Size)	01
Speaker Box Type (0.5WMax.8E)(SP-3 CYBER)	01
ACR-13/0 Displacement Jig	01
ACR-12/0 Angular Jig	01
ACR-16/0 EMI Jig	01
Screw driver Big (5" , 3mm Flat Head)	01
Steel Ruler (15cm/6")	01
Mirror (45x56)mm	01
Sharp Razor (TOPAZ)	01
Plastic fiber Step Index, Multimode, 1000um ,15cm	01
Plastic fiber Step Index, Multimode, 1000um,1meter	01
Plastic fiber Step Index, Multimode, 1000um with fiber	02
Plastic fiber Step Index, Multimode, 1000um with Fiber optic simplex plug connector at one end, 0.5 mtr	02
Plastic fiber Step Index, Multimode, 1000um,3meter	01
Plastic fiber Step Index. Multimode, 1000um with Fiber optic simplex plug connector at one end, 3meter	01
Plastic fiber Step Index. Multimode, 1000um , 20meter	01
Fiber optic Simplex Plug (single Position Plug [N2400228087-1])	02
Simplex Receptance (connecting sleeve) (single position bulkhead receptacle)	01
Fiber Polish Paper 8X5.5cm [0.3um (Alum. Oxide)]White	01
Jumper caps small (5mm)	05
Copper Link Copper Patch Chord 10" with strip at both end	01

FUNCTIONAL BLOCK





FG-01



INTRODUCTION

FIBER OPTICS TECHNOLOGY is rapidly becoming a familiar & indispensable part of human life. You have undoubtedly heard of Fiber Optics. All giant telephone companies like AT & T, sprint have saturated the airways & print media with advertisements heralding this bright & new technology, Futurists, science writers & entrepreneurs forecast fantastic growth for Fiber Optics applications.

And their enthusiasm is valid. Today medical fiber optic systems allow physicians to peer inside the human body without surgery. Military commanders demand portable battlefield communication systems, which use superior fiber optic transmissions. The massive bundles of copper wiring which once carried telephone conversations between continents now are being replaced with the optical fibers from coast to coast.

Very few technologies ever come close to the fantastic growth rate that market analyst love for them to predict. Fiber optics has exceeded those rates. The next decade promises even more wonders. The influences of fiber optics will provide our homes & workplaces recreational facilities. Your television reception will take on startling clarity. Newsprint may largely become a thing of the past as environmentally friendly electronic newspapers become feasible. Two-way fiber optic will allow you to “attend college” in the comfort of your home.

Anyone who takes an interest in fiber optic & pursues a career in the field today is on the ground floor of opportunity. The technology is sufficiently new that few “experts” exist. Many science colleges, universities & engineering colleges have included fiber optics in their syllabus. The future beckons exciting with the prospect of making new discoveries finding new applications for the technology.

This kit is an introduction to fiber optics communication technology for instructors, students & hobbyists. Welcome to the fascinating & expanding world of fiber optics. We hope that you find the kit challenging, stimulating & - yes – even fun.

ABOUT FIBER OPTICS

Before fiber optics came along, the primary means of real time data communication was electrical in nature. It was accomplished using copper wire or by transmitting electromagnetic (radio) waves through free space. Fiber optics changed that by providing an alternative means of sending information over significant distances – using light energy. Although initially a very controversial technology fiber optics has today been shown to be very reliable & cost effective. Light as utilized for communication has a major advantage because it can be manipulated (modulated) at significant higher frequencies than electrical signals can. For example, a fiber optic cable can carry up to 100 million times more information than telephone line! The fiber optic cable has lower energy loss & wider bandwidth capabilities than copper wire.

As you will learn, fiber optic communication is a quite simple technology, closely related to electronics. In fact, it was research in electronics that set the stage for fiber optics to develop into the communication giant that it is today. Fiber optics became reality when several technologies came together at once. It was not an

immediate process, nor was it easy, but it was most impressive when it occurred. An example of one critical product, which emerged from that technological merger, was the semiconductor LED that we used in our kit.

How Optical Fiber Works?

The principle of operation of optical fiber lies in the behavior of light. Light travels in straight line through most optical materials, but that's not necessarily the case at the junction (interface) of two materials of different refractive indices. Air & water are a case in point as shown in fig. 1. The light ray traveling through air actually is bent as it enters the water. The amount of bending depends on the refractive indices of the two materials involved & also on the angle of the incoming (incident) ray of light as it strikes the interface. The angle of the incident ray is measured from a line drawn perpendicular to the surface. The same is true for the angle of the outgoing (transmitted) ray of light after it has been bent.

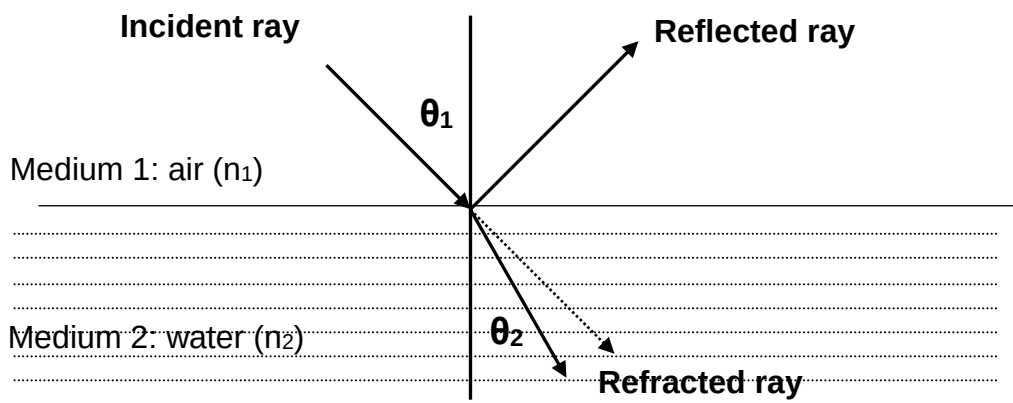


FIG. 1: The different portions of a light ray at a material interface

The mathematical relationship between the incident ray & the refracted ray is explained by **Snell's Law**

$$n_1 \cdot \sin \Theta_1 = n_2 \cdot \sin \Theta_2$$

in which n_1 & n_2 are the refractive indices of the initial and secondary materials respectively and Θ_1 and Θ_2 are the incident and transmitted angles.

Snell's Law says that refraction (bending) of light cannot take place when the angle of incidence grows too large (as when light travels from a material with a high refractive index to one with a low refractive index). If the angle of incidence exceeds a certain critical value (in which the product of n_1 & the sine of the angle equals or exceeds one.) light cannot exit.

If light cannot exit the material, it is reflected. The angle that is reflected is equal to the angle of incidence. The phenomenon just described is called Total Internal Reflection & it is what keeps light inside an optical fiber.

Types of Optical Fiber

The simplest fiber optic cable consists of two concentric layers of transparent materials. The inner portion (the core) transports the light. The outer covering (the cladding) must have a lower refractive index than the core, so the two are made of different materials. The cable used in this kit also has a jacket to protect the optical properties of the core and cladding.

Optical fiber is generally made from either plastic or glass. This kit uses plastic fiber, which is very easy to terminate & does not require special tools. Plastic fiber is generally limited to uses involving distance of less than 100 meters because of high attenuation (higher loss).

Glass Fiber on Other Hand Has Very Low Attenuation (Low Loss) & Hard To Cut, Requires Special End Connections & Is More Expensive.

The core of fiber is typically made of silica doped with impurities, which increases the refractive index relative to pure silica. The cladding, which surrounds the fiber core & creates an optical interface, is typically made from pure silica. The outer buffer coating is a plastic cover.

The first type of fiber optic cable put to use was called Step Index. In this design the cladding has a different index of refraction from the core. The light bounces off the sides & is reflected back into the fiber core. The problem with this design is that the reflected light must travel a slightly longer distances than that which travels down the center of the fiber, thus limiting the max. transmission rate. This design was improved with the use of Graded Index Fiber. In this design, the index of refraction decreases in proportion to the distance away from the center of the fiber core. The light moves more quickly in the outer portion, thus compensating for the additional distance. The change in index also has the effect of 'bending'.

The light reflects back towards the core. This change increases the transmission capacity. In the newest Single Mode design, the diameter of the fiber core is made so small that all the light travels in a straight line. Even the latest fiber optic facility in use today uses less than 5% of the max. Theoretical capacity of a Single Mode Fiber.

Single Mode V/S Multimode

There are essentially two different types of fiber optic transmission schemes in use. Multimode fiber optic systems are typically used in short haul applications such as LANS & FDDI. Recent growth in multimode installations has been driven by LAN/FDDI applications. The term Multimode means that the diameter of the fiber optic core is large enough to propagate more than one mode (electromagnetic wave). Because of the multimode modes, the pulse that is transmitted down the fiber tends to become stretched over distance. This is referred to as Modal Dispersion & has the effect of reducing available bandwidth.



FIG. 2: Modal dispersion of light

Typical wavelengths used in multimode applications are 850 nm & 1300 nm. There are other wavelengths used (e.g. 660 nm for plastic fiber, 820/870 nm) in specialized applications such as the military. The vast majority of multimode applications are 850 or 1300 nm wavelengths. Most multimode cables in use today are graded index in type with a cladding diameter of 125 microns & fiber core diameter of 62.5 or 50 microns (Typically referred to as 62.5/125 or 50/125 fiber)

Single Mode Systems are generally used in long haul applications (10's or 100's of kilometers) & local loop applications in both the carrier & CATV markets. Some large utilities such as electrical power companies also utilize single mode fiber optic backbones. As the name implies single mode fiber is designed to propagate only one mode of light. It is therefore not affected by modal dispersion & has higher bandwidth capacity. It is considered to be higher quality transmission system. Wavelengths used in single mode applications are 1300 & 1550 nm. The majority of today's installed systems uses 1300 nm wavelength technology & is designed for higher speed applications. These systems are more sensitive to back reflections from connectors & sharp cable bends. They often require additional tests, such as Optical Return Loss, to be run. Most single mode cables in use today are step index in type with cladding diameter of 125 micron & fiber core diameter of 9 micron (Typically referred to as 9/125 fiber).

Advantages of Fiber Optics over Conventional Copper cables

- Much greater bandwidth (Several orders of magnitude are theoretically possible).
- Immunity to electrical disturbances, ground loops, cross talks etc. in addition, no EMI radiation is generated)
- Much lighter. A 3-inch bundle of 900 twisted copper pairs can carry about 21,000 channels of traffic & weighs about 25,000 pounds/mile. A ½ inch fiber link with 12 fiber strands can carry about 3,00,000 channels & weighs about 200 pounds/mile. In addition less duct space is required to make the cable within building.
- Better in hostile environment, not affected as much by temperature, water etc.
- Lower transmission loss. Typical loss on a fiber link is 0.2 dB/Km. On a copper based facility, one can usually expect at least 5 db/km. For longer links, fewer repeaters are required.
- Better security as it is not possible to simply bridge onto the facility & monitor the traffic.

With all these properties of optical fiber, it is becoming very important in sensing field also. Most of the properties of optical fiber are useful for sensors. Further optical fiber sensors offer extreme sensitivity & can be used to sense almost any physical parameters like temperature, velocity, current, voltage etc.

ABOUT FCL-01 AND FCL-02

The purpose of ADVANCE FIBER OPTIC COMMUNICATION LAB is to provide hands on experience on the various Fiber Optic & Digital Communications Techniques. In preparing this manual, we assumed you have a basic grasp of digital & transistor circuits. The experiments that can be performed with this Lab have been designed in a ladder fashion.

You have to start from the first step & then go for the next. Experiments start with basic principles of fiber optics & then go on to more sophisticated techniques. Every advanced experiment uses concept, procedures, and results of the previous experiments. So you are advised not to skip in between experiments. In this manual before starting the procedure of every experiment, objective of the experiment & basic theory related to that is given. You are advised to read it before starting the experiment because it will help you to understand the experiment very well. Also wherever necessary refer the datasheets of various components provided in the backside of the circuit description manual.

Before starting the experiments check following things.

- FCL-01 and FCL-02
- Plastic fiber of 15cm (1No.), 1Meter (2 Nos.), 3 Meter (1 No.) & 20 Meter (1 No.) lengths
- N.A. Jig (Precision Jig No.1) for holding fiber (1 No.)
- Precision Jig No.2 for holding fiber (1 No.)
- EMI Jig for holding fiber (1 No.)
- Power Supply (1 No.)
- Steel Rule (1 No.)
- Mirror (1 No.)
- Function Generator FG-01
- Polish Paper
- Connecting Plug
- Link set containing Patch chords, Jumpers, crocodile clips wires.
- Experimental & Circuit Description Manual.

Optical Fiber Preparation Instructions

The fibers provided with this Lab are multimode, step index, 1000-micron plastic fiber & are factory prepared for use. They are not require to prepared again even after several operations, if handled carefully. In case the fiber is broken or damaged you have to prepare each end of the optical fiber cable very carefully so it transmits light effectively.

- Cut off the ends of the cable with a single-edge razor blade or sharp knife. Try to obtain a precise 90-degree angle.
- Wet the polishing paper with water or light oil & place it on flat, firm surface. Hold the optical fiber upright, at right angle to the paper & polish the fiber tip with a gentle “figure8” motion. You may get the best result by supporting the upright fiber against some flat object such as a portion of the printed circuit board.
- Using an 18-gauge wire stripper, remove 3 mm of the jacket from the end of the fiber. Be careful not to nick the buffer in the process or you will be losing light later.

CIRCUIT DESCRIPTION

FCL-01

1.1 Clock Generation

The timer U3 (IC 555) is used to generate the clock waveform of 32 KHz. This is given as clock input to U4 (IC 74HC193) which is operated in UP counting mode. All parallel load inputs are tied to ground so that at power on, it is in the reset condition. Hence outputs Q0, Q1, Q2 and Q3 are all low. The counter increments from first clock pulse at the input to get the 16KHz, 8KHz and 4KHz frequencies.

1.2 Analog Buffer

This is a unity gain amplifier built around op-amp IC LF357 (U15 & U16). Output voltage is limited to 4.7Vpp using zener diode at the output post of buffer. It is used to form Fiber optic Analog Link.

1.3 Reference Pulse Generator

Reference pulse (PULSE TX) of 2V amplitude and 150ns wide is generated using high frequency timer U17 (IC 555) & monostable multivibrator U19 (IC 74123). Time shift in the reference pulse is introduced using Dual monostable multivibrator U20 (IC 74LS221). Pot P1 is provided to adjust the position of the reference pulse on time scale for observation on oscilloscope.

1.4 Fiber Optics Transmitter

Fiber optic transmitters are typically composed of a buffer, driver & an optical source. The buffer section provides both electrical connection & isolation between the optical transmitter & analog information source. The driver section provides electrical power to the optical source in a fashion that duplicates the pattern of data being fed to the transmitter. Finally the optical source (LED) converts the electrical current to light energy with the same pattern.

FCL-01 kit supply two optical LED transmitters namely SFH450V & SFH756V. The LED SFH450V operates outside the visible light spectrum. Its optical output is centered at near infrared wavelength of 950nm. The LED SFH756V operates on the visible light spectrum. Its optical output is centered at visible red light wavelength of 660nm. These LED's are coupled to the transistor driver in common emitter mode. In the absence of input signal half of the supply voltage appears at the base of the transistor. This biases the transistor near midpoint within the active region for linear applications. Thus LED emits constant intensity of light at this time. When the signal is applied to transmitter it overrides the DC level at the base of the transistor, which causes the Q point of the transistor to oscillate about the midpoint. So the intensity of the LED varies about its previous constant value. This variation in the intensity has linear relation with the input electrical signal. Optical signal is then carried over by the optical fiber.

Jumper JP1 & JP3 provides VCC signal to anode of SFH756V & SFH450V respectively. Jumper JP4 is used to select either SFH756V or SFH450V. Switch SW2 selects either input signal or fixed bias signal to be applied to base of driver transistor. Fixed bias signal can be varied using pot P4.

1.5 Pulse Amplitude Modulation

Pulse Amplitude Modulation is a technique of communication in which the amplitude of pulse varies with respect to the amplitude of analog input signal. The modulating signal is sampled by the pulses. The PAM signal is nothing but the high frequency square wave in which the amplitude of each pulse is equal to that of the information signal at the respective sampling instant. Switch SW1 is used to select either 16KHz or 32KHz signal as clock for sampling the applied analog signal and to separate out sampled signal at the output of modulator. U23 (IC 4066) is used as Analog switch. Using clk out you will get sample and hold output form of applied signal and using clk out inverted signal hold portion of sampled signal is extracted to get flat top pulses with varying amplitude thus generating Pulse Amplitude Modulation.

1.6 Amplitude Modulation

The integrated circuit LM1496 (U25) consists of a differential amplifier, driven by a further differential stage. The output signal consists of a constant, which multiplies the two input signals CARRIER IN and SIGNAL IN. The pot CARRIER NULL enables the balancing or unbalancing of the circuit: in the first case (balance state) the output is exactly the product of the input signals, and the circuit operates as balanced modulator; in the second case (unbalance state) the output contains a fixed component of the signal CARRIER and the circuit operates as amplitude modulator.

1.7 Time Division Multiplexing

Analog inputs are applied to the analog switch U31 (4066) through unity gain amplifiers U30 (IC KF347). A monostable multivibrator U26 (IC 74LS123) is used to generate enable pulse for decoder and D flip-flops U27 (IC 74LS74) are used to generate channel selection pulses for decoder. A 2:4 decoder U28 (IC 74LS139) decodes the counter output to generate enable signals for analog switches (IC 4066). The four analog switches form a 4:1 multiplexer with four different input lines and a single o/p line.

This technique enables simultaneous transmission of number of signals along the same communication channel. The basic concept behind this is that the signals from different sources are multiplexed by allowing only a specific time slot for each signal.

1.8 Audio Pre-Amplifier

Audio Pre-amplifier is built around Q4 (BC548) transistor in common emitter mode. Signal from the microphone is applied to the base of this transistor which is amplified at the output.

1.9 EMI Section

EMI section consists of EMI coil, which is used to observe effect of electromagnetic interference on transmitting medium, for optical link and copper link.

1.10 Switch Faults

Four switch faults are provided to insert logical faults in various sections of the circuit.

FCL-02

2.1 Time Division Demultiplexing

Clock for Demultiplexing is taken from FCL-01. This clock is given to D Flip Flop U3 (IC 74LS74) to generate channel selection signal for decoder U5 (74LS139). A 2:4 decoder U5 (74LS139) decodes the counter output to generate enable signals for analog switches U9 (4066). Analog input from detector is applied to the analog switches U9 (4066). The demux section is a 1:4 demultiplexer with one common input and four different outputs, which are selected with respect to four enable signals. Demultiplexed signals available at CH1, CH2, CH3 and CH4 posts. Synchronization signal is separated out using U2 (IC LM311).

2.2 Fiber Optics Receiver1 (Signal Conditioning)

At the receiver, light is converted back into electrical form reproducing the actual signal originally fed to the transmitter.

The function of the receiver is to convert the optical energy into electrical form, which is then conditioned to reproduce the transmitted electrical signal in its original form. The detector SFH250V used in the FCL-02 has a diode type output. The parameters usually considered in the case of detector are its responsivity at peak wavelength & response time. SFH250V has responsivity of about $4\mu\text{A}$ per $10\mu\text{W}$ of incident optical energy at 950 nm and it has rise & fall time of $0.01\mu\text{sec}$.

PIN photodiode is normally reverse biased. When optical signal falls on the diode, reverse current start to flow, thus diode acts as closed switch and in the absence of light intensity, it acts as an open switch. Since PIN diode usually has low responsively, a transimpedance amplifier is used to convert this reverse current into voltage. This voltage is then amplified with the help of another amplifier circuit. This voltage is the duplication of the transmitted electrical signal.

Using Switch SW1 received signal can be either fed to Signal Strength section or taken directly as analog output.

2.3 Fiber Optics Receiver2 (SFH 350 Amplifier)

At the receiver, light is converted back into electrical form with the same pattern as originally fed to the transmitter. Detected signal from SFH350 Phototransistor detector is amplified with fixed gain using op-amp U1 (IC LF353).

2.4 Signal Strength Indicator

Signal strength indicator section consists of voltage comparator stage built around op-amp U13 & U14 (IC LM324). Depending on the input signal strength LED's at the output of voltage comparator glow to indicate signal strength as a visual effect. The number of LED's glowing corresponds to the signal strength.

2.5 Amplitude Demodulation

This section consists of diode envelope detector. Positive envelope from the input signal is separated out and it is further filtered out using low pass filter circuit built around U15 (IC LF353), to recover the actual analog information signal transmitted.

2.6 Pulse Amplitude Demodulation

Fourth Order Butterworth Low Pass Filter U18 (IC LM324) is used to Demodulate Pulse Amplitude Modulated Signal

2.7 Audio Amplifier

Refer to the circuit diagram as shown later in this manual sheet **6 of 6** of **FCL-02**. Detected electrical signal is given to U19 (IC LM386), which amplifies this signal. For Volume control potentiometer P4 is provided.

2.8 Switch Faults

Four switch faults are provided to insert logical faults in various sections of the circuit.

FG-01

3.1 Variable Frequency Generator

The integrated circuit U1 (ICL8038) generates sine (pin 2) and triangular waveform (pin 3). The capacitors which can be connected to pin 10 enable to select the range (10Hz, 100Hz, 1KHz, 10KHz or 100 KHz max). The frequency is varied with the pot **P1 FREQ.** (P7 limits the inferior frequency). The trimmers P9 and P10 adjust the symmetry and the form of the sine signal.

With the Jumpers JP1 & JP2 the sine and the triangular signals are applied to the amplifier U2 (LF353) and then to the output. The pot **P2** varies the signal amplitude

With Jumper JP3 the square signal can be applied to the output amplifier. Frequency range can be selected with help of switch SW1 and can be observed at **OUT** post.

3.2 Clock Generator

The timer U4 (IC NE555) is used to generate the clock waveform of 32 KHz. This is given as clock input to U5 (IC 74LS193), which is operated in UP counting mode. All parallel load inputs are tied to VCC so that at power on, it is in the reset condition. Hence outputs Q0, Q1, Q2 and Q3 are all High. The counter increments from first clock pulse at the input to get the 16KHz, 8KHz, 4KHz and 2KHz frequencies.

3.3 Fixed Frequency Sine Wave Generator

This basically provides four amplitude variable (0 - 5 Vpp) synchronized sine waves, each 250Hz, 500Hz, 1KHz and 2KHz.

The 32KHz, 16KHz, 8KHz and 4KHz frequencies generated in clock and timing section are fed to serial to parallel shift register U6, U10, U11, U13 (IC 74HC164), which generates the sine wave, by serial shift operations. The serial to parallel shift registers (IC 74HC164) have resistive ladder network at the output. For 16 shifts of the register, one stare case sine wave is produced. So if a 16KHz clock is fed to the shift register, 1Khz sine wave is generated. Thus by using the clock 32khz, 16khz, 8Khz, 4Khz, sine waves of 2KHz, 1KHz, 500Hz and 250Hz are generated respectively. The active filter U7 (IC KF347) at the output suppresses the ripple and also takes care of the impedance matching. This gives smooth sine waves. The potentiometers P3, P4, P5, P6 are used to vary amplitudes of 2KHz, 1KHz, 500Hz and 250Hz sine waves respectively.

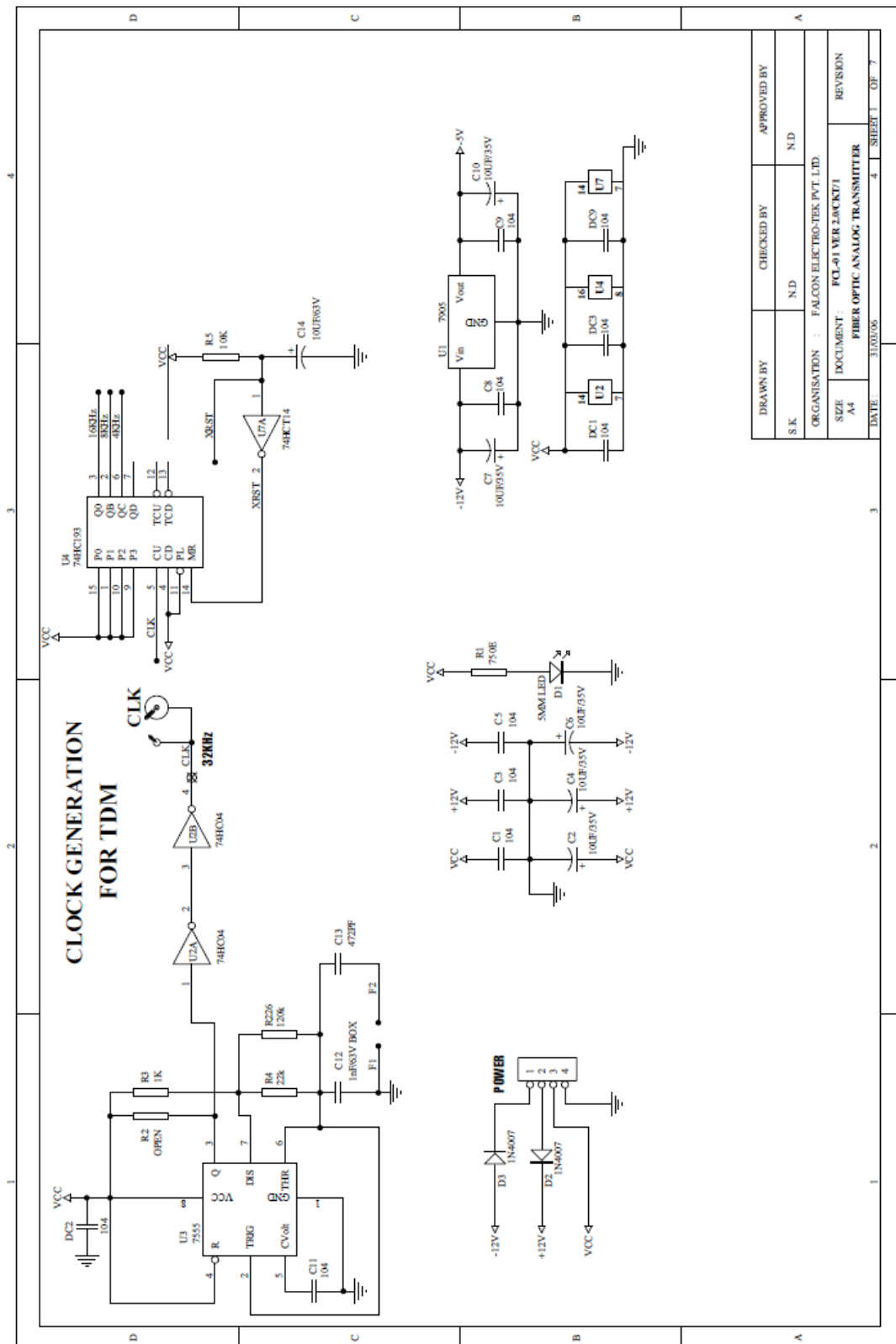
SWITCH FAULTS

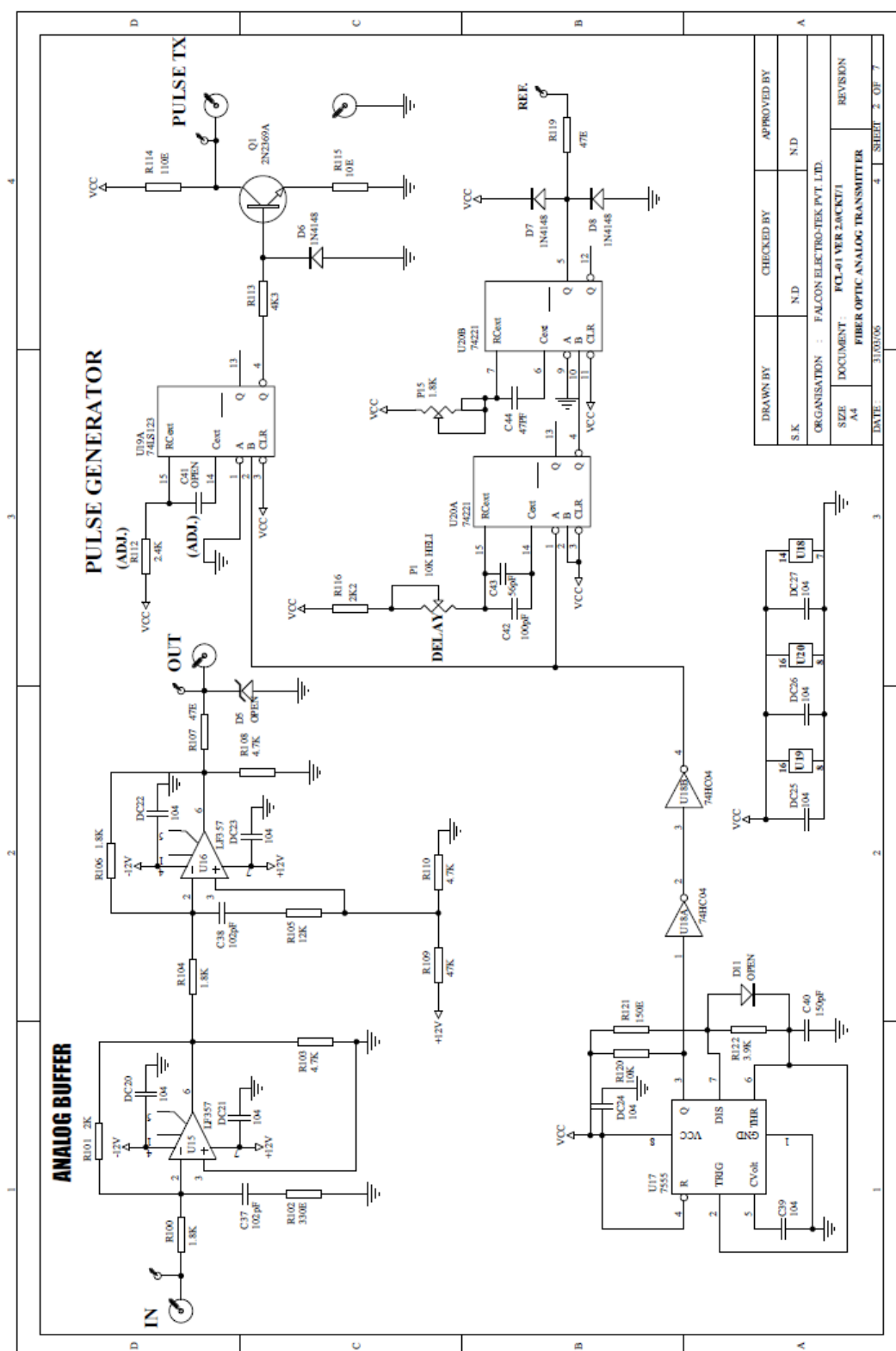
Switch fault section on FCL-01 is provided with 4 switches & switch fault section on FCL-02 is provided with 4 switches. These switches can be used to simulate fault conditions in various parts of the circuit. In order to keep FCL-01 & FCL-02 kits fully operational, all switches should be set to OFF position before use. Switch faults are introduced one at a time.

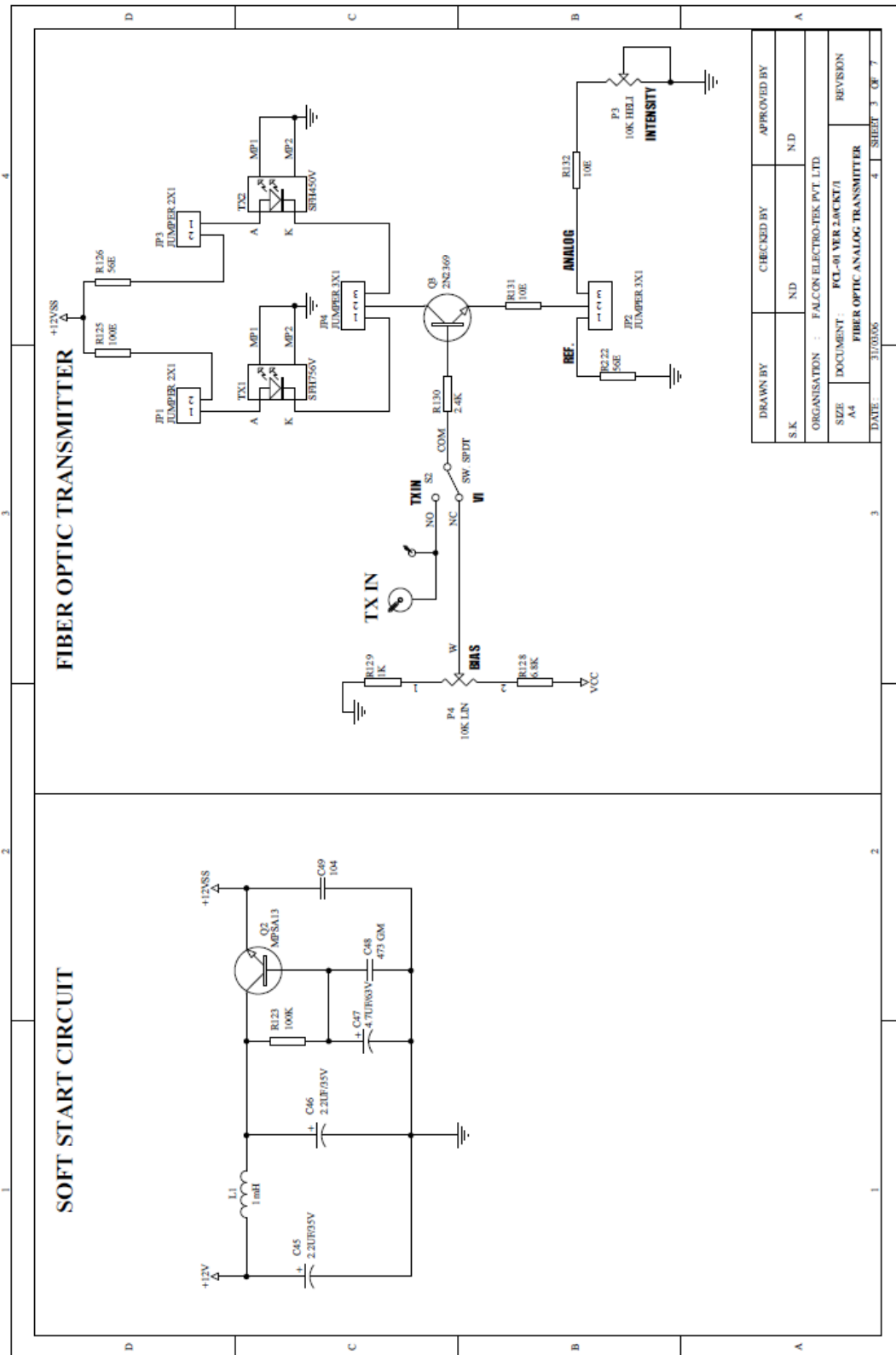
- Put switch **1** of **SF1** in Switch Fault section of FCL-01 to ON position. This will short capacitor C12 & C13 in Clock generator section. RC Time increases and clock frequency reduces. It affects TDM & PAM sections.
- Put switch **2** of **SF1** in Switch Fault section of FCL-01 to ON position. This will short resistor R168 & R221 at the bias pin of Balanced Modulator from Amplitude Modulator block. Amplitude Modulator output gets distorted.
- Put switch **3** of **SF1** in Switch Fault section of FCL-01 to ON position. This will short capacitor C63 & C64 in Time Division Multiplexing section. Synchronization pulse duty cycle get increases. TDM output distorts and similarly Demodulator output also gets distorted.
- Put switch **4** of **SF1** in Switch Fault section of FCL-01 to ON position. This will short CH3 & CH4 signals going for TDM. Mixed signal appears for CH3 & CH4 positions at Multiplexer and demultiplexer outputs.
- Put switch **1** of **SF1** in Switch Fault section of FCL-02 to ON position. This will open capacitors C43 & C44 from Pulse Amplitude Demodulator section. Pulse Amplitude Demodulator output approximates zero.
- Put switch **2** of **SF1** in Switch Fault section of FCL-02 to ON position. This will open capacitors C37 from fourth filter. Fourth filter output gets distorted.
- Put switch **3** of **SF1** in Switch Fault section of FCL-02 to ON position. This will open capacitors C19 & C20 in Amplitude demodulator section. Output of Amplitude Demodulator gets distorted.
- Put switch **4** of **SF1** in Switch Fault section of FCL-02 to ON position. This will open decoder pulse for CH3 in Time Division Demultiplexing section. Output for third channel will be absent.

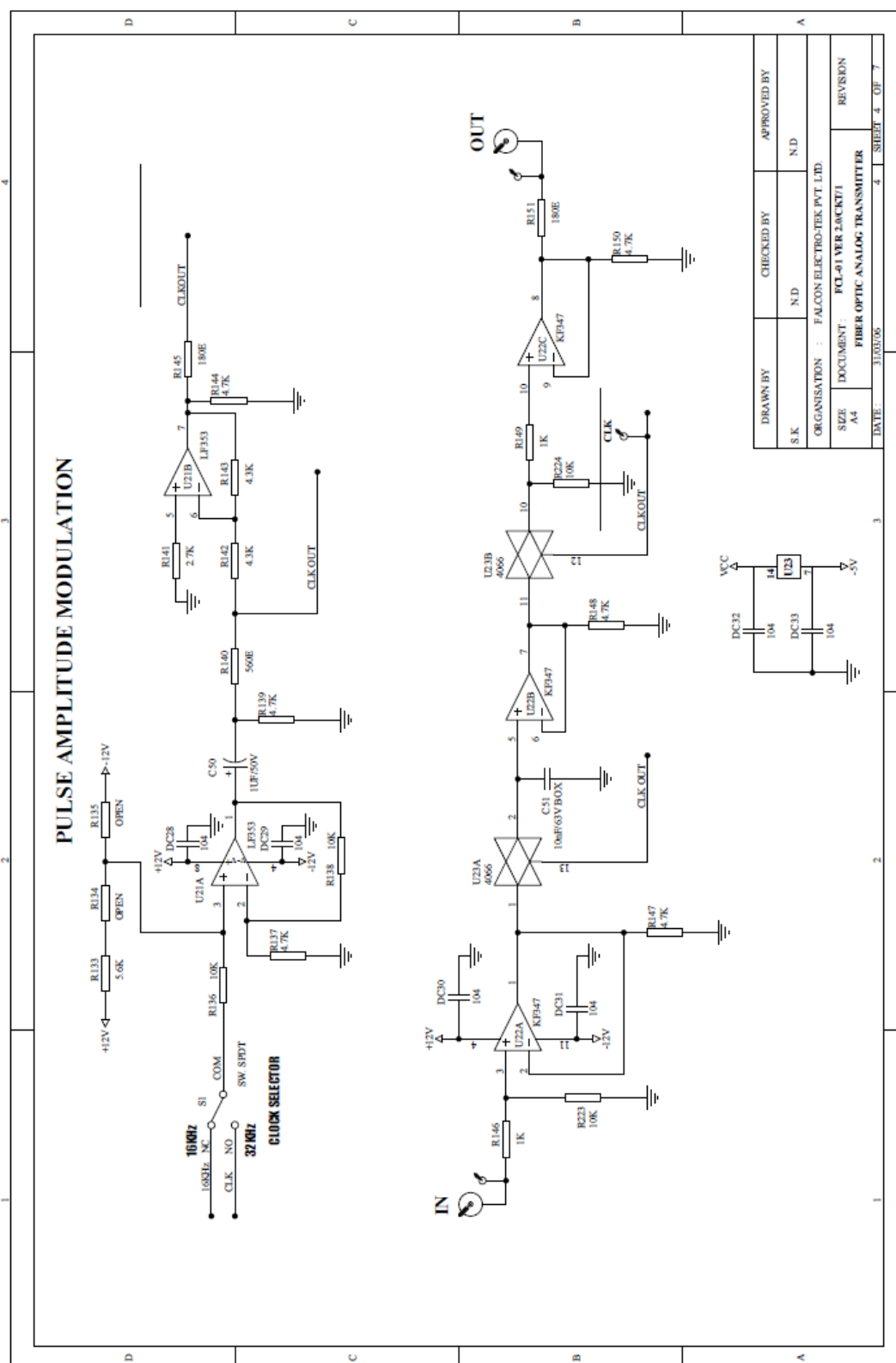
CIRCUIT DIAGRAM

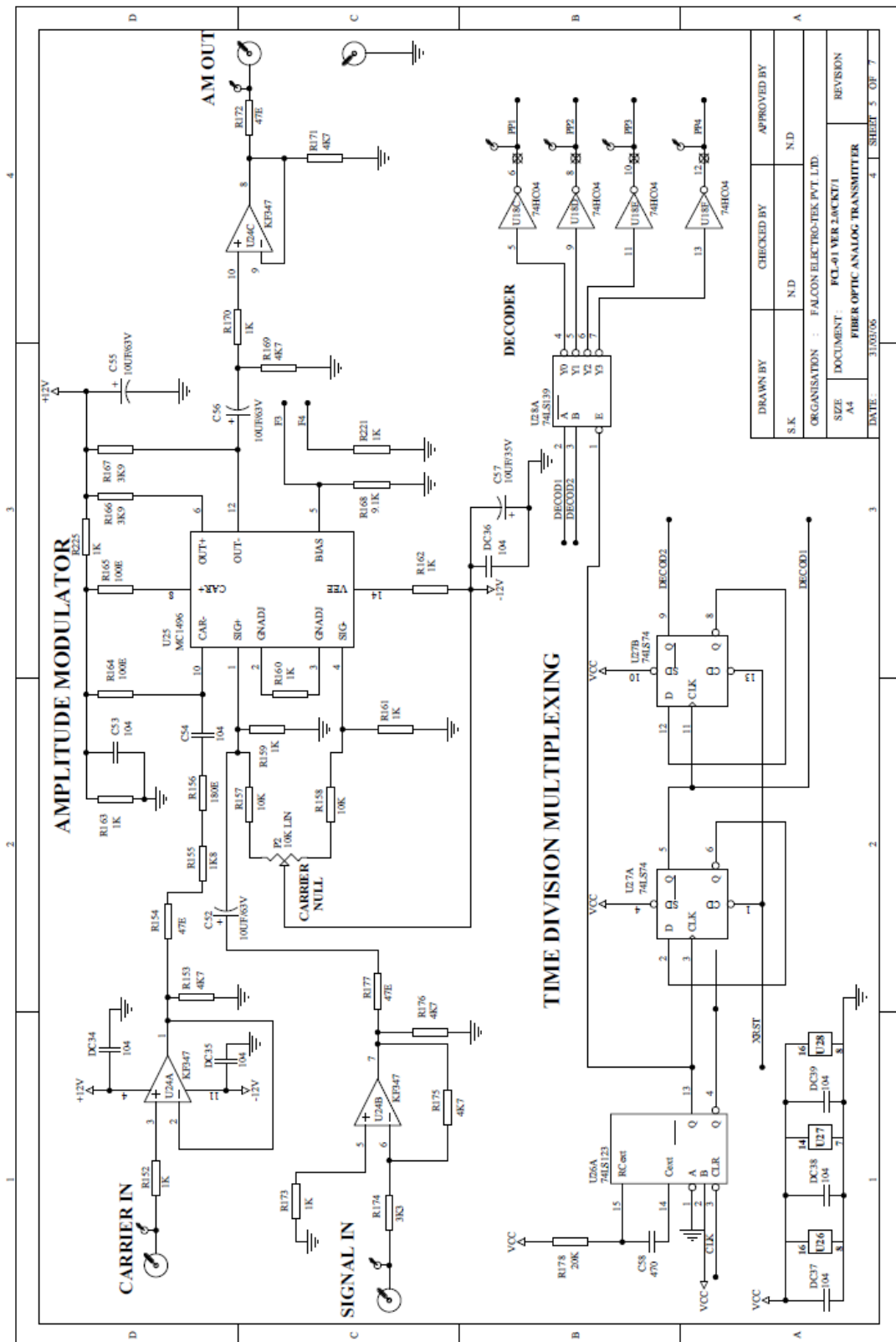
FCL-01

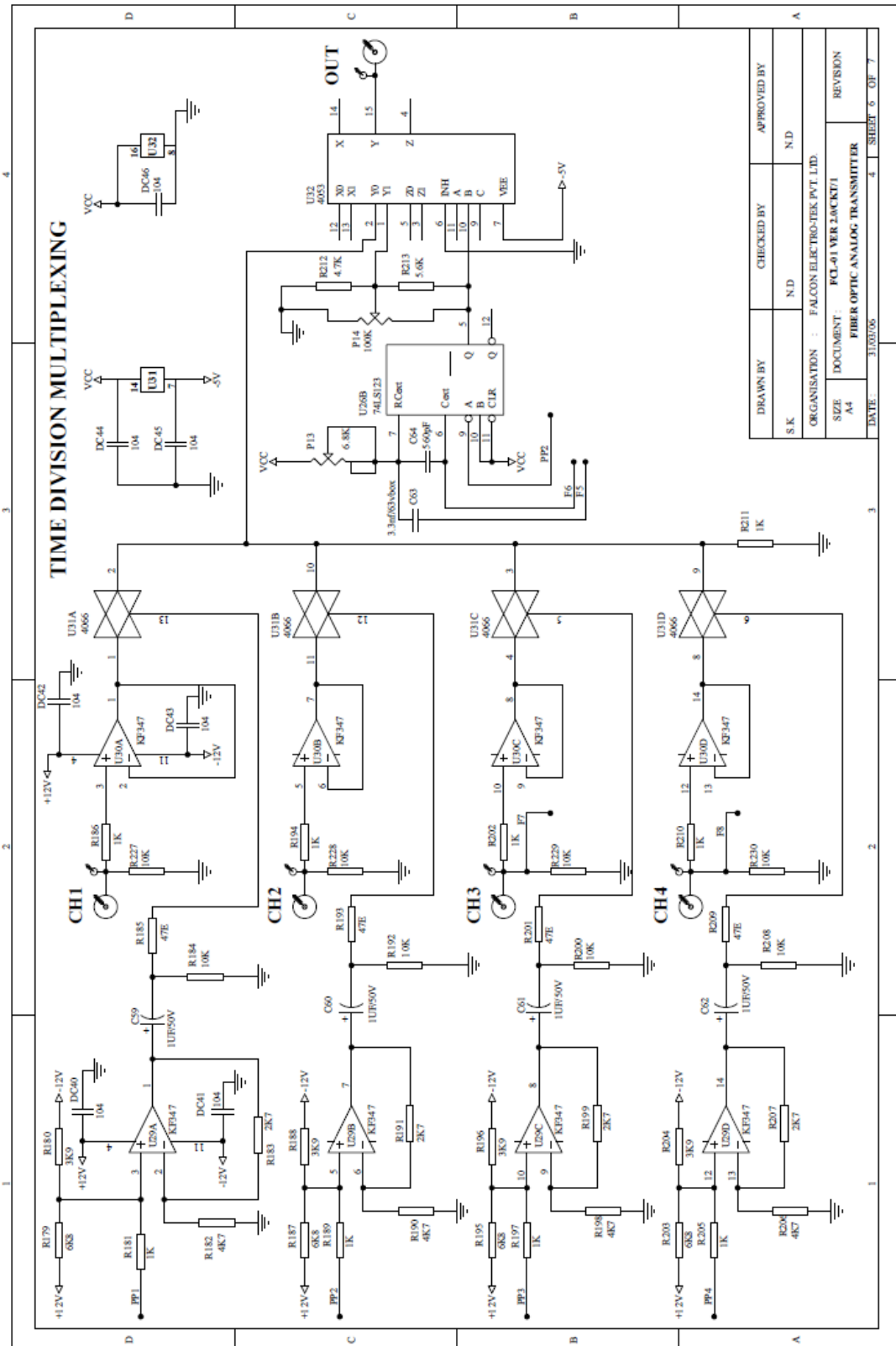


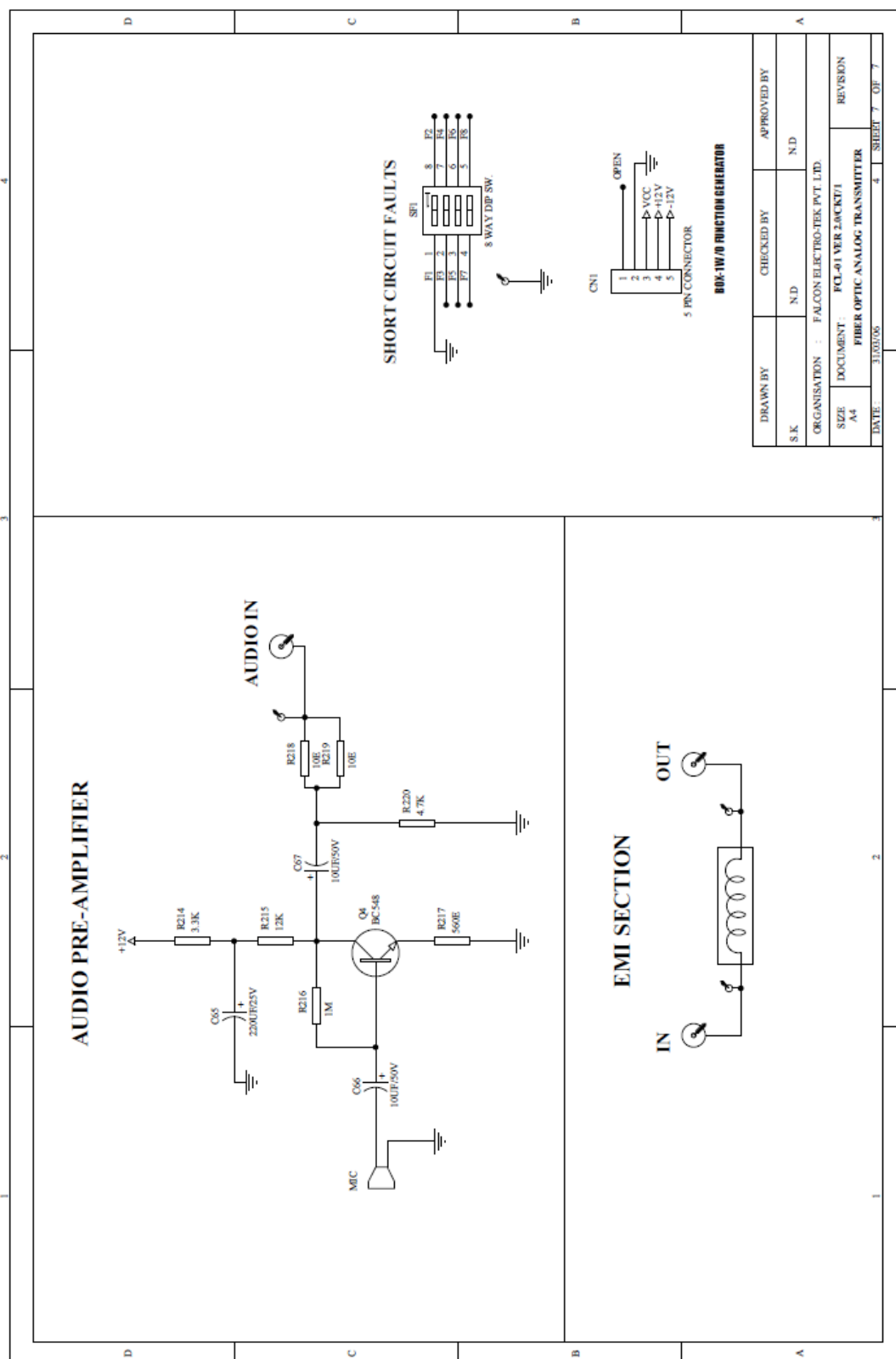




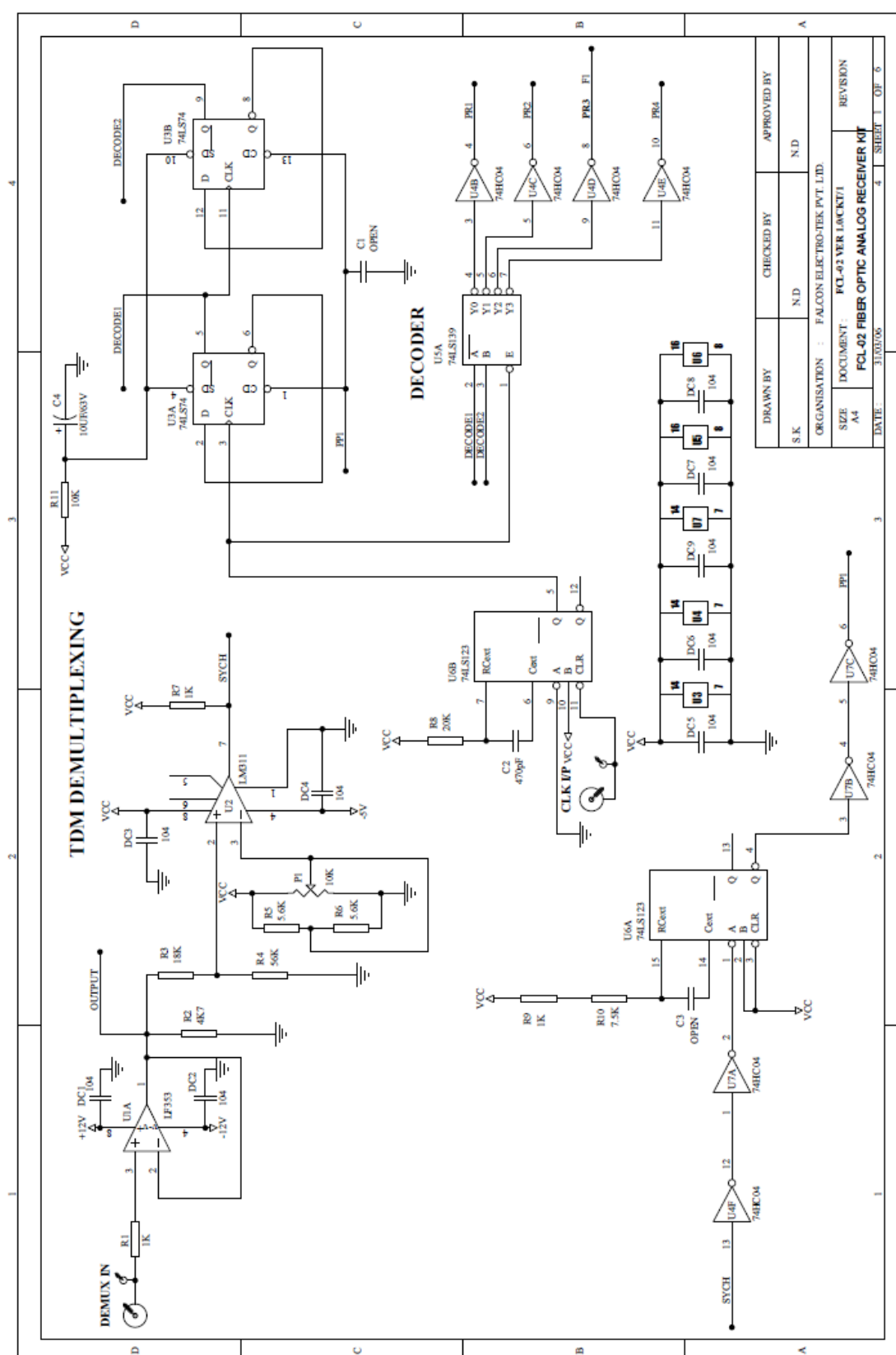


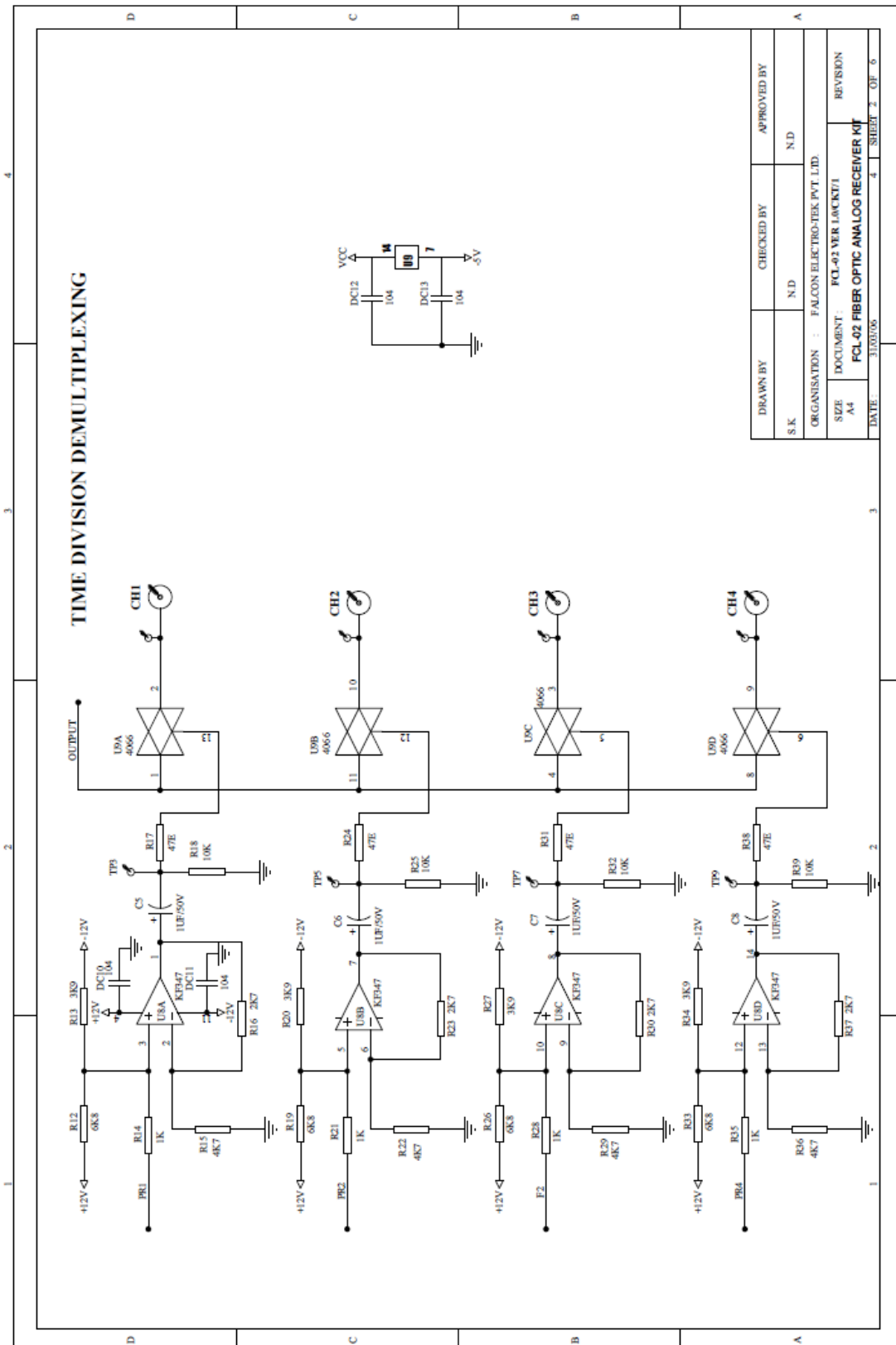


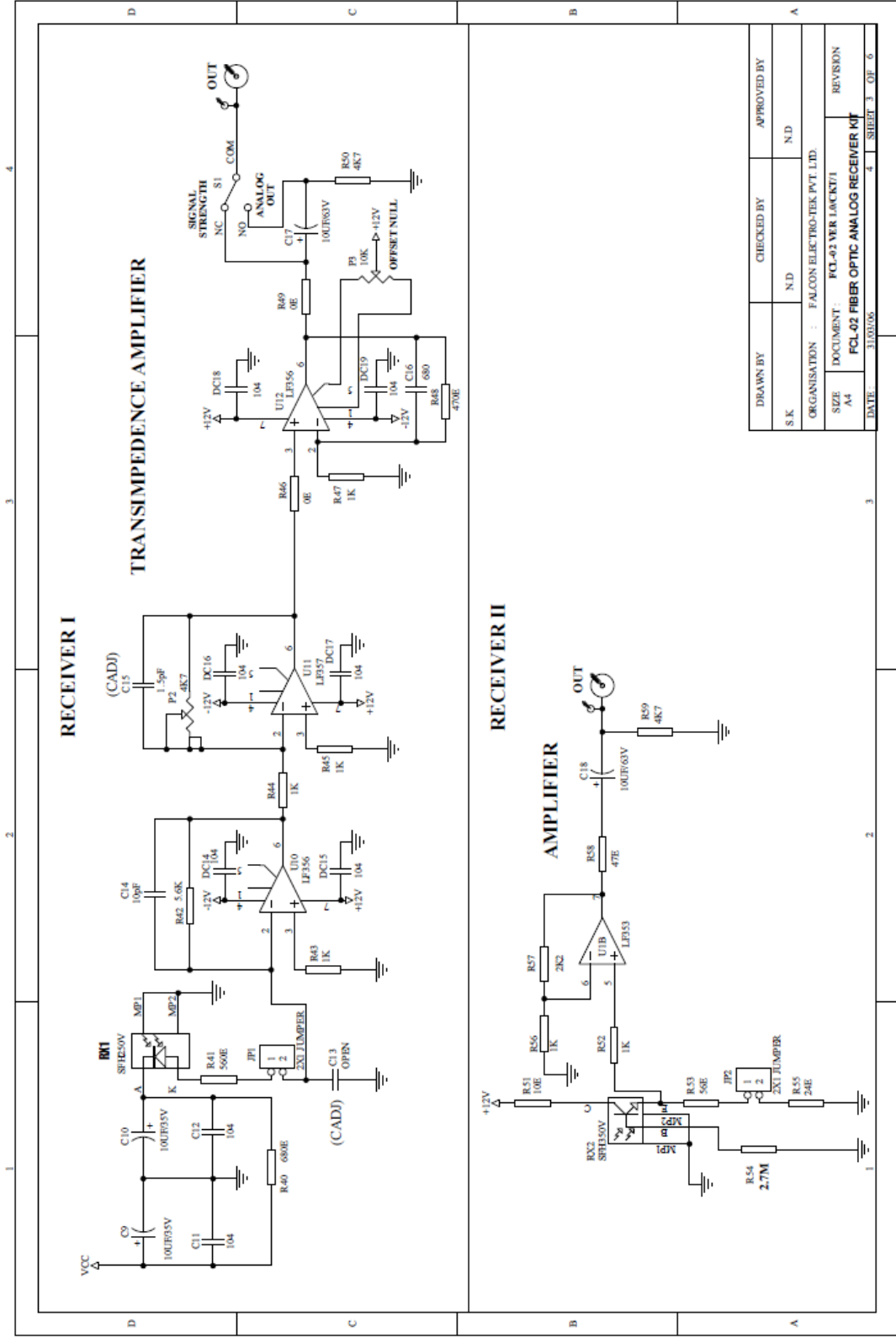


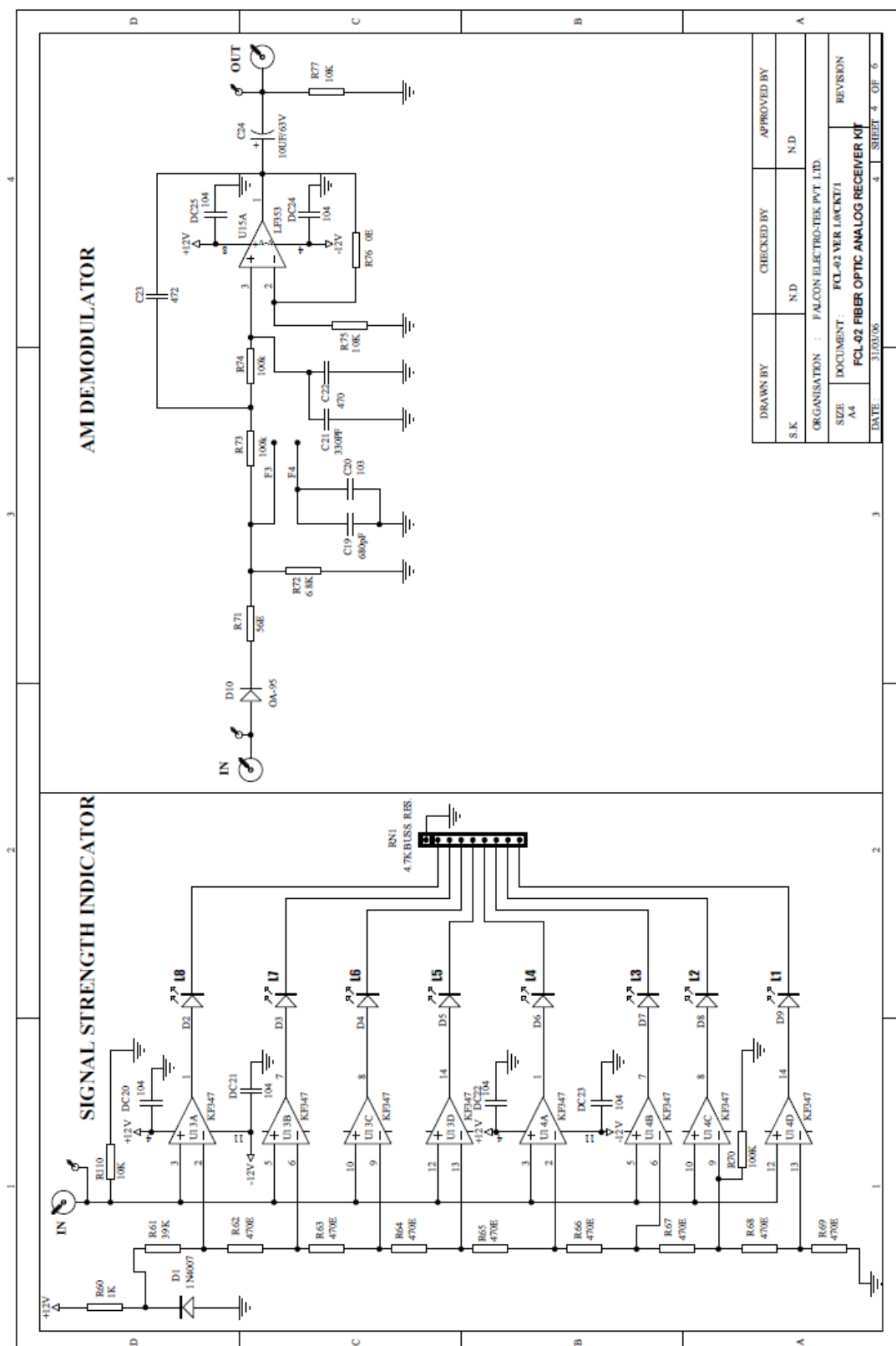


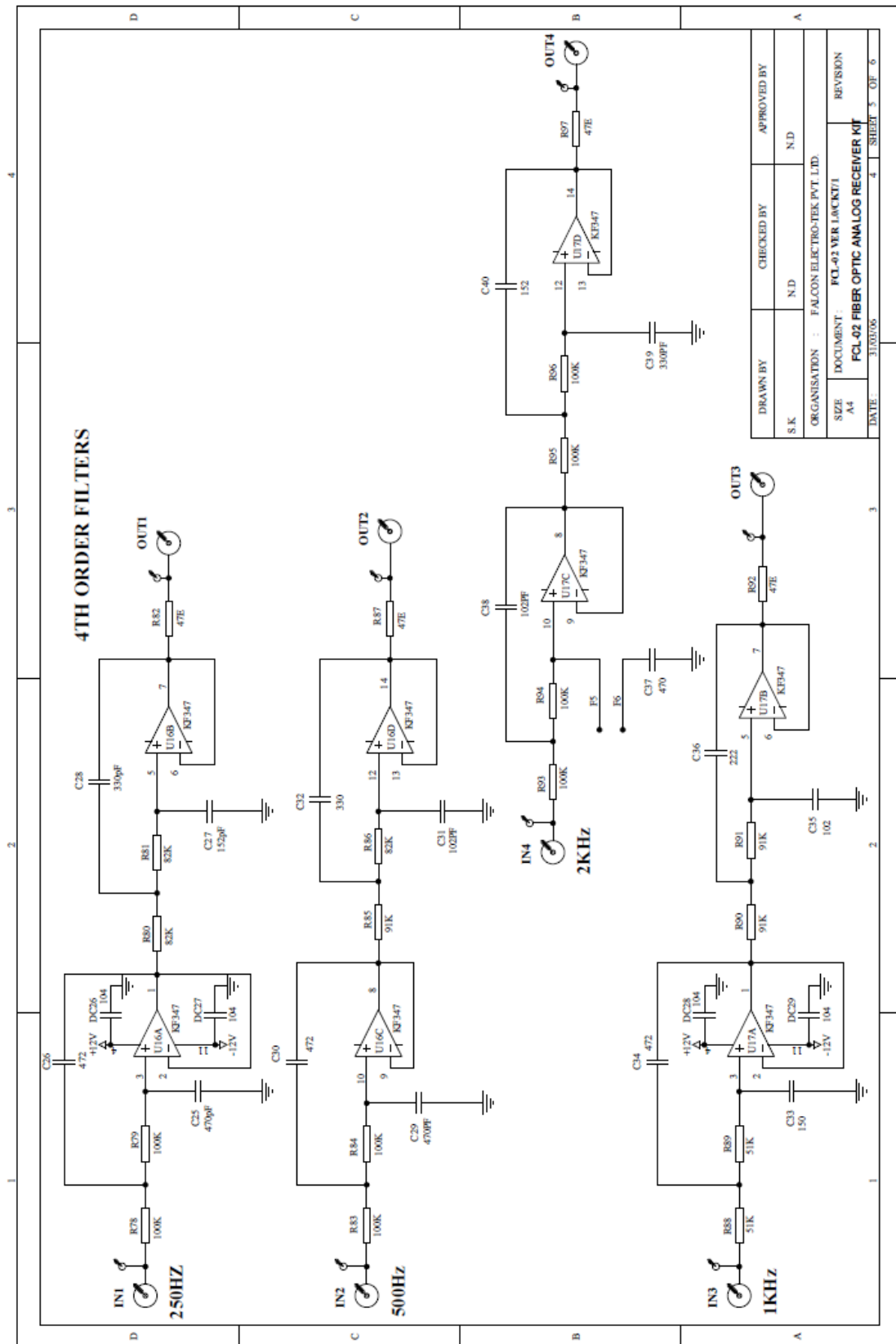
FCL-02

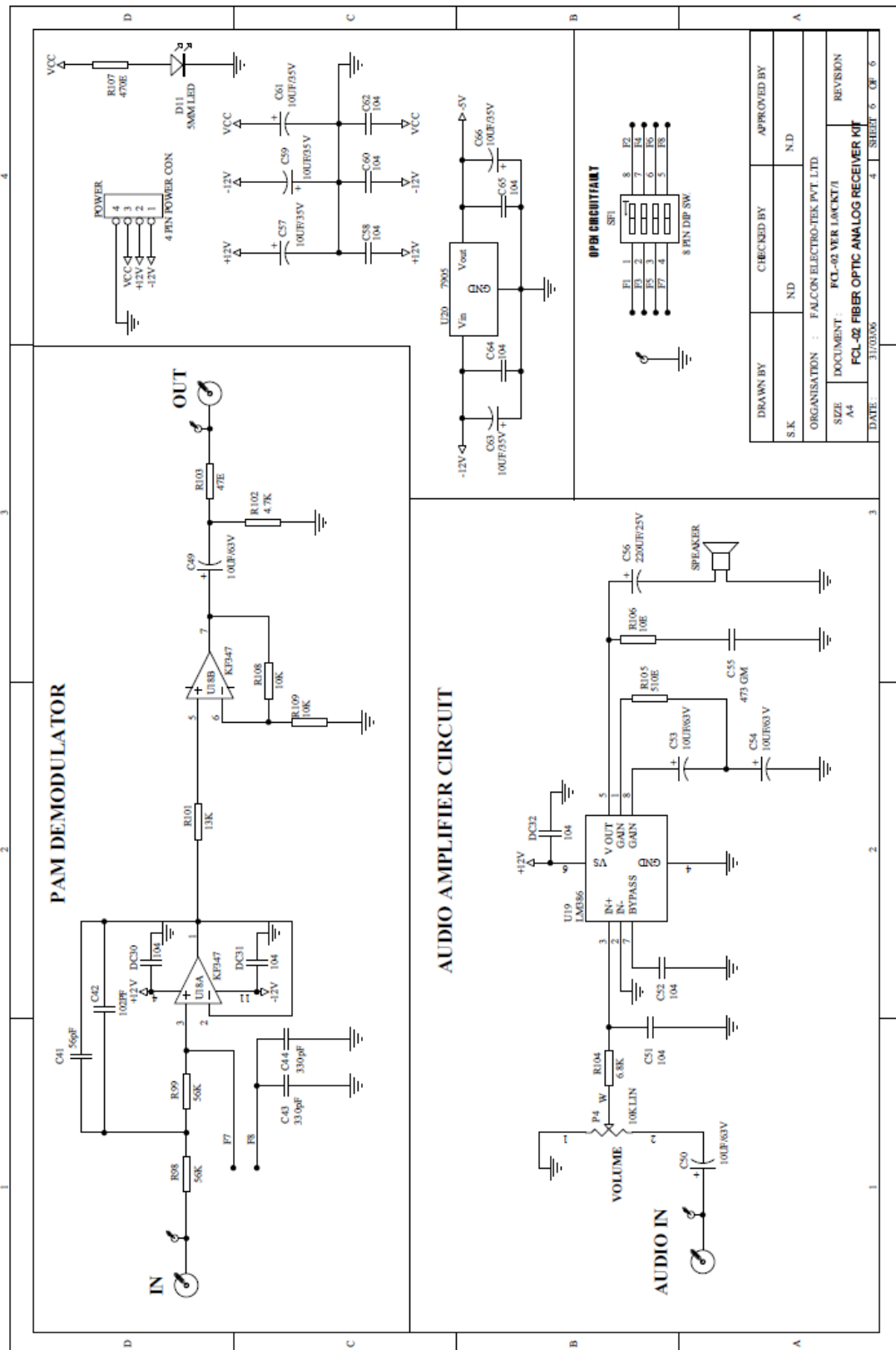




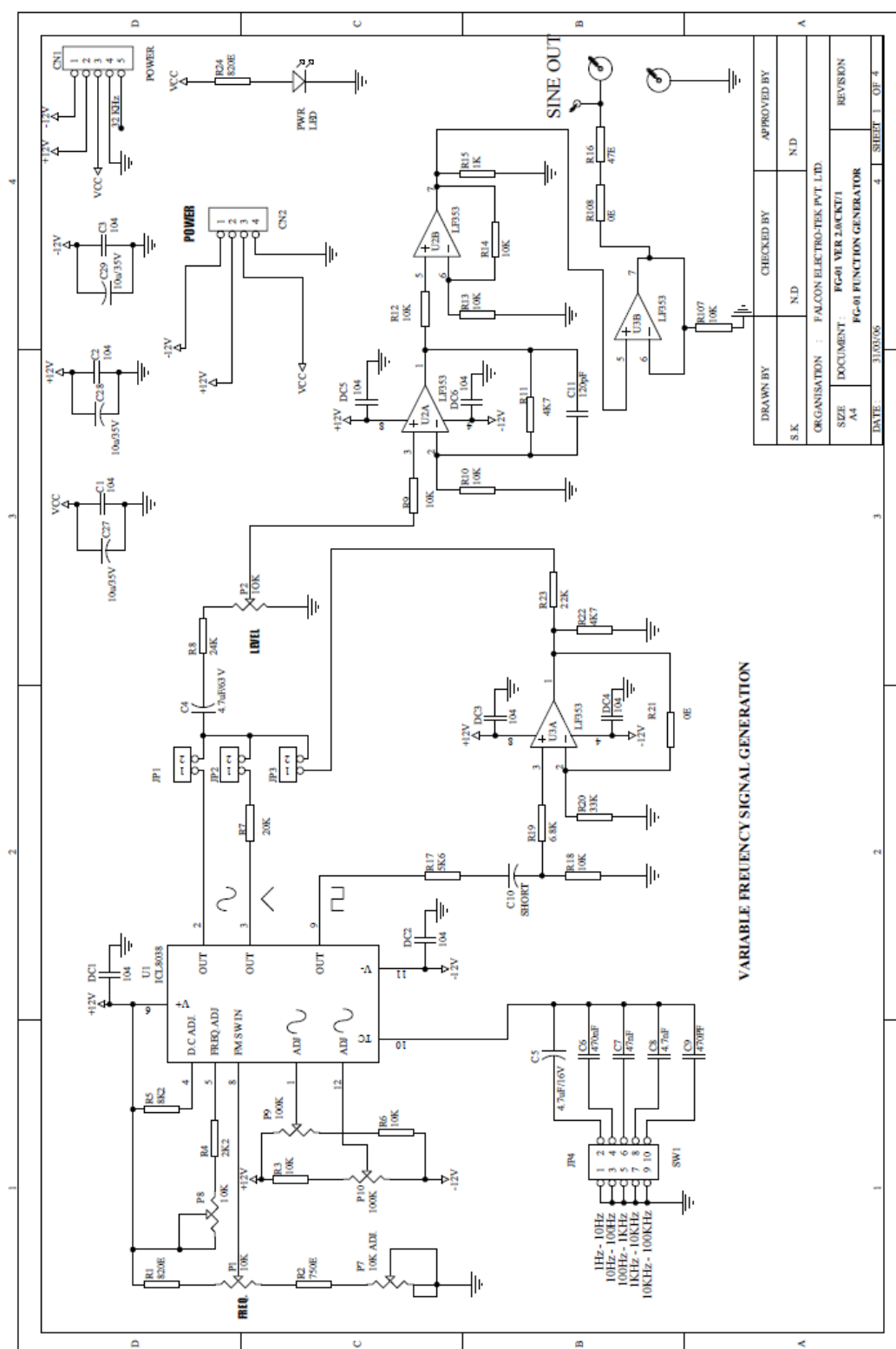


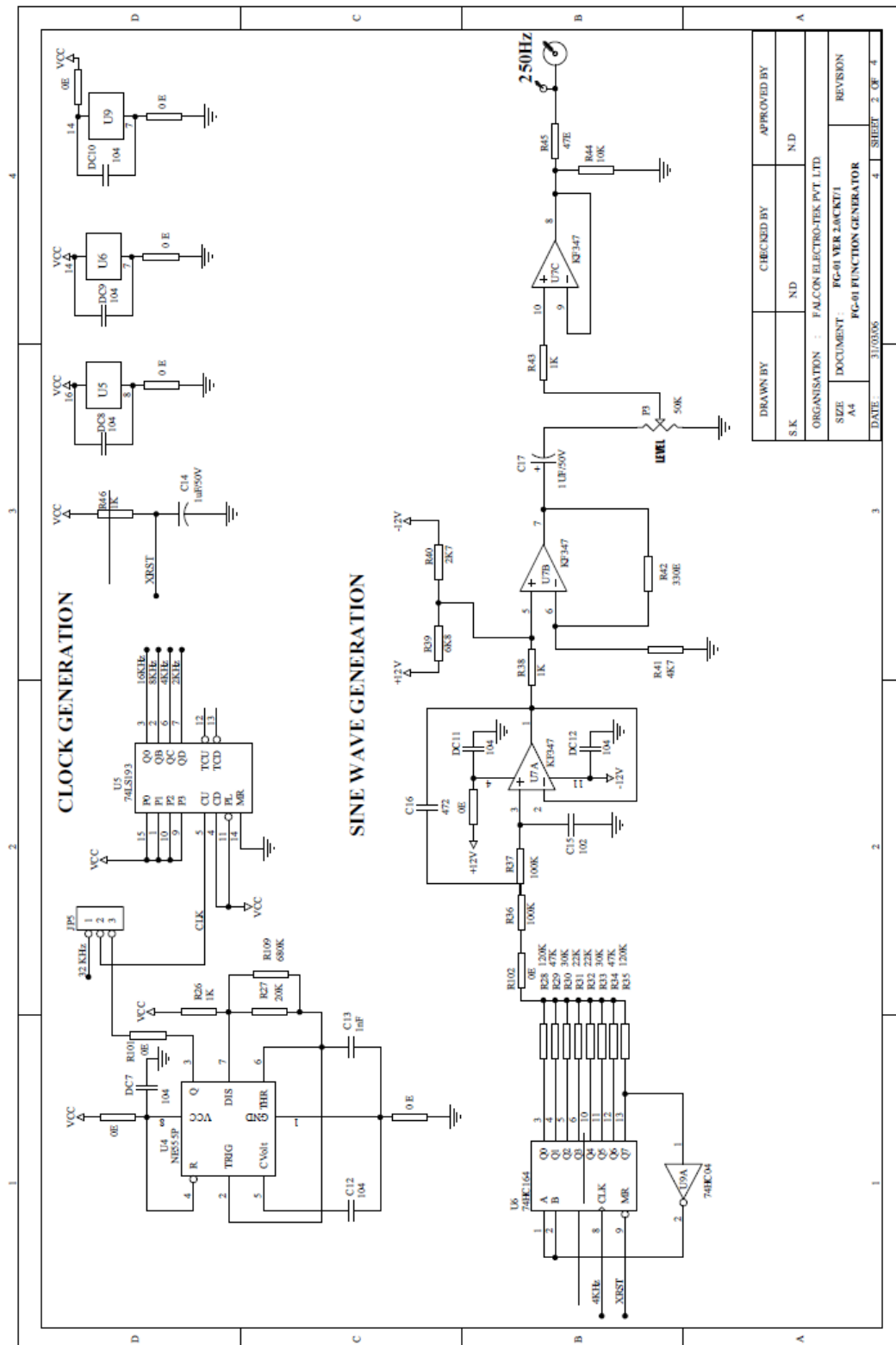


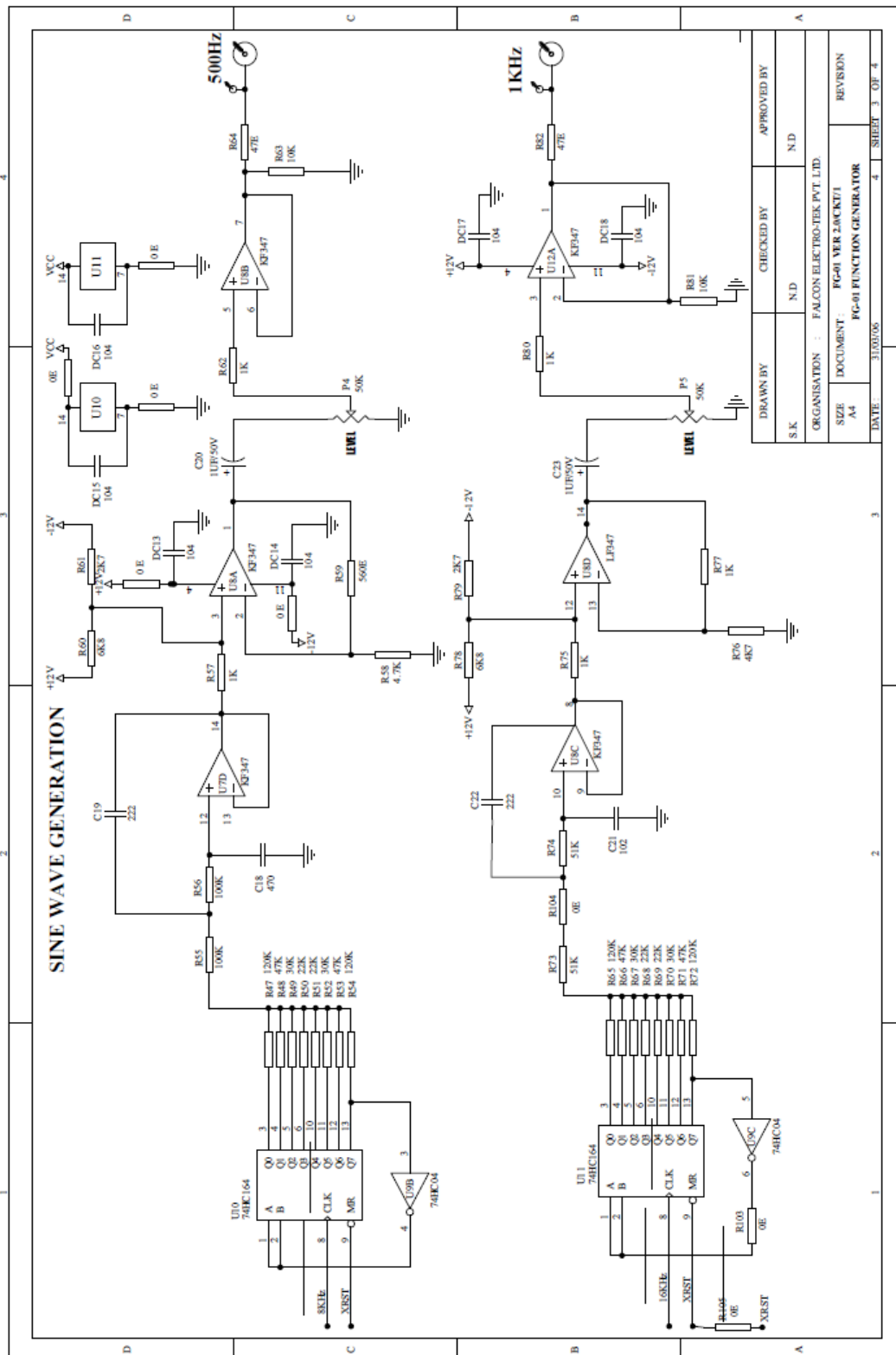


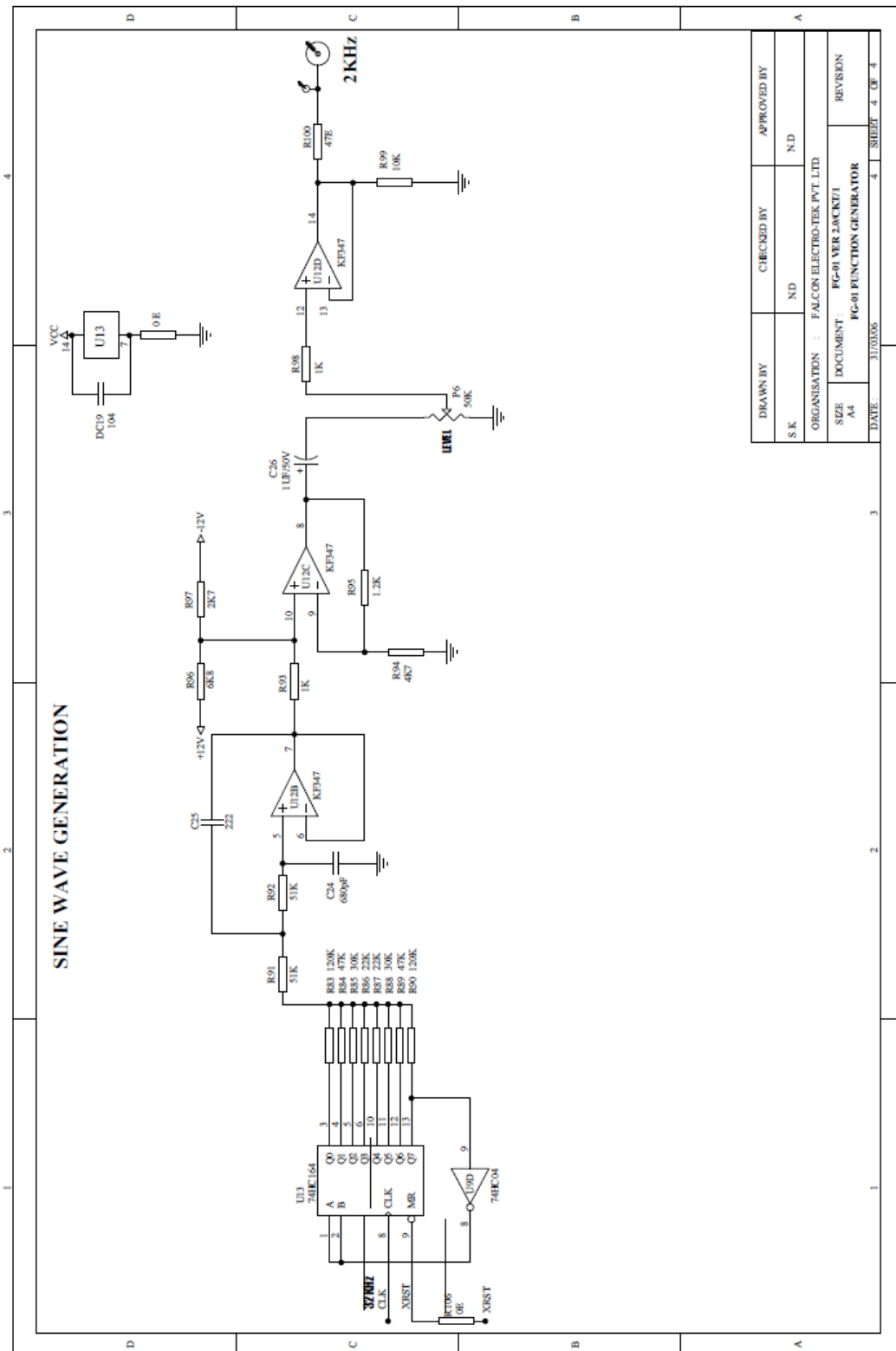


FG-01

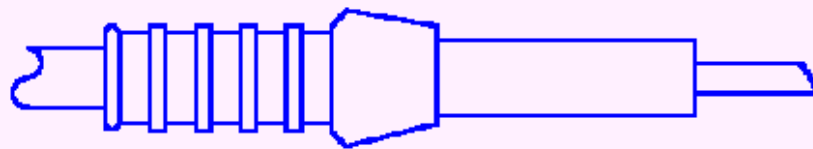
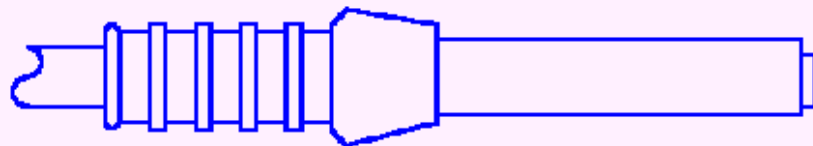








EXPERIMENT NO.1

**Figure 1.1****Figure 1.2****Figure 1.3**

EXPERIMENT NO: 1

NAME

Initial Fiber End Preparation (Connectorization)

OBJECTIVE

Study of initial fiber end preparations and connecting plastic fiber to the connector

THEORY

In order to use the pieces of fiber provided in the trainer kit along with coupling jigs, the ends of the fiber have to be prepared so that the light can enter and leave the fiber easily. In this experiment, you will just prepare the ends so that the fibers can be used in the experiment follows.

EQUIPEMENTS

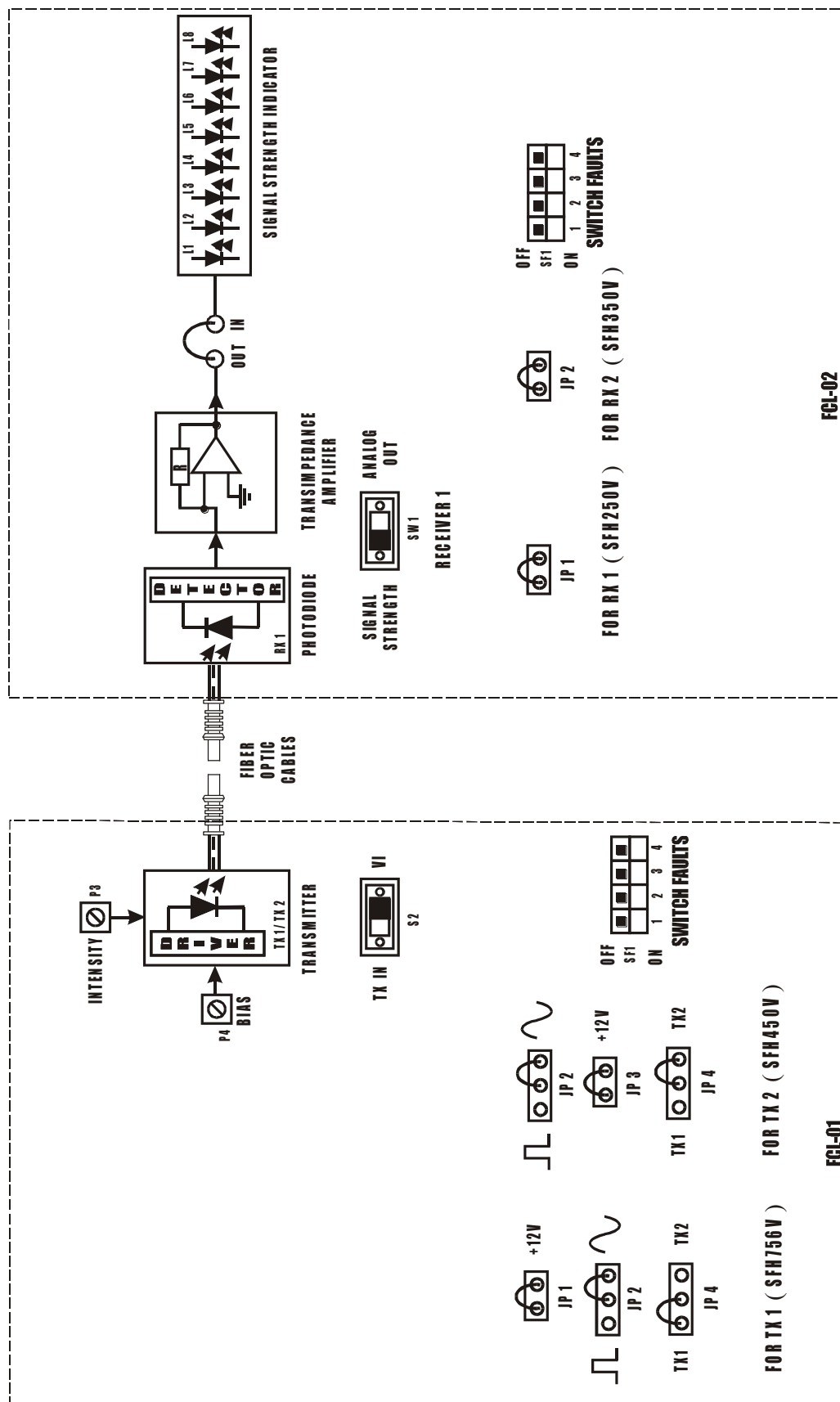
- 1 meter Fiber cable
- Connecting Plug - 02
- Polish papers
- Sharp knife or blade

PROCEDURE

- Remove 1 meter piece of fiber and two connecting plugs from the kit. Also remove the polishing papers to polish fiber ends & sharp knife to cut the fiber and remove the outer coating.
- Cut the fiber in two equal parts i.e. 0.5-meter length each.
- Cut off the very ends of the fibers (about 1mm or 1/16") with scissors or a knife (The end will probably be quite uneven). Cut around the outside of the fiber about 1/2" or 10mm from the end so that you cut through the black plastic but NOT in to the clear fiber. Pull the cut piece of black plastic off the ends of the fibers (see fig. 1.1). Insert a connector onto each of the fiber ends. Firmly push the fiber up through the connector plugs until a small portion appears at the ferrule's end (see fig. 1.2) Note that the connector housing has a metal sleeve in the lower ringed portion. This sleeve grips the fiber and holds it in the connectors.
- Now take the polishing papers with the polishing surfaces on it from the kit. Using the rough side, polish the ends by moving the fiber in the figure 8

- motions until they are almost flush with the ends of the connectors (see fig. 1.3). Take the smooth surface of the polish paper and polish the ends. It often helps to put a small drop of water on the ends of the fiber while polishing. When you complete polishing, the ends of the fibers should be smooth and look 'clear' with no scratches. Before you begin the experiments test your connectors to ensure they are properly polished.
- Keep un-connectorized end of the fiber in to the natural light and observe the other end, core should illuminate complete bright spot without any dark spot. If any dark spot appears, polish the ends of the fiber again. You might only need to use the smooth side of the polishing paper.

EXPERIMENT NO.2



EXPERIMENT NO: 2

NAME

Transmission through a Gap Between Fibers

OBJECTIVE

The objective of this experiment is to study transmission of light through a gap between two fibers.

THEORY

While connecting two fibers together, the connectors and connection sleeve do not always align with the fiber perfectly. Sometimes there is a gap between the fibers. The experiment will look at light transmission from one fiber to another when there is a gap between the fibers.

EQUIPEMENTS

FCL-01 & FCL-02

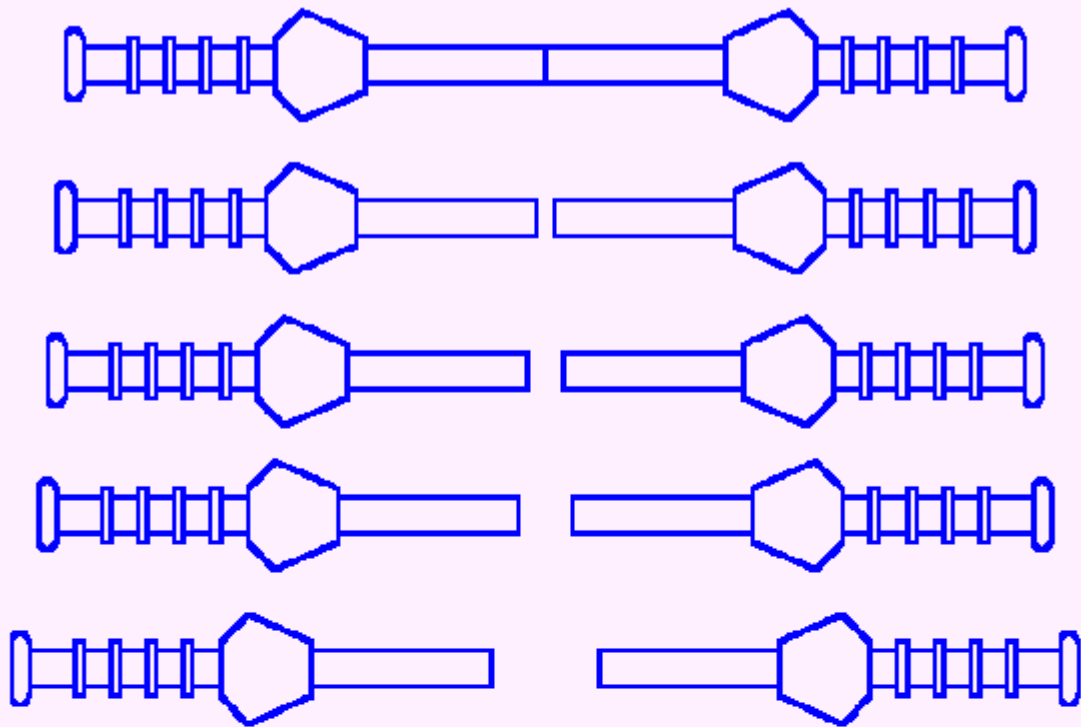
- 0.5 Meter Fibers with connectorized ends
- Patch chords
- Connection sleeve (Splicing unit)
- Power Supply (Use only one provided)

NOTE: Keep All Switch Faults In Off Position.

PROCEDURE

- Make connections as shown in fig.2.1. Connect the power supply with proper polarity to FCL-01 & FCL-02 Kits. While connecting this, ensure that the power supply is OFF.
- Keep pot **P4** fully clockwise position and pot **P3** fully anticlockwise position.
- Keep the jumpers **JP1, JP2, JP3 & JP4** on FCL-01 as shown in fig.2.1.
- Keep the jumpers **JP1 & JP2** on FCL-02 as shown in fig.2.1.
- Keep switch **S2** in **VI** position on FCL-01.
- Switch on the power supply.

- Slightly unscrew the cap of LED SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the 0.5Meter fiber unconnectorized end into the cap. Now tighten the cap by screwing it back.



Cap	Signal Strength
Touching	
1 mm	
2 mm	
3 mm	
4 mm	

Fig.2.2

- Insert the connectorized end of fiber into connecting sleeve.
- Take another fiber and Insert the connectorized end of fiber into other side of connecting sleeve.

Slightly unscrew the cap of Photo Diode SFH250V. Do not remove the cap from the connector. Once the cap is loosened, insert the unconnectorized end of second 0.5Meter fiber into the cap. Now tighten the cap by screwing it back.

- Keep switch **SW2** in **SIGNAL STRENGTH** position on FCL-02.
- Connect the output of Photo Diode detector post **OUT** to post **IN** of Signal Strength Indicator block.
- Observe the signal strength LEDs, adjust the TRANSMITTER LEVEL using Intensity control pot **P3** until you get the reading of all LED's glow. Now remove the fibers from the connection sleeve.
- Line up the connectors as per the fig.2.2 with the connectors touching. Move the connectors so that you get the maximum signal. What reading you get on the signal strength now? Note down the reading. Then increase the gap between connectors in steps of 1mm and note down the readings for every step.
- Plot the graph of Gap against LED glow.

EXPERIMENT NO.3

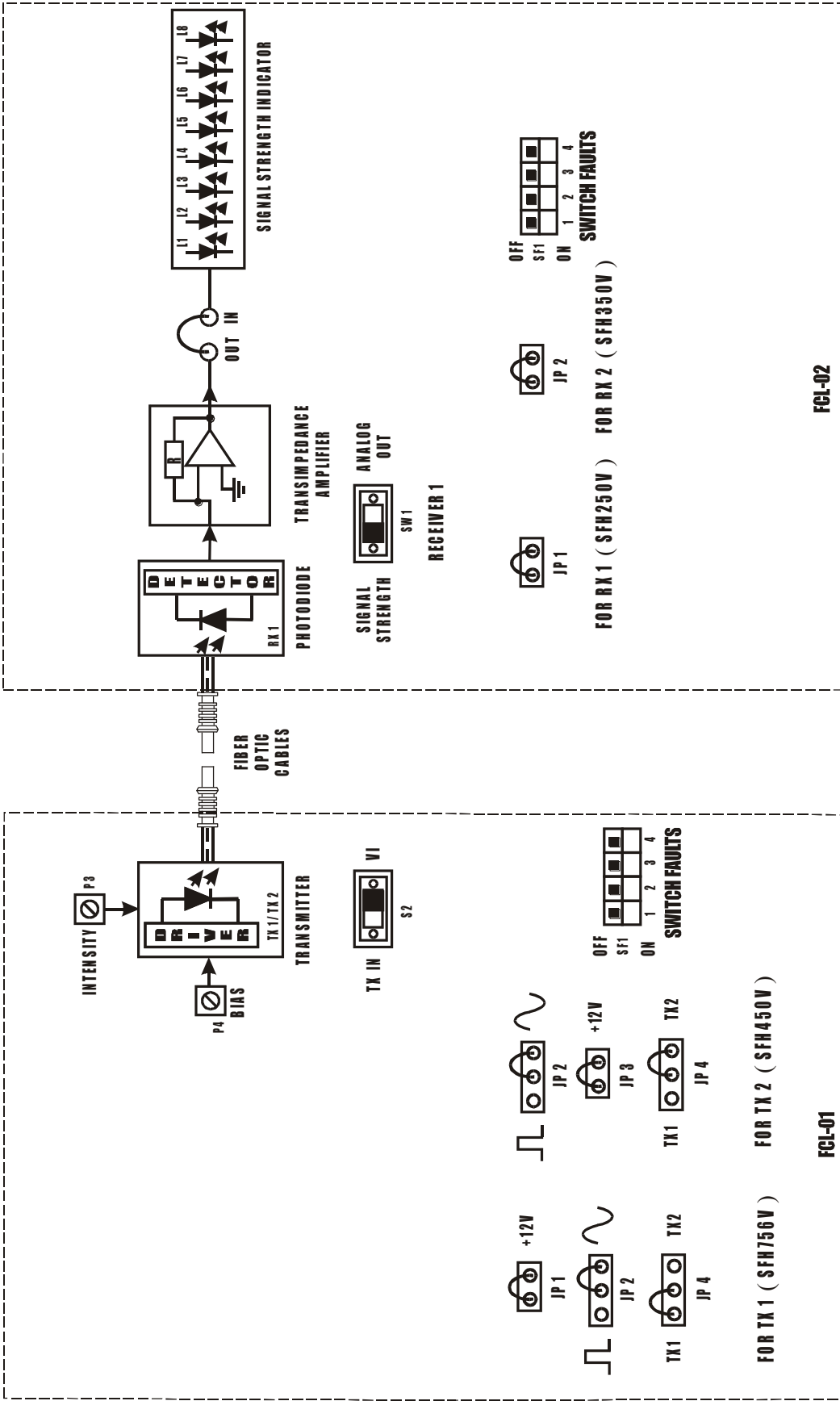


FIG. 3.1 BLOCK DIAGRAM FOR FIBER OPTIC TRANSMISSION SENSOR

EXPERIMENT NO: 3

NAME

Fiber Optic Transmission Sensor

OBJECTIVE

The objective of this experiment is to study fiber optic transmission sensor.

THEORY

Fiber optics and light are used extensively in sensing applications. On a production line, a fiber sensor may be used to detect the presence of an object on a conveyer belt. The advantage of using light including the sensor doesn't have to touch the object and light cannot create a spark, so it can be used in hazardous environments.

In this experiment, you will look at a transmission sensor system for detecting objects. It is called a transmission system because the object to be sensed affects the transmission of light. The other type of sensor, which is discussed in the next experiment, is the reflection sensor.

EQUIPEMENTS

FCL-01 & FCL-02

- 0.5 meter Fibers with connectorized ends
- Patch chords
- Power Supply (Use only one provided)

NOTE: Keep All Switch Faults In Off Position.

PROCEDURE

- Make connections as shown in fig.3.1. Connect the power supply cables with proper polarity to FCL-01 & FCL-02 Kits. While connecting this, ensure that the power supply is OFF.
- Keep pot **P4** fully clockwise position and pot **P3** fully anticlockwise position.
- Keep the jumpers **JP1**, **JP2**, **JP3** & **JP4** on FCL-01 as shown in fig.3.1.
- Keep the jumpers **JP1** & **JP2** on FCL-02 as shown in fig.3.1.
- Keep switch **S2** in **VI** position on FCL-01.

- Switch on the power supply.
- Slightly unscrew the cap of LED SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the 0.5 Meter fiber unconnectorized end into the cap. Now tighten the cap by screwing it back.
- Slightly unscrew the cap of Photo Diode SFH250V. Do not remove the cap from the connector. Once the cap is loosened, insert the unconnectorized end of second 0.5meter fiber into the cap. Now tighten the cap by screwing it back.
- Keep switch **SW2** in **SIGNAL STRENGTH** position on FCL-02.
- Connect the output of Photo Diode detector post **OUT** to post **IN** of Signal Strength Indicator block.
- Hold the connectorized fiber ends a short distance apart (approx. 3mm). You can hold the fiber ends in place using a stack of books.
- Adjust the transmitter level with help of **INTENSITY** pot **P3** to a maximum value and The SIGNAL STRENGTH indicator should read about 2 or 3. If it doesn't then try to adjust the position of the connectors. Now when you pass an object between the fiber ends, the SIGNAL LED will go out, indicating there is an object between the fiber ends. Check how far apart can you make the ends and still able to sense objects?

EXPERIMENT NO.4

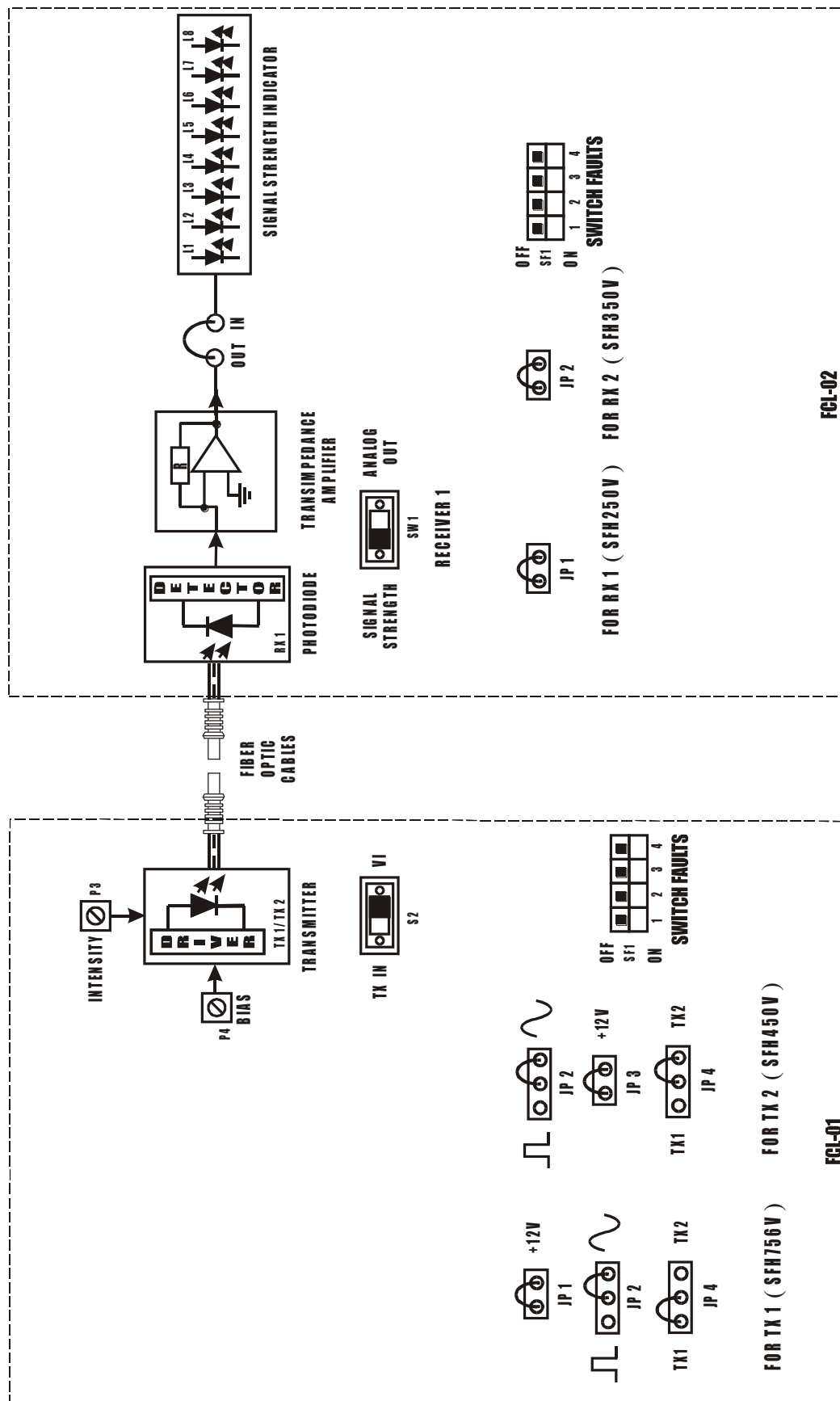


FIG. 4.1 BLOCK DIAGRAM FOR FIBER OPTIC REFLECTION SENSOR

EXPERIMENT NO: 4

NAME

Fiber Optic Reflection Sensor

OBJECTIVE

The objective of this experiment is to study fiber optic reflection sensor.

THEORY

In this experiment you will look at the REFLECTION SENSOR system for detecting objects. It is called a reflection system because the object to be sensed reflects or doesn't reflect light.

EQUIPEMENTS

FCL-01 & FCL-02

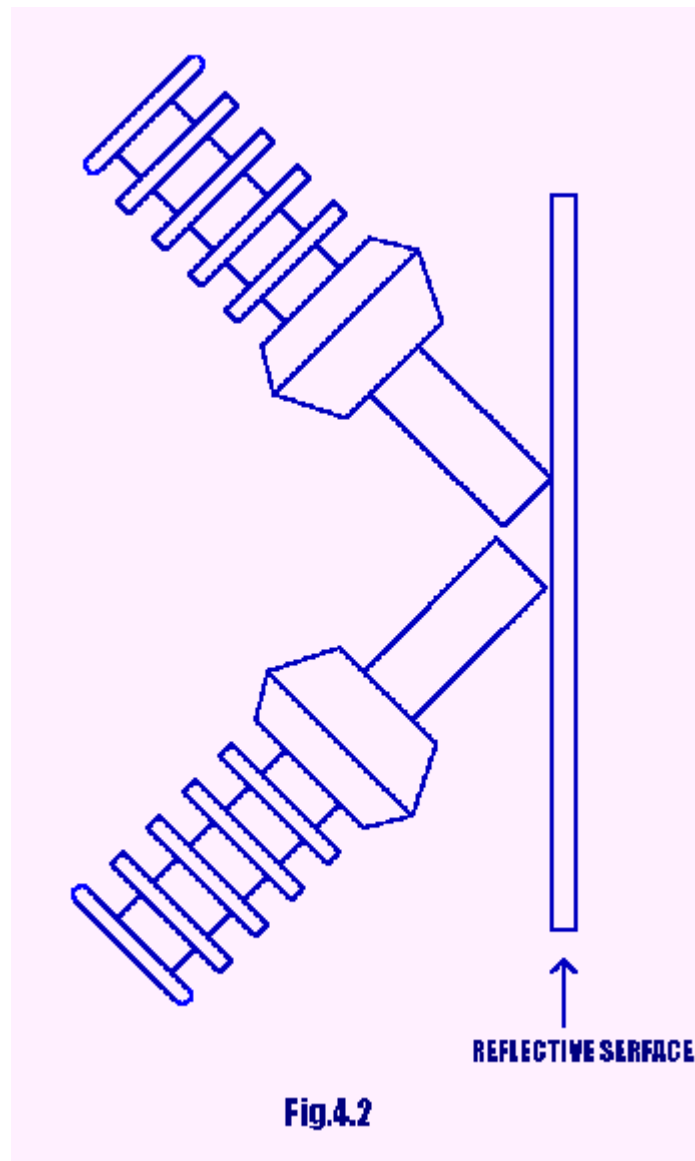
- 0.5 meter Fibers with connectorized ends
- Patch chords
- Mirror
- Power Supply (Use only one provided)

NOTE: Keep All Switch Faults In Off Position.

PROCEDURE

- Make connections as shown in fig.4.1. Connect the power supply cables with proper polarity to FCL-01 & FCL-02 Kits. While connecting this, ensure that the power supply is OFF.
- Keep pot **P4** fully clockwise position and pot **P3** fully anticlockwise position.
- Keep the jumpers **JP1**, **JP2**, **JP3** & **JP4** on FCL-01 as shown in fig.4.1.
- Keep the jumpers **JP1** & **JP2** on FCL-02 as shown in fig.4.1.
- Keep switch **S2** in **VI** position on FCL-01.
- Switch on the power supply.
- Slightly unscrew the cap of LED SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the 0.5meter fibers un-connectorized end into the cap. Now tighten the cap by screwing it back.

- Slightly unscrew the cap of Photo Diode SFH250V. Do not remove the cap from the connector. Once the cap is loosened, insert the unconnectorized end of second 0.5meter fibers into the cap. Now tighten the cap by screwing it back.



- Keep switch **SW2** in **SIGNAL STRENGTH** position on FCL-02.
- Connect the output of Photo Diode detector post **OUT** to post **IN** of Signal Strength Indicator block.
- The signal reflected from objects is usually quite small. In order to do this experiment you need to have very shiny surfaces to reflect enough light from one fiber to another. Align the two fibers up as shown in the fig.4.2. Turn the INTENSITY pot **P3** fully clockwise. Place an object mirror one provided or like a shiny piece of metal at the place shown in the figure. Adjust the fiber position relative to the surface until you get a reading on the SIGNAL STRENGTH. Now the presence of the object will turn the one of the SIGNAL STRENGTH LEDs ON and the absence will turn the LED OFF. Why is this the reverse of the last experiment, where the presence of the object turned the LED off?

EXPERIMENT NO.5

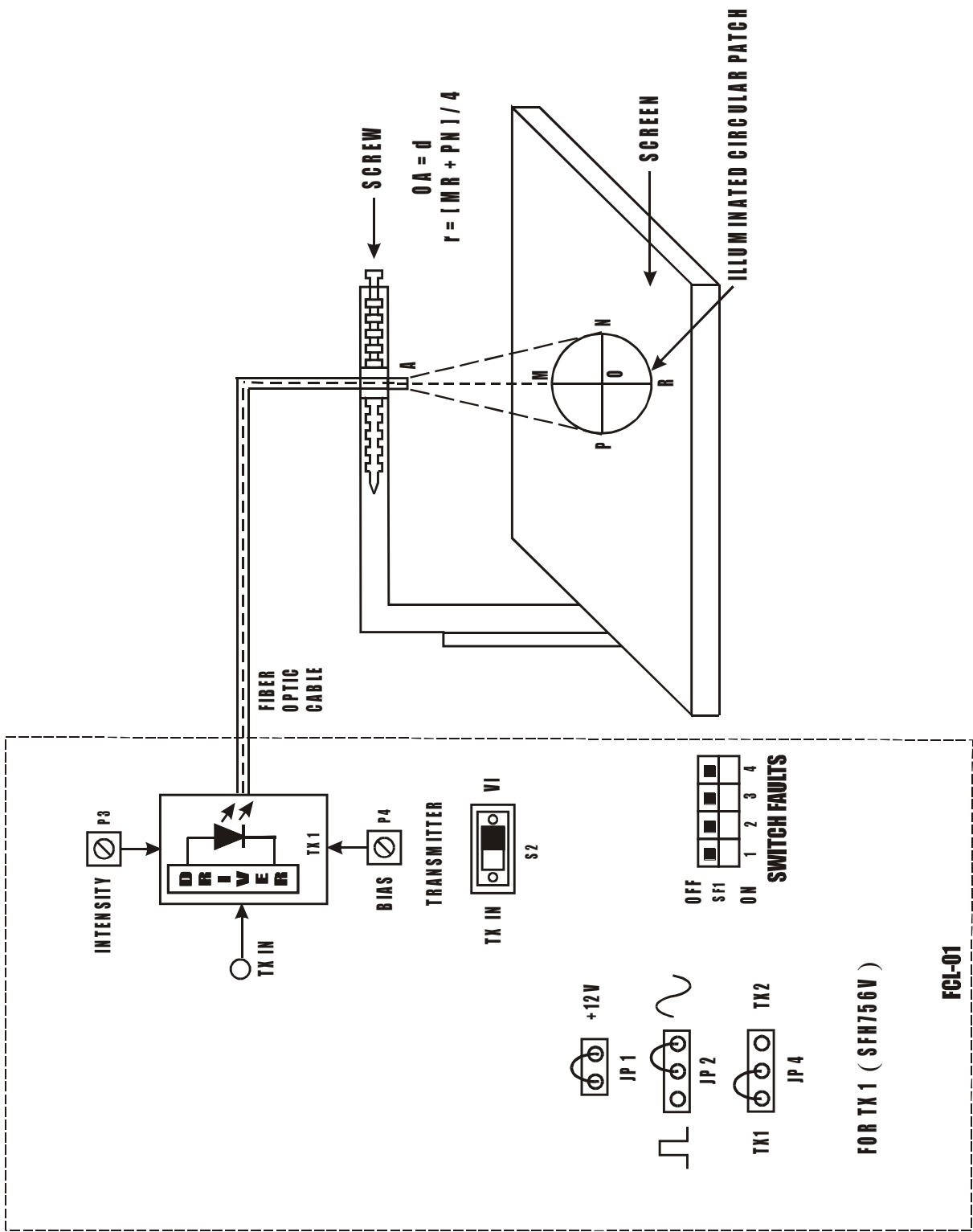


FIG. 5.1 BLOCK DIAGRAM FOR NUMERICAL APERTURE MEASUREMENT

EXPERIMENT NO: 5

NAME

Study of Numerical Aperture of Optical Fiber

OBJECTIVE

The objective of this experiment is to measure the numerical aperture of the plastic fiber provided with the kit using 660 nm wavelength LED.

THEORY

Numerical aperture refers to the maximum angle at which the light incident on the fiber end is totally internally reflected and is transmitted properly along the fiber. The cone formed by the rotation of this angle along the axis of the fiber is the cone of acceptance of the fiber. The light ray should strike the fiber end within its cone of acceptance, else it is refracted out of the fiber core.

Considerations in Na Measurement

- It is very important that the optical source should be properly aligned with the cable & the distance from the launched point & the cable should be properly selected to ensure that the maximum amount of optical power is transferred to the cable.
- This experiment is best performed in a less illuminated room.

EQUIPMENTS

FCL-01

- 1-meter fiber cable
- NA JIG (Precision Jig No.1)
- Ruler
- Power Supply (Use only one provided)

NOTE: Keep All Switch Faults In Off Position.

PROCEDURE

- Make connections as shown in fig.5.1. Connect the power supply cables with proper polarity to FCL-01 Kit. While connecting this, ensure that the power supply is OFF.
- Slightly unscrew the cap of LED SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the fiber into the cap. Now tighten the cap by screwing it back.
- Keep the jumpers **JP1**, **JP2&JP4** on FCL-01 as shown in fig.5.1.
- Keep switch **S2** in **VI** position on FCL-01.
- Switch on the power supply.
- Insert the other end of the fiber into the numerical aperture measurement Jig. Hold the white sheet facing the fiber. Adjust the fiber such that its cut face is perpendicular to the axis of the fiber.
- Keep the distance of about 10 mm between the fiber tip and the screen. Gently tighten the screw and thus fix the fiber in the place.
- Observe the bright red light spot on the screen by varying Intensity pot **P3** and Bias pot **P4**.
- Measure exactly the distance **d** and also the vertical and horizontal diameters **MR** and **PN** as indicated in the fig. 5.1.
- Mean radius is calculated using the following formula

$$r = (MR + PN) / 4$$
- Find the numerical aperture of the fiber using the formula

$$NA = \sin\theta_{\max} = r / \sqrt{d^2 + r^2}$$

where $\theta_{\max}$ is the maximum angle at which the light incident is properly transmitted through the fiber.

EXPERIMENT NO.6

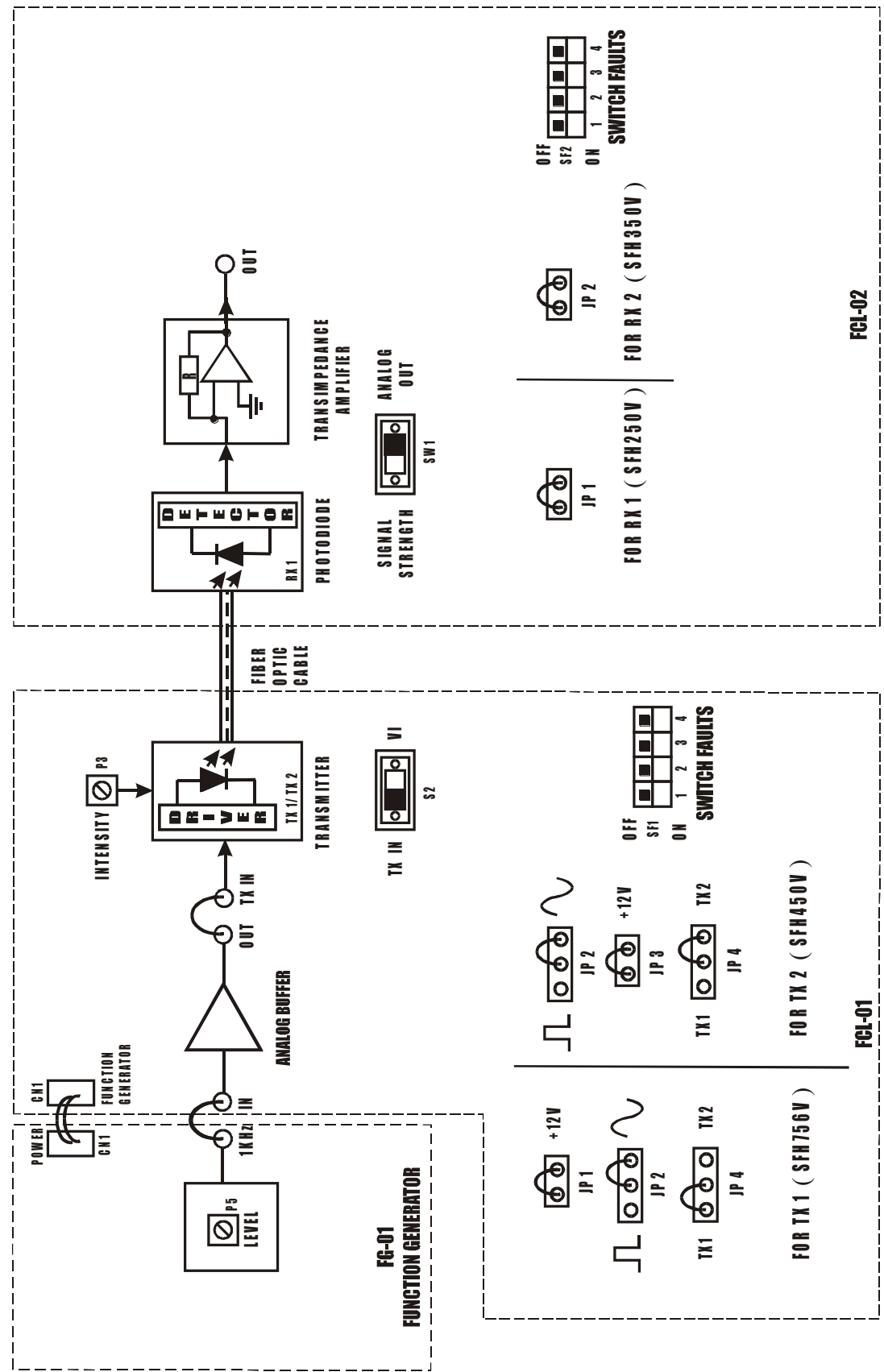


FIG. 6.1 BLOCK DIAGRAM FOR SETTING UP FIBER OPTIC ANALOG LINK

EXPERIMENT NO: 6

NAME

Setting Up a Fiber Optic Analog Link

OBJECTIVE

The objective of this experiment is to study Fiber Optic Analog Link. In this experiment, you will study the relationship between the input signal & received signal.

THEORY

Fiber Optic Links can be used for transmission of digital as well as analog signals. Basically, a fiber optic link contains three main elements, a transmitter, an optical fiber & a receiver. The transmitter module takes the input signal in electrical form & then transforms it into optical (light) energy containing the same information. The optical fiber is the medium, which carries this energy to the receiver. At the receiver, light is converted back into electrical form with the same pattern as originally fed to the transmitter.

EQUIPMENTS

FCL-01 & FCL-02

FG-01 with power cable

- 1 Meter Fiber cable
- Patch chords
- Power Supply (Use only one provided)
- 20 MHz Dual Channel Oscilloscope

NOTE: Keep All Switch Faults In Off Position.

PROCEDURE

- Make connections as shown in fig.6.1. Connect the power supply cables with proper polarity to FCL-01 & FCL-02 Kits. While connecting this, ensure that the power supply is OFF.
- Connect Function Generator FG-01 to FCL-01 using power cable.
- Keep the jumpers **JP1**, **JP2**, **JP3** & **JP4** on FCL-01 as shown in fig.6.1
- Keep the jumpers **JP1** & **JP2** on FCL-02 as shown in fig.6.1
- Keep switch **S2** in **TX IN** position on FCL-01.
- Switch on the power supply.
- Connect the **1KHz, 2Vpp** Signal from FG-01 as a constant signal to the **IN** post of Analog Buffer on FCL-01.
- Connect the output of Analog Buffer post **OUT** to post **TX IN**.
- Slightly unscrew the cap of LED SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the fiber into the cap. Now tighten the cap by screwing it back.
- Now rotate the Optical Power Control pot **P3** in FCL-01 in anticlockwise direction. This ensures minimum current flow through LED.
- Slightly unscrew the cap of Photo Diode SFH250V. Do not remove the cap from the connector. Once the cap is loosened, insert the other end of fiber into the cap. Now tighten the cap by screwing it back.
- Keep Switch **SW1** to **ANALOG OUT** position in FCL-02.
- Observe the output signal from the detector at **OUT** post on Oscilloscope by adjusting Optical Power Control Pot **P3** on FCL-01 and you should get the reproduction of the original transmitted signal.
- To measure the analog bandwidth of the link, keep the same connections and vary the frequency of the input signal from 100 Hz onwards. Measure the amplitude of the received signal for each frequency reading.
- Plot a graph of Gain/Frequency. Measure the frequency range for which the response is flat.
- Perform the above procedure again for all the combinations of Transmitter & Receiver.

EXPERIMENT NO.7

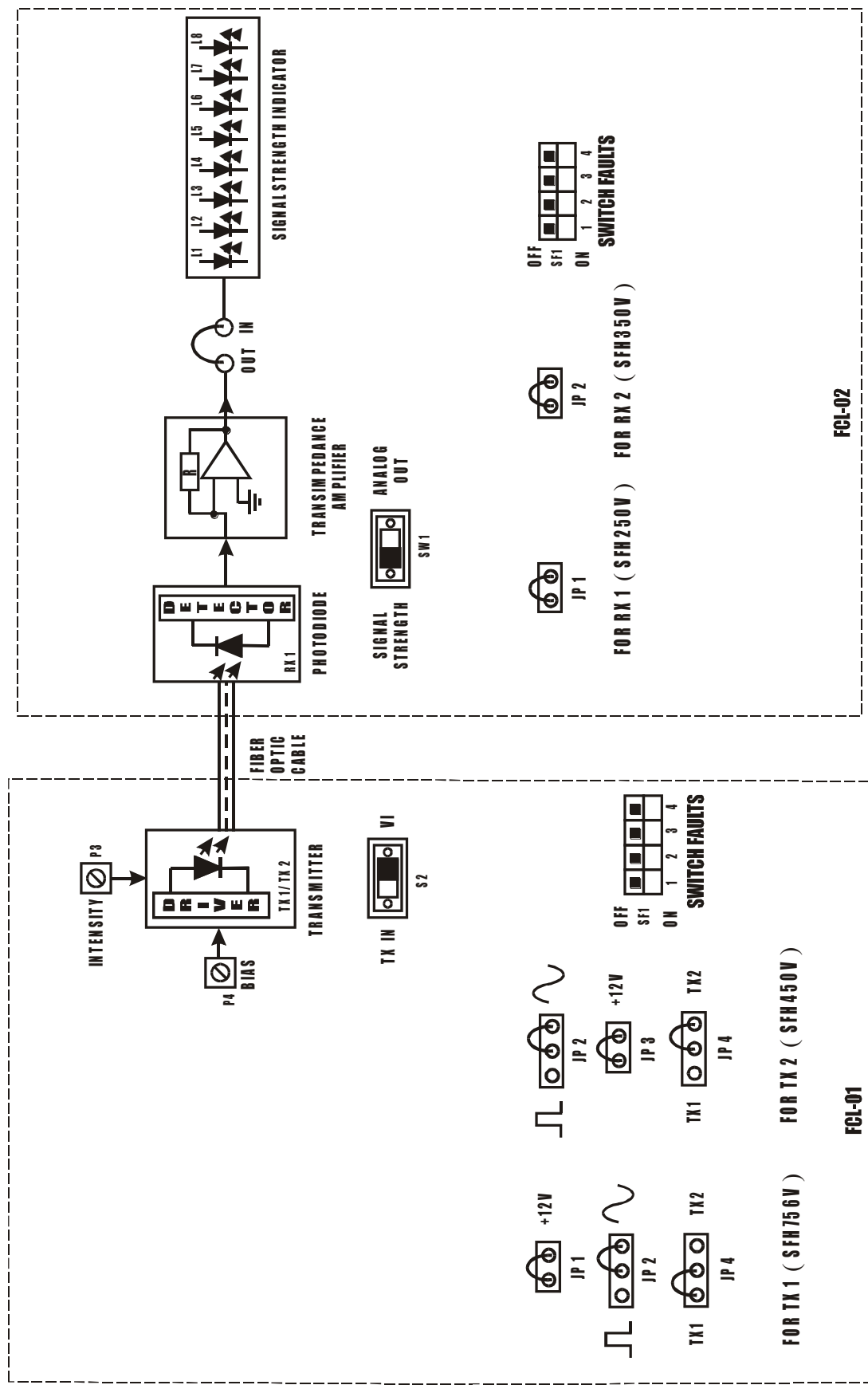


FIG. 7.1 BLOCK DIAGRAM FOR MEASUREMENT OF FIBER OPTIC LOSSES

EXPERIMENT NO: 7

NAME

Measurement of Propagation Loss In The Fiber

OBJECTIVE

The objective of this experiment is to measure loss in the fiber.

- Propagation Loss
- Bending Loss
- Connector Loss

THEORY

Optical fibers are available in different variety of materials. These materials are usually selected by taking into account their absorption characteristics for different wavelengths of light. In case of optical fiber, since the signal is transmitted in the form of light, which is completely different in nature as that of electrons, one has to consider the interaction of matter with the radiation to study the losses in fiber. Losses are introduced in fiber due to various reasons. As light propagates from one end of fiber to another end, part of it is absorbed in the material exhibiting absorption loss. Also part of the light is reflected back or in some other direction from the impurity particles present in the material contributing to the loss of the signal at the other end of the fiber. In general terms, it is known as propagation loss. Plastic fibers have higher loss of the order of 180 dB/Km. Whenever the condition for angle of incidence of the incident light is violated, the losses are introduced due to refraction of light. This occurs when fiber is subjected to bending. Lower the radius of curvature more is the loss. Other losses are due to the coupling of fiber at LED & photo detector ends.

When light travels down optical fibers, some of the light is absorbed by the glass or plastic. This means that the light coming out of the end of the fiber is not as strong as the light going into the fiber. While designing a fiber communication system, you need to know this loss, to calculate the maximum distance the signal will travel. In this experiment, you will try one way of measuring the loss in the fiber.

EQUIPEMENTS

FCL-01 & FCL-02

- 1 & 3 Meter Fiber cable
- 0.5-meter connectorized fibers

- Patch chords
- Power Supply (Use only one provided)
- 20 MHz Dual Channel Oscilloscope

NOTE: Keep All Switch Faults In Off Position.

PROCEDURE

A. Measurement of Propagation Loss

- Make connections as shown in fig.7.1. Connect the power supply cables with proper polarity to FCL-01 & FCL-02 Kits. While connecting this, ensure that the power supply is OFF.
- Keep the jumpers **JP1**, **JP2**, **JP3** & **JP4** on FCL-01 as shown in fig.7.1.
- Keep the jumpers **JP1** & **JP2** on FCL-02 as shown in fig.7.1.
- Keep switch **S2** in **VI** position on FCL-01.
- Switch on the power supply.
- Slightly unscrew the cap of LED SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the 1meter fiber into the cap. Now tighten the cap by screwing it back.
- Now rotate the Optical Power Control pot **P3** in FCL-01 in anticlockwise direction. This ensures minimum current flow through LED.
- Slightly unscrew the cap of Photo Diode SFH250V. Do not remove the cap from the connector. Once the cap is loosened, insert the other end of fiber into the cap. Now tighten the cap by screwing it back.
- Keep Switch **SW1** to **SIGNAL STRENGTH** position in FCL-02.
- Connect the output of Photo Diode detector post **OUT** to post **IN** of Signal Strength Indicator block.
- Observe the signal strength LEDs, adjust the TRANSMITTER LEVEL using Intensity control pot **P3** until you get the reading of all LEDs glow.
- We will measure the light output using the SIGNAL STRENGTH section of the kit. The loss will be larger for a longer piece of fiber, so you will measure the loss of the long piece of fiber. In order to measure the loss in the fiber you first need a reference of how much light goes in to the piece of fiber from the LIGHT TRANSMITTER. You will use the short piece of fiber to measure this reference.
- Now remove the 1meter fiber and insert 3meter fiber.
- What reading do you get? Loss in optical fiber systems is usually measured in dBs. Loss of fiber itself is measured in dBs per meter. Subtract the length of the short fiber from the length of the long fiber to get the difference in the fiber lengths (3m-1m). The extra length of two meters is what created the extra loss you measured. Then take the signal strength reading you obtained for the loss of the long fiber and convert it to dB using the **Equation 1.0**. Finally divide the dB reading by the length to get the loss in dB per meter.

Equation 1.0:

$$\begin{aligned}\text{Power} &= 10 \log (P_2 / P_1) \text{ dB} \\ &= 10 \log (8/ 6) \text{ dB} \\ &= 1.25 \text{ dB}\end{aligned}$$

P₂ : Reference reading by 1 meter fiber.

P₁ : Reading obtained after replacing 3 meter fiber.

For example your signal strength reading is 6 and then your loss in dB would be 1.25 dB. Taking 1.25/ 2 gives 0.625 dB per meter

The reason for converting to dB per meter is that now in order to find the loss of any length of fiber you just have to multiply the dB per meter by the length of the fiber. For e.g. If you have a 10 meter long piece of fiber the loss will be

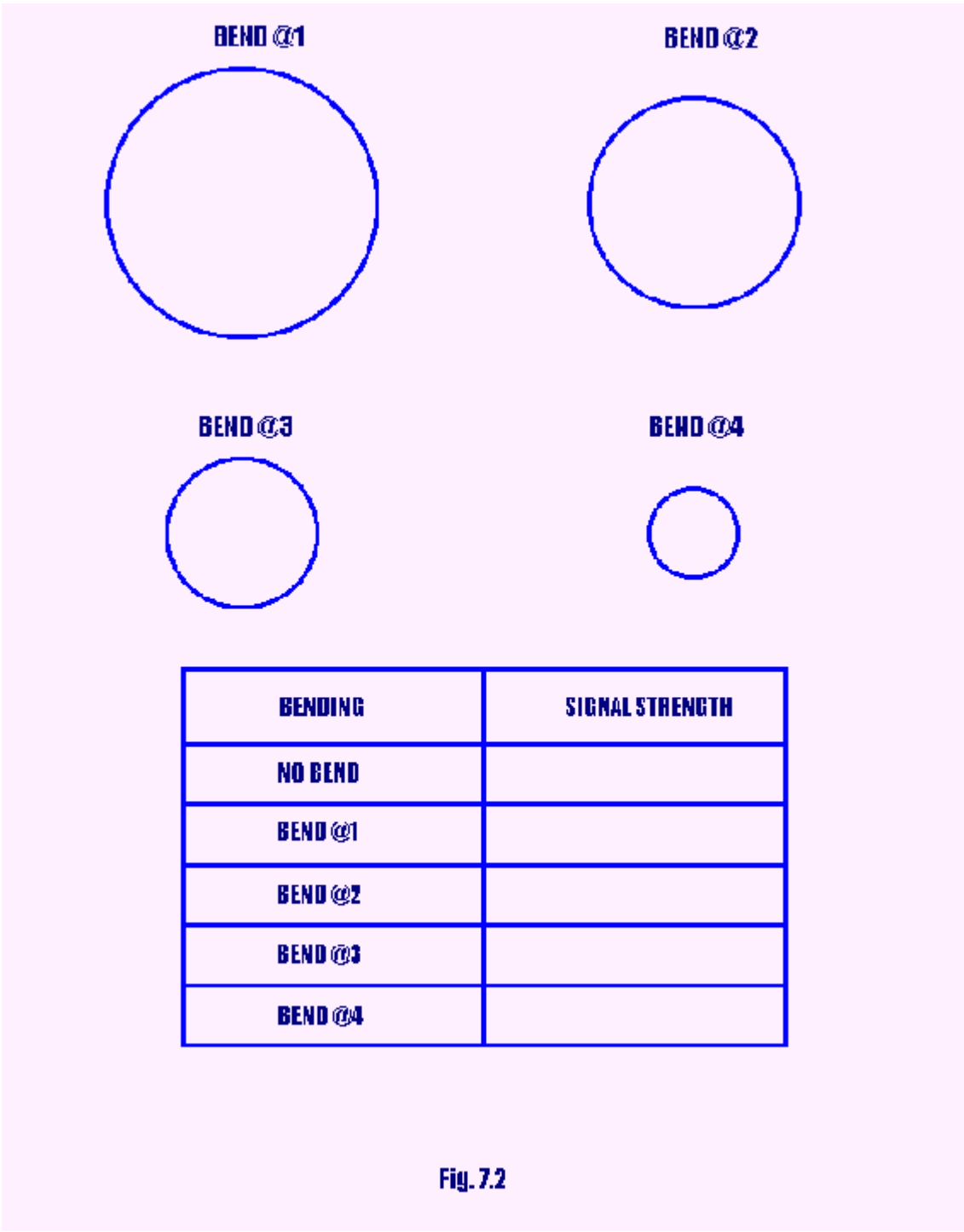
$$0.625 \text{ dB per meter} * 10 \text{ meters} = 6.25 \text{ dB}$$

B. Measurement of Bending Loss

- Keep the connections with 1-meter fiber as per the above procedure.
- Adjust the transmitter power so that the SIGNAL STENGTH reading is 8. Now take the portion of the fiber and loop it to match the bends as shown in fig.7.2. As you match each bends write down the reading from SIGNAL STRENGTH indicator. What happens as bends the fibers? Don't bend the fiber too tightly or it may not come back to shape.
- If you were designing the fiber optic communications system, you would need to know the relationship between the size of the bend and the light loss from the bend. In order to describe this relationship, you can measure the loss for a number of different bends and plot them on a graph. Do you notice a pattern to the numbers? Do they steadily increase or decrease? Can you predict what the readings would be for other bends?

C. Measurement of Connector Loss

- Keep the connections with 1-meter fiber as per the above procedure.
- Adjust the transmitter power so that the SIGNAL STENGTH reading is 8.
- Remove the 1-meter fiber and insert 0.5-meter connectorized fibers through connecting sleeve. What reading you get on the signal strength? Now take your loss from this measurement, say 7.0dB and subtract it from 8dB. Your connector loss is then 8.0dB-7db = 1dB this is actual connector loss.



EXPERIMENT NO.8

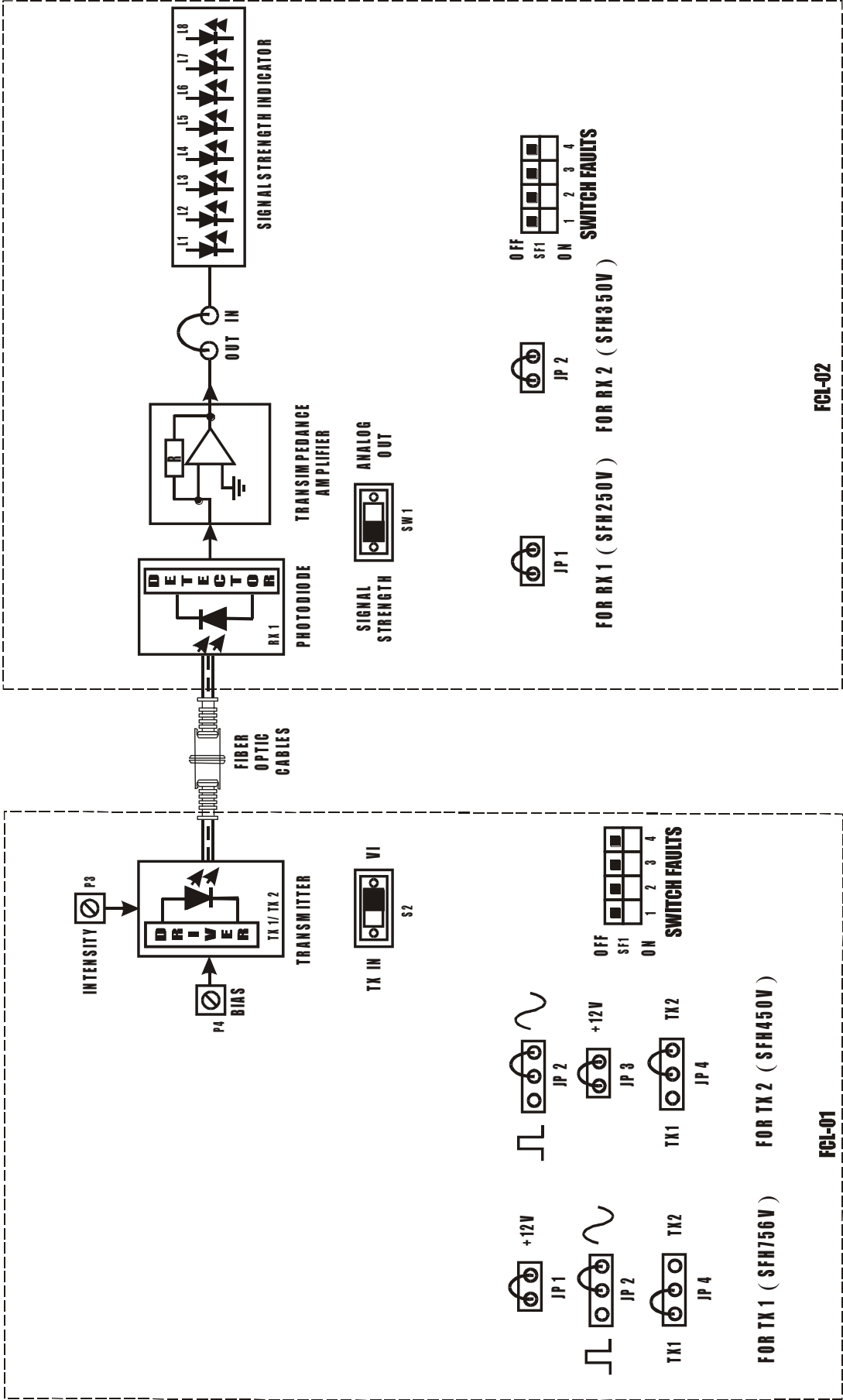


FIG. 8.1 BLOCK DIAGRAM FOR LATERAL, LONGITUDINAL AND ANGULAR DISPLACEMENT MEASUREMENT

EXPERIMENT NO: 8

NAME

Effect of Lateral, Longitudinal Angular Displacement

OBJECTIVE

The objective of this experiment is to study effect of lateral and longitudinal displacement. In this experiment, you will study the relationship between the input signal & received signal through angular and displacement Jig.

THEORY

Mechanical Misalignment In Fibers

Mechanical misalignment is a major problem while joining two fibers because of their microscopic size. It results in the radiation loss because the radiation cone of the emitting fiber does not match the acceptance cone of the receiving fiber. The magnitude of the radiation loss depends upon the degree of misalignment. There are three types of mechanical misalignments as shown in Fig. 8.5. These are lateral (coaxial), longitudinal (end separation) and angular. The lateral misalignment results when the axes of the fibers are separated (fig.8.5a). The longitudinal misalignment occurs when the fibers have the same axis but have some gap between their end faces (Fig.8.5b). The angular misalignment occurs when the fiber end faces are no longer parallel (Fig.8.5c).

The most common misalignment that occurs in practice is the lateral misalignment. This misalignment reduces the overlap area of the two fiber core end faces and consequently reduces the amount of power that can be coupled from one fiber into the other. The power loss due to this type of misalignment is usually more than that due to the other types of misalignment.

In the case of step-index fibers, the loss from lateral misalignment can be found by simply calculating the overlap area of the core because the irradiance is constant over the face of a step-index fiber. The loss due to longitudinal misalignment can be calculated by comparing the area of the receiving fibers with the area of the beam from the first fiber at a distance z , where z is the separation between the end faces. The loss from the angular misalignment has the same form as the longitudinal misalignment loss with $\sin \theta / \text{NA}$ replacing z/a (see Fig.8.5c). For graded index fibers, the calculation is much more complicated because the irradiance varies as a function of the position within the fiber core.

In this experiment fiber loss due to mechanical misalignments are measured. The fiber could be step index or graded index type.

Fiber Optic Links can be used for transmission of digital as well as analog signals. Basically, a fiber optic link contains three main elements, a transmitter, an optical fiber & a receiver. The transmitter module takes the input signal in electrical form & then transforms it into optical (light) energy containing the same information. The optical fiber is the medium, which carries this energy to the receiver. At the receiver, light is converted back into electrical form with the same pattern as originally fed to the transmitter. If we transmit signal between two fibers by putting them through coupling jig, depending on the extent of coupling, we will get appropriate signal at the detector side.

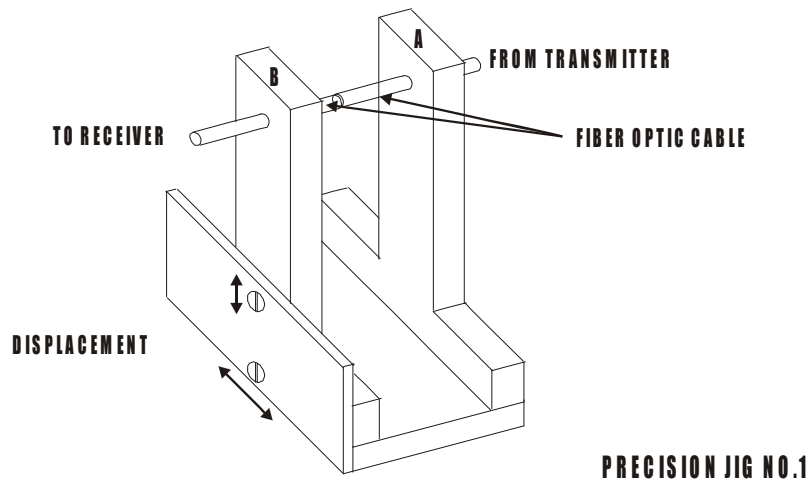


FIG.8.2 LATERAL DISPLACEMENT

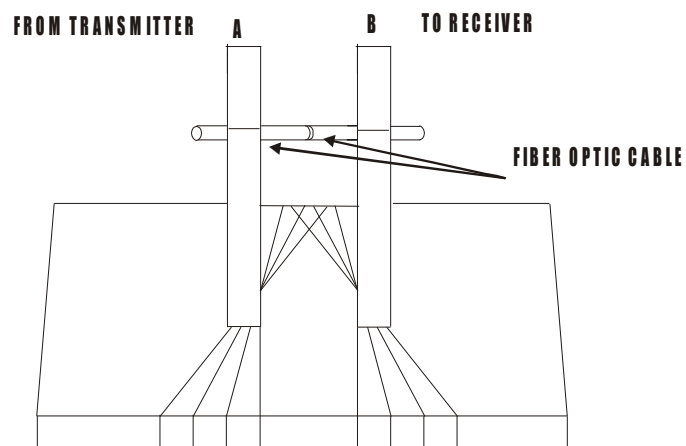


FIG.8.3 ANGULAR DISPLACEMENT

PRECISION JIG NO.2

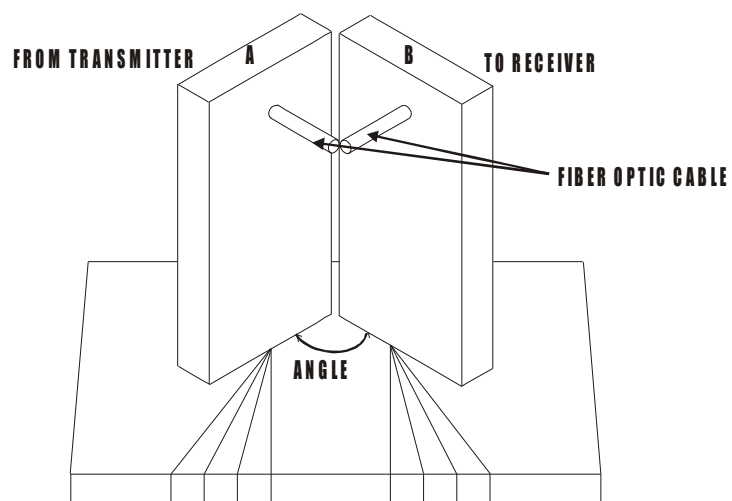


FIG.8.4 ANGULAR DISPLACEMENT

EQUIPMENTS

- FCL-01 & FCL-02
- FG-01 with power cable
- 20 MHz Dual Channel Oscilloscope
- 0.5-meter connectorized Fiber cables
- Connecting Sleeve
- Precision Displacement Jigs
- Patch chords
- Power supply (Use only one provided)

NOTE: Keep All Switch Faults In Off Position

PROCEDURE

- Refer to fig 8.1 and make the following connections.
- Connect the power supply with proper polarity to FCL-01 and FCL-02. While connecting this, ensure that the power supply is OFF.
- Keep the jumpers **JP1, JP2, JP3 & JP4, S2** on FCL-01 as shown in fig.8.1
- Keep switch **S2** on FCL-01 to **VI** position.
- Keep switch **SW2** in **SIGNAL STRENGTH** position on FCL-02.
- Keep pot **P3** fully anticlockwise & P4 fully clockwise position.
- Connect the output of Photo Diode detector post **OUT** to post **IN** of Signal Strength Indicator block.
- Switch on the power supply.
- Take two fibers of 0.5meters length after connectorization as explained in Exp.No.1. Slightly unscrew the cap of LED SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the 0.5meter fiber into the cap. Now tighten the cap by screwing it back.
- Connect 0.5meter fiber with plug form receiver to part A of Precision Displacement Jig.
- Connect 0.5meter fiber with plug form receiver to part B of Precision Displacement Jig.
- Check that vertical and horizontal alignment is proper as shown in fig. 8.2.
- Place the other end of this fiber into Photo Diode SFH250V of FCL-02 by Slightly unscrewing the cap. Do not remove the cap from the connector. Once the cap is loosened, insert the other end of fiber into the cap. Now tighten the cap by screwing it back.
- Observe the signal strength LEDs, adjust the TRANSMITTER LEVEL using Intensity control pot **P3** until you get the reading of all LEDs glow.

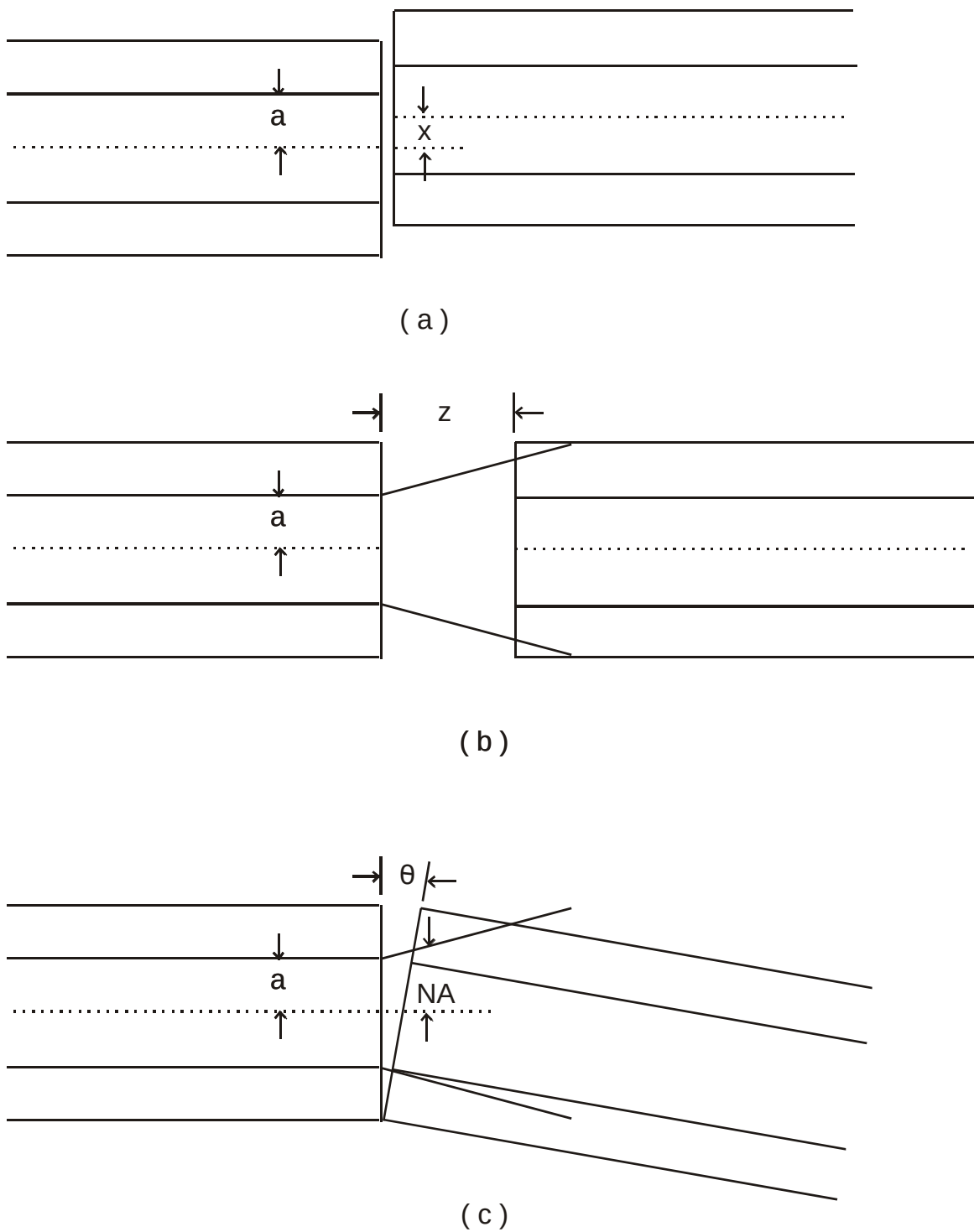


FIG. 8.5 Misalignment of fibers
(a) Lateral(axial)
(b) Longitudinal(end separation) and
(c) Angular

Lateral Displacement

- Introduce some lateral misalignment with the help of adjusting screws. For horizontal displacement fix the vertical displacement screw SC1, vary the horizontal displacement between part A & part B from 0 to 2 mm in steps of 0.5mm.
- Also For vertical displacement fix the horizontal displacement screw SC1, Vary the vertical displacement by adjusting screw SC2 between part A & part B from 0 to 2 mm in steps of 0.5mm.
- The amount of misalignment effect can be observed with the help of signal strength meter coupled into the second fiber. Measure the detected signal at every step and compare it with the earlier reading. And realign the fiber ends.

Longitudinal Displacement

- Check that the alignment of part A and part B of the precision jig is proper, if not make it proper by adjusting appropriate screws.
- Now introduce longitudinal misalignment with the help of loosening-tightening screw at the upper end of the precision jig. Then move the front end of fiber connecting plug away from each other around 1 to 2 mm.
- The amount of misalignment effect can be observed with the help of signal strength meter coupled into the second fiber. Measure the detected signal at every step and compare it with the earlier reading and then realign the fiber ends.

Angular Displacement

- Now replace the precision jig no.1 with precision jig no. 2.
- Do the connections as shown in fig. 8.3.
- Now introduce angular misalignment by 5,10,15 degree with the help of part A & part B of the precision jig no.2. The amount of misalignment effect can be observed with the help of signal strength meter coupled into the second fiber. Measure the excess loss as a function of angular misalignment, using the rotational stage to change the angular orientation of fiber ends and observe the detected signal on signal strength meter at every angle.

EXPERIMENT NO.9

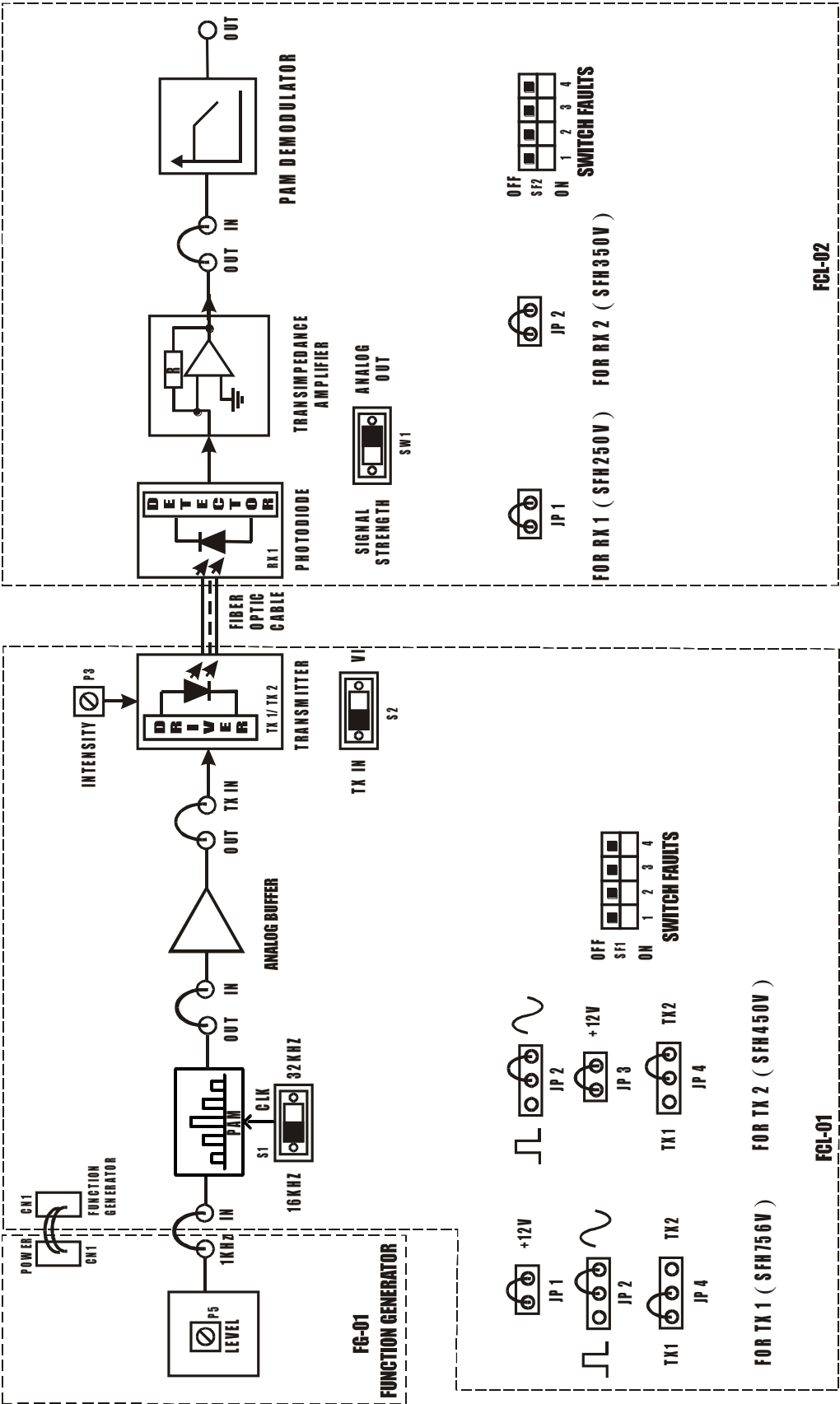


FIG. 9.1 BLOCK DIAGRAM FOR PULSE AMPLITUDE MODULATION

EXPERIMENT NO: 9

NAME

Study of Pulse Amplitude Modulation Technique

OBJECTIVE

The objective of this experiment is to obtain Amplitude Modulation of the analog signal, transmit it over a fiber optic cable and demodulate the same at the receiver end to get back the original signal.

THEORY

Pulse Amplitude Modulation is a technique of communication in which the high frequency square wave is modulated by the low frequency signal. The modulating signal is sampled by the pulses. The PAM signal is nothing but the high frequency square wave in which the amplitude of each pulse is equal to that of the information signal at the respective sampling instant.

The block diagram clearly indicates the pulse amplitude modulation and demodulation concept.

EQUIPMENTS

- FCL-01 & FCL-02
- FG-01 with power cable
- 20 MHz Dual Channel Oscilloscope
- 1 meter Fiber cable
- Patch chords
- Power supply (Use only one provided)

NOTE: Keep All Switch Faults In Off Position.

PROCEDURE

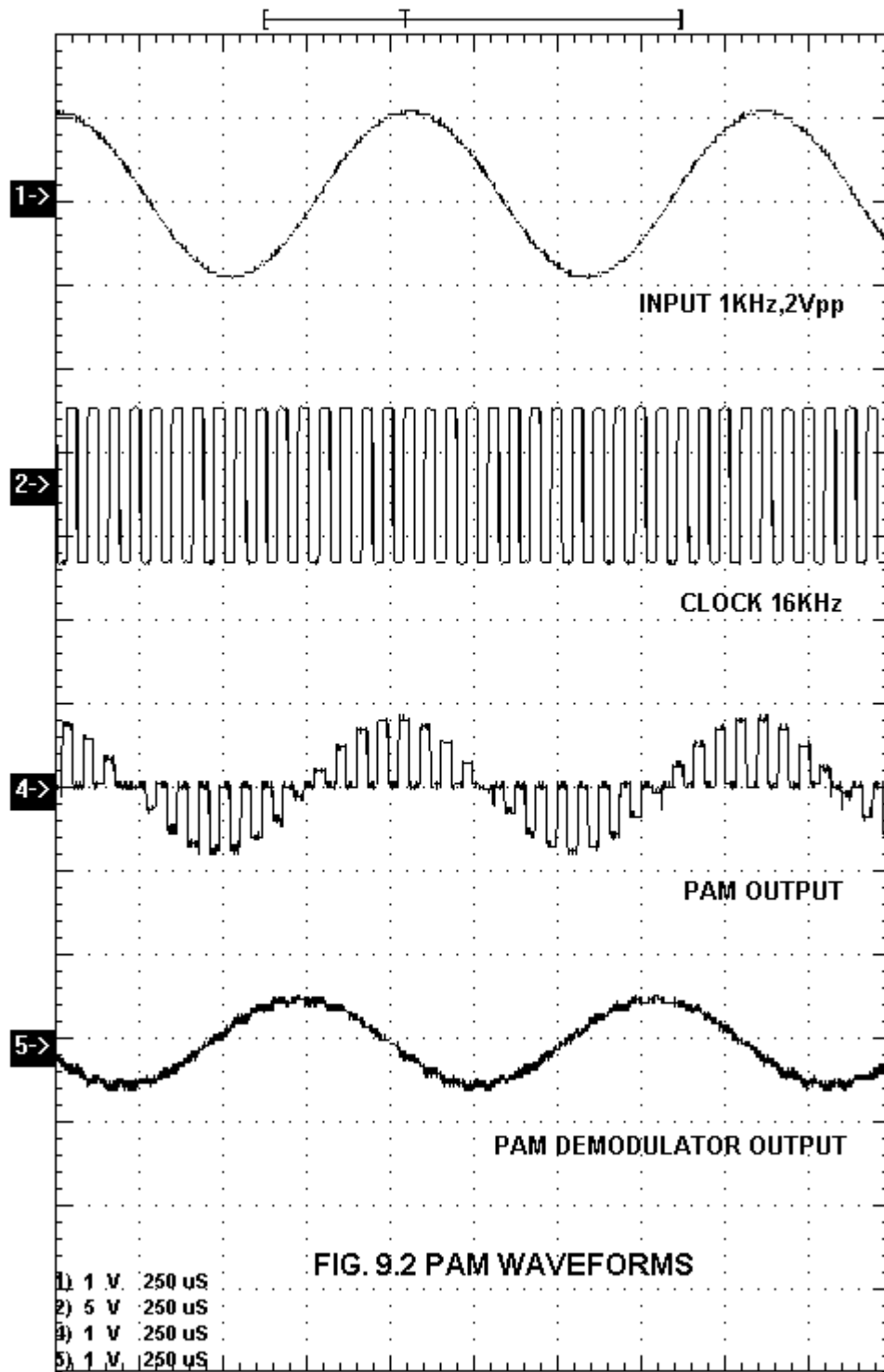
- Refer to fig 9.1 and make the following connections.
- Connect the power supply with proper polarity to FCL-01 and FCL-02. While connecting this, ensure that the power supply is OFF.
- Connect Function Generator FG-01 to FCL-01 using power cable.

- Keep the jumpers **JP1, JP2, JP3 & JP4** on FCL-01 as shown in fig.9.1.
- Keep the jumpers **JP1 & JP2** on FCL-02 as shown in fig.9.1.
- Keep switch **S1** at **16KHz** position on FCL-01.
- Keep switch **S2** in **TX IN** position on FCL-01.
- Switch on the power supply.
- Connect the **1KHz, 2Vpp** Signal from FG-01 as a constant signal to the **IN** post of PAM section on FCL-01.
- Connect the output of PAM section post **OUT** to post **IN** of Analog Buffer on FCL-01.
- Connect the output of Analog Buffer post **OUT** to post **TX IN**.
- Slightly unscrew the cap of LED SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the 1-meter fiber into the cap. Now tighten the cap by screwing it back.
- Slightly unscrew the cap of Photo Diode SFH250V. Do not remove the cap from the connector. Once the cap is loosened, insert the other end of fiber into the cap. Now tighten the cap by screwing it back.
- Keep switch **SW2** in **ANALOG OUT** position on FCL-02.
- Connect the output of Photo Diode detector post **OUT** to post **IN** of PAM Demodulator Section.
- Now observe the waveform at various test points. You will find fantastic pattern of square wave whose amplitude (i.e. pulse height) is varying according to the sine wave input as shown in fig.9.2
- Observe the output in FCL-02, you will receive the same sine wave at the output. In this way, the signal is pulse amplitude modulated, transmitted, received and again demodulated successfully as shown in fig.9.2
- Observe the effect of sampling rate by changing the sampling clock from 16KHz to 32KHz.
- Observe the effect of varying signal frequency by applying different signals from FG-01 Function Generator or External signals.
- Perform the above procedure again for all the combinations of Transmitter & Receiver.

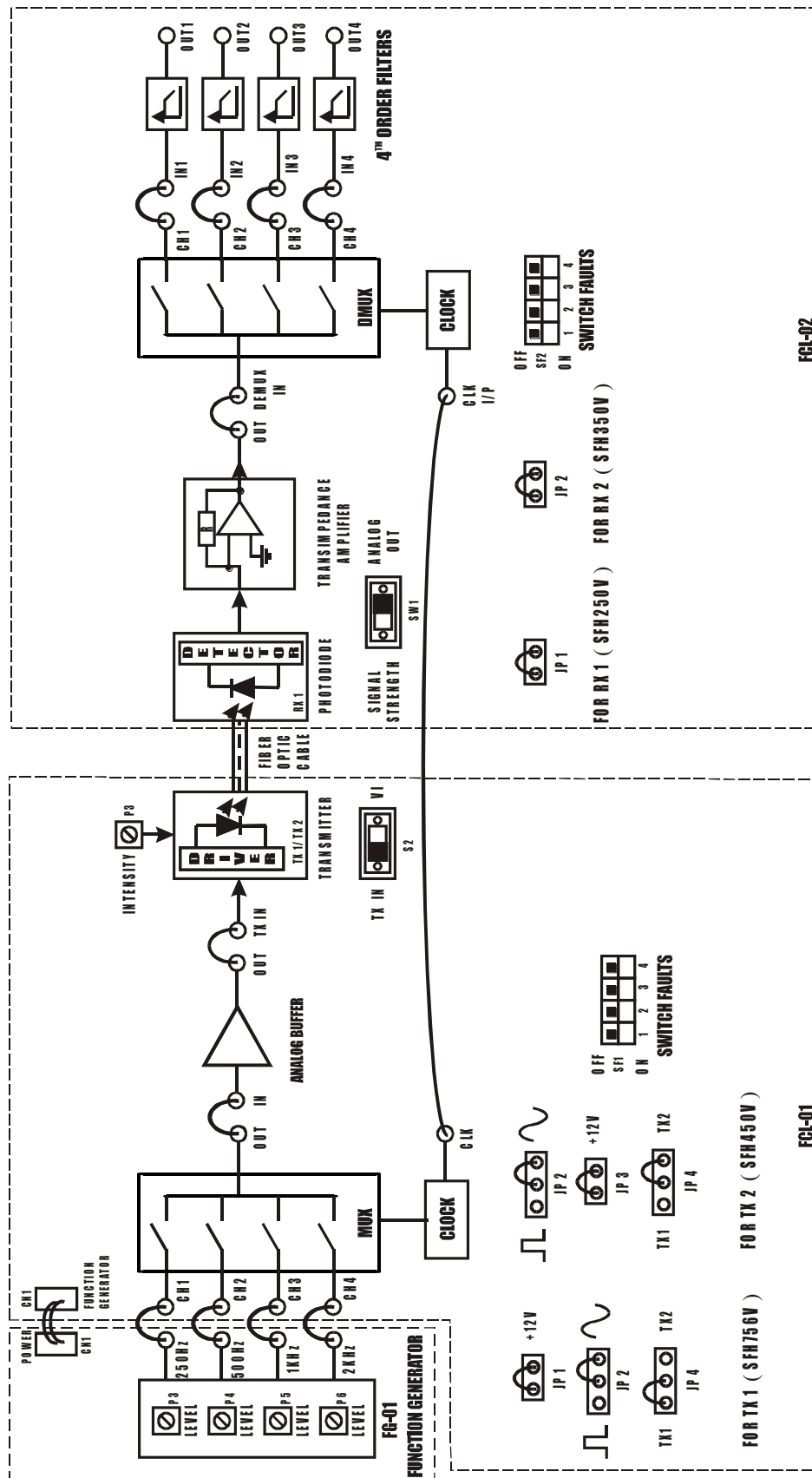
SWITCH FAULTS

NOTE: Keep the connections as per the procedure. Now switch **ON** corresponding fault switch button to **ON** position & observe the different effects on the output. The faults are normally used one at a time.

- Put switch **1** of **SF1** in Switch Fault section of FCL-01 to ON position. This will short capacitor C12 & C13 in Clock generator section. RC Time increases and clock frequency reduces. It affects TDM & PAM sections.
- Put switch **1** of **SF1** in Switch Fault section of FCL-02 to ON position. This will open capacitors C43 & C44 from Pulse Amplitude Demodulator section. Pulse Amplitude Demodulator output gets distorted.



EXPERIMENT NO.10



EXPERIMENT NO: 10

NAME

Study of Simple Time Division Multiplexing (Analog)

OBJECTIVE

The objective of this experiment is to study of Time Division Multiplexing for 4 analog channels fiber optic analog link.

THEORY

This technique enables simultaneous transmission of number of signals along the same communication channel. The basic concept behind this is that the signals from different sources are multiplexed by allowing only a specific time slot for each signal. At the Receiver side, the multiplexed data is distributed to the corresponding channels. The block diagram shows different parts used in the TDM circuit. The circuit consists of three parts:

- A.** Clock Pulse Generator.
- B.** Four stage Up/Down Binary Counter.
- C.** Analog Multiplexer.

EQUIPMENTS

- FCL-01 & FCL-02
- FG-01 with power cable
- 20 MHz Dual Channel Oscilloscope
- 1 meter Fiber cable
- Patch chords
- Power supply (Use only one provided)

NOTE: Keep All Switch Faults In Off Position.

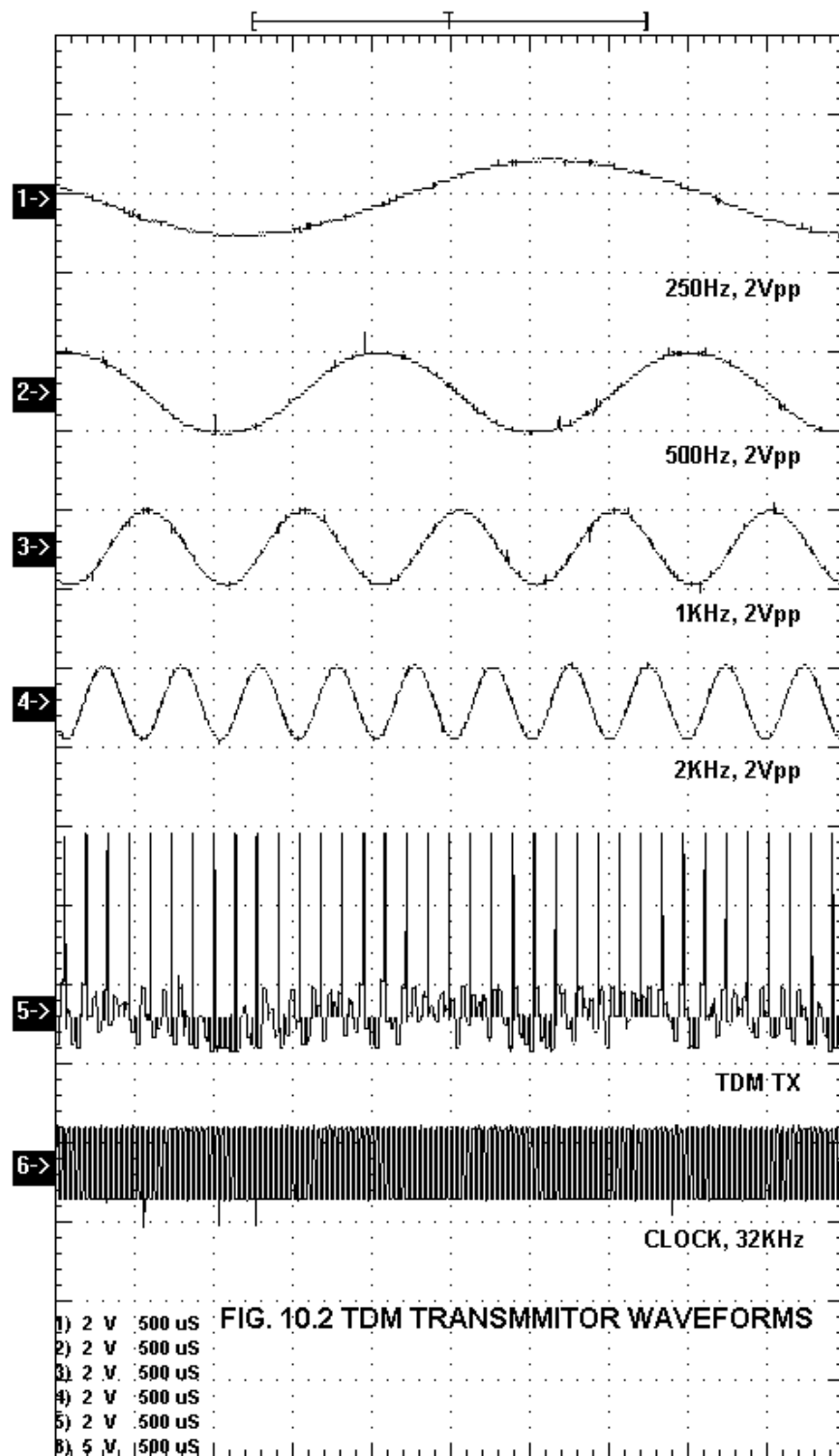
PROCEDURE

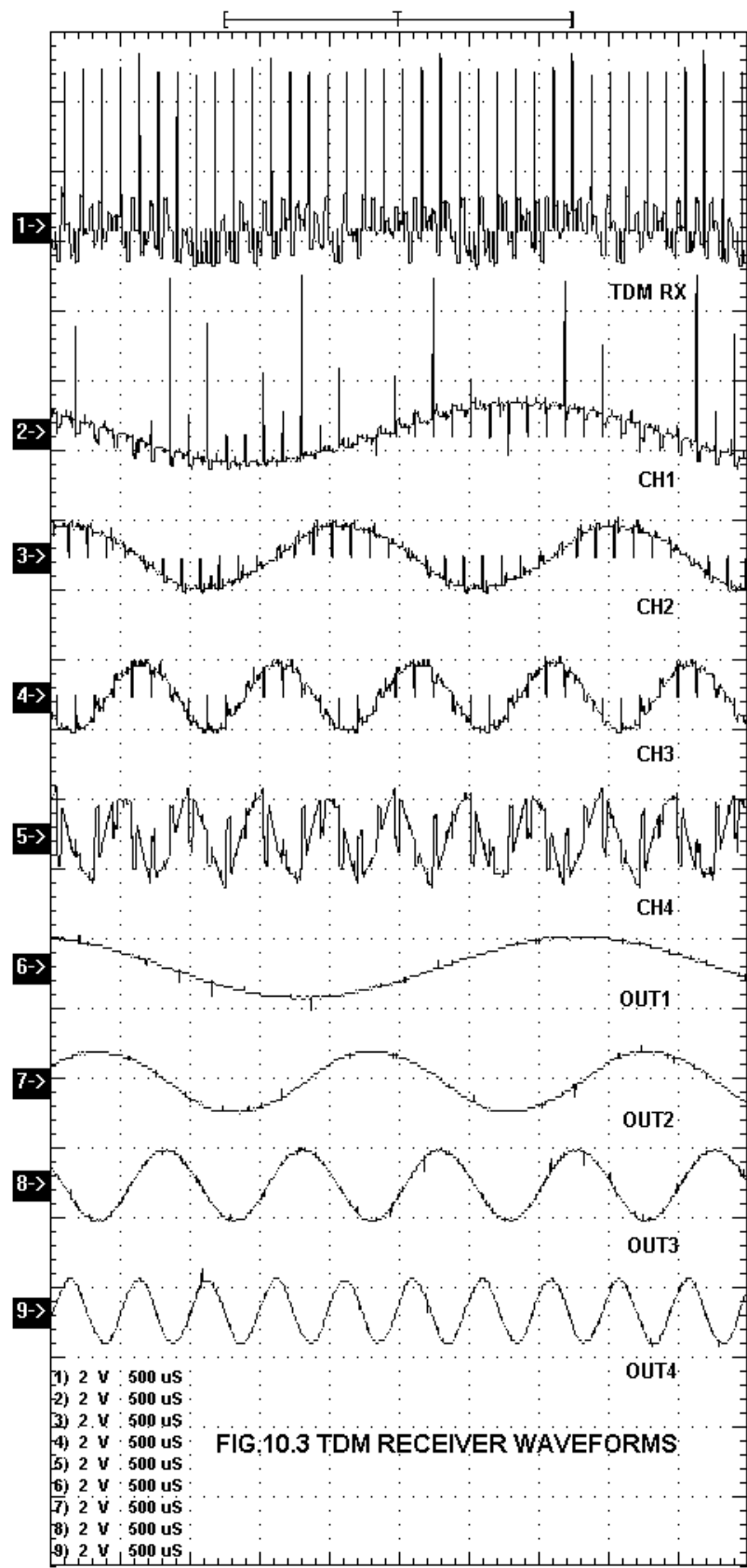
- Refer to fig 10.1 and make the following connections.
- Connect the power supply with proper polarity to FCL-01 and FCL-02. While connecting this, ensure that the power supply is OFF.
- Connect Function Generator FG-01 to FCL-01 using power cable.
- Keep the jumpers **JP1, JP2, JP3 & JP4** on FCL-01 as shown in fig.10.1.
- Keep the jumpers **JP1 & JP2** on FCL-02 as shown in fig.10.1.
- Keep switch **S2** in **TX IN** position on FCL-01.
- Switch on the power supply.
- Connect four sinusoidal signals of different frequencies as **250 Hz, 500 Hz, 1 KHz & 2 KHz** generated on FG-01 to **CH1, CH2, CH3 & CH4** inputs of Time Division Multiplexing section on FCL-01 respectively. The amplitude of these signals can be varied with the help of potentiometers **P3, P4, P5** and **P6**. Observe these signals on Oscilloscope and adjust their amplitude to **2 Vp-p**.
- Observe the TDM output at **OUT** post of Time Division Multiplexing section as shown in Fig. 10.1.
- Connect the output of Time Division Multiplexing section post **OUT** to post **IN** of Analog Buffer on FCL-01.
- Connect the output of Analog Buffer post **OUT** to post **TX IN**.
- Connect post **CLK** on FCL-01 to post **CLK I/P** on FCL-02.
- Slightly unscrew the cap of LED SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the 1 meter fiber into the cap. Now tighten the cap by screwing it back.
- Slightly unscrew the cap of Photo Diode SFH250V. Do not remove the cap from the connector. Once the cap is loosened, insert the other end of fiber into the cap. Now tighten the cap by screwing it back.
- Keep switch **SW2** in **ANALOG OUT** position on FCL-02.
- Connect the output of Photo Diode detector post **OUT** to post **DEMUX IN** of Time Division Demultiplexing Section.
- Connect the output of Time Division Demultiplexing Section **CH1, CH2, CH3 & CH4** to input of 4th order low pass filters **IN1, IN2, IN3 & IN4** respectively.
- Observe four different reconstructed signals at the output channels marked as **OUT1, OUT2, OUT3 & OUT4** of 4th order low pass filters in FCL-02 as shown in fig 10.3.
- Perform the above procedure again for all the combinations of Transmitter & Receiver.

SWITCH FAULTS

NOTE: Keep the connections as per the procedure. Now switch **ON** corresponding fault switch button to **ON** position & observe the different effects on the output. The faults are normally used one at a time.

- Put switch **3** of **SF1** in Switch Fault section of FCL-01 to ON position. This will short capacitor C63 & C64 in Time Division Multiplexing section. Synchronization pulse duty cycle get increases. TDM output distorts and similarly Demodulator output also gets distorted.
- Put switch **4** of **SF1** in Switch Fault section of FCL-01 to ON position. This will short CH3 & CH4 signals going for TDM. Mixed signal appears for CH3 & CH4 positions at Multiplexer and de-multiplexer outputs.
- Put switch **2** of **SF1** in Switch Fault section of FCL-02 to ON position. This will open capacitors C37 from fourth filter. Fourth filter output gets distorted.
- Put switch **4** of **SF1** in Switch Fault section of FCL-02 to ON position. This will open decoder pulse for CH3 in Time Division Demultiplexing section. Output for third channel will be absent.





EXPERIMENT NO.11

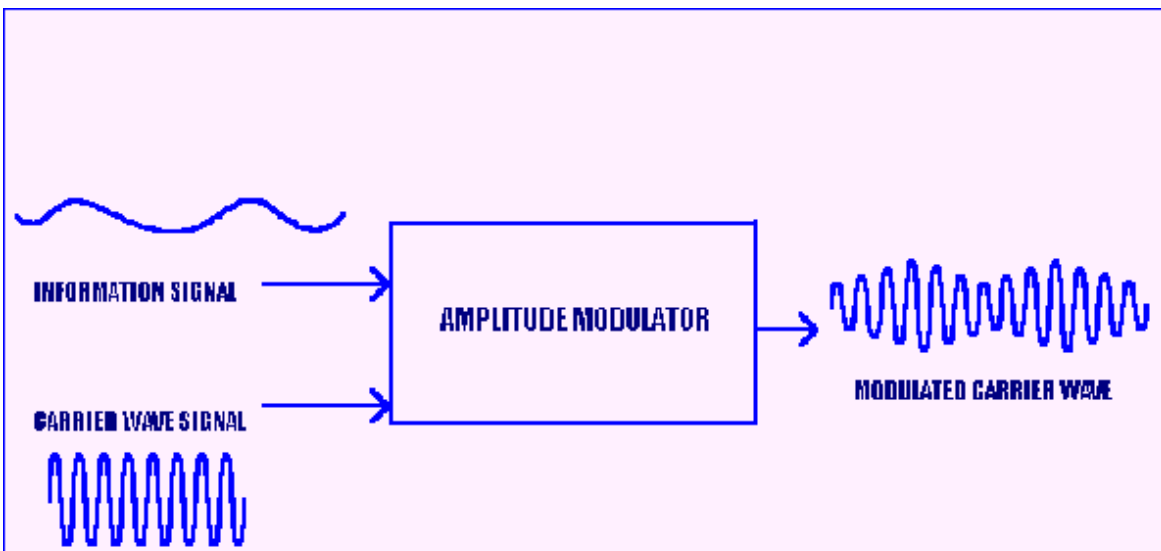


FIG. 11.1

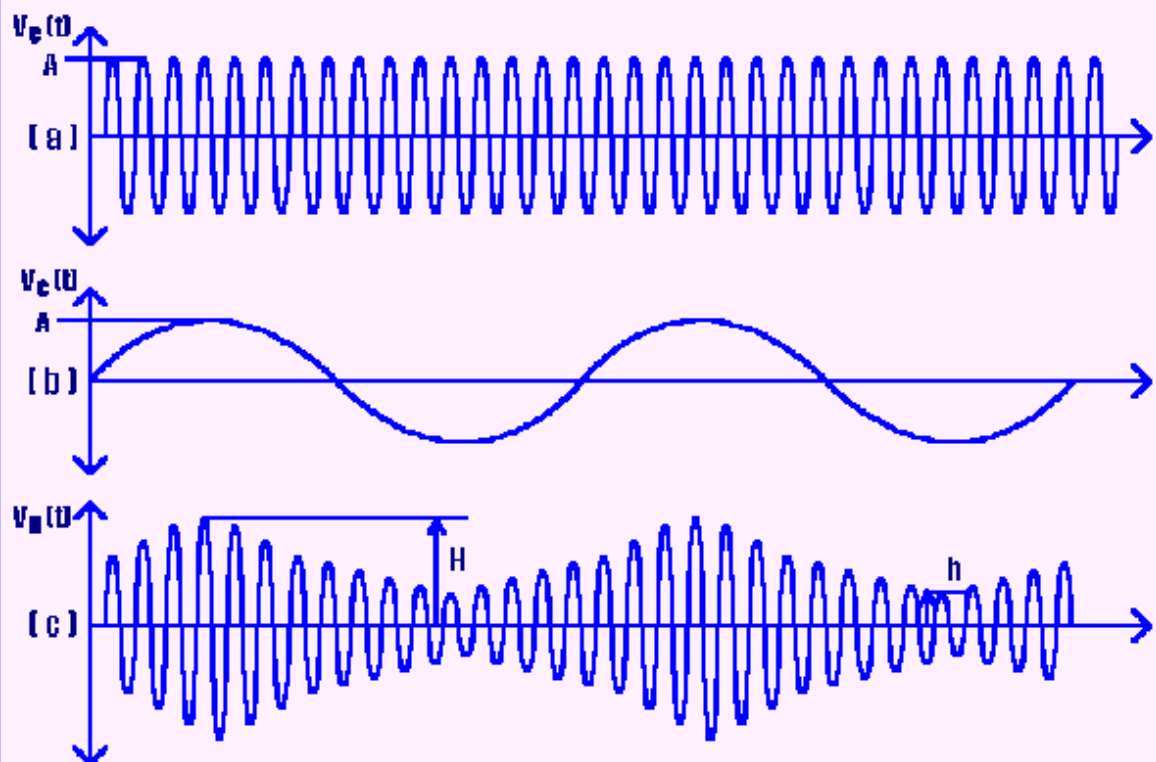


FIG. 11.2 A) CARRIER SIGNAL B) MODULATING SIGNAL C) MODULATED SIGNAL

EXPERIMENT NO: 11

NAME

Study of Amplitude Modulation

OBJECTIVE

The objective of this experiment is to study Amplitude Modulation and Demodulation through fiber optic analog link.

THEORY

Amplitude Modulation

In Amplitude Modulation the amplitude of high frequency sine wave (carrier) is varied in accordance with the instantaneous value of the modulating signal. Refer Fig.11.1.

Consider a sine signal $v_m(t)$ with frequency f (Fig.11.2).

$$V_m(t) = B \cdot \sin(2\pi f \cdot t)$$

And another sine signal $v_c(t)$ is called modulating signal, the signal $v_c(t)$ is called carrier signal.

$$V_c(t) = A \cdot \sin(2\pi F \cdot t)$$

The signal $v_m(t)$ is called modulating signal; the signal $v_c(t)$ is called carrier signal.

Vary the amplitude of the carrier $v_c(t)$ adding the modulating signal $v_m(t)$ to A . You obtain a signal $v_M(t)$ amplitude modulated, which can be expressed by:

$$v_M(t) = [A + k \cdot B \cdot \sin(2\pi f \cdot t)] \cdot \sin(2\pi F \cdot t) = A \cdot [1 + m \cdot \sin(2\pi f \cdot t)] \cdot \sin(2\pi F \cdot t)$$

With k = constant of proportionality.

Percentage modulation signal is defined as the value:

$$m = \frac{K \cdot B}{A} \cdot 100$$

With reference to FIG.11.2, the modulation index m can be calculated in this way

$$m = \frac{H-h}{H+h} \cdot 100\%$$

Envelope Detector

The extraction of the modulating signal from an AM signal can be carried out using an envelope detector. Consider the AM signal shown in Fig.11.2, and note that the modulating signal constitutes the envelope of the waveform reported.

The most common envelope detector consists of a diode followed by a RC filter. Its operation is analogous to the half-wave rectifier, as the output voltage follows the maximum values of the carrier. As the amplitude of the carrier is variable, by properly choosing R and C , the output of the detector can faithfully reproduce faithfully these variations.

Distortions Of The Detected Signal

The demodulated signal can have two kinds of distortions:

- If the time constant $R \cdot C$ is too small with respect to the period of the carrier, the envelope is approximated to a waveform affected by ripple, the more evident the lower is the value of $R \cdot C$
- If the time constant $R \cdot C$ is too high with respect to the period of the modulating signal, the detected signal always follows the behavior of the envelope, but sometimes it takes an exponential decreasing law (distortion by diagonal cutting).

The distortion presented by the detected signal is known as Distortion by diagonal cutting.

The maximum value of $R \cdot C$ is calculated supposing that the capacitor C discharges on the resistor R to a higher or equal rhythm to the one in which the envelope of the modulated signal drops. The detected condition is as follows:

$$R \cdot C \leq \frac{1}{f_{\max}} \cdot \frac{\sqrt{(1-m \cdot 2)}}{m}$$

with m = modulation index
 $f_{\max}$ = maximum frequency of the modulating signal.

Note that for $m=1$ the condition cannot be satisfied and so the detected signal will be certainly distorted.

Detection Efficiency

The efficiency η of the diode envelope detector is defined as the ratio between the amplitude of the output signal of the detector and the amplitude of the envelope of the input AM signal. Supposing that at the carrier frequency, the reactance of the capacitor $[1/(2\pi \bullet F \bullet C)]$ is much smaller than the resistance R (hypothesis checked if $RC \gg 1/F$ to reduce the ripple), the efficiency η practically depends on the relation between R and the differential resistance r_d of the diode.

Last Considerations On The Choice Of $R \bullet C$ In The Envelope Detector

- To obtain high detection efficiency the resistance R must be much higher than the differential resistance r_d of the diode ($R \gg r_d$).
- To minimize the ripple in the detected signal the product $R \bullet C$ must be much higher than the period T of the carrier ($R \bullet C \gg T$).
- To avoid distortion by diagonal cutting the product $R \bullet C$ must be smaller than or equal to a value, which depends on the modulating frequency and on the modulation index. In case of the modulating signal with variable amplitude and frequency, the maximum values of the frequency as well as of the modulation index must be considered.

EQUIPMENTS

- FCL-01 & FCL-02
- FG-01 with power cable
- 20 MHz Dual Channel Oscilloscope
- 1 meter Fiber cable
- Patch chords
- Power supply (Use only one provided)

NOTE: Keep All Switch Faults In Off Position.

PROCEDURE

- Refer to fig 11.3 and make the following connections.
- Connect the power supply with proper polarity to FCL-01 and FCL-02. While connecting this, ensure that the power supply is OFF.
- Connect Function Generator FG-01 to FCL-01 using power cable.
- Keep the jumpers **JP1** & **JP4** on FG-01 as shown in fig.11.3.
- Keep the jumpers **JP1**, **JP2**, **JP3** & **JP4** on FCL-01 as shown in fig.11.3.
- Keep the jumpers **JP1** & **JP2** on FCL-02 as shown in fig.11.3.
- Keep switch **S2** in **TX IN** position on FCL-01.
- Switch on the power supply.

- Connect the **1KHz, 0.5Vpp** Signal from FG-01 as a constant signal to the **SIGNAL IN** post of Amplitude Modulator section on FCL-01.
- Keep **SINEOUT** signal of about **50KHz** with amplitude of **2Vpp**.
- Connect the post **SINEOUT** of FG-01 to post **CARRIER IN** of Amplitude Modulator section on FCL-01.
- Connect the output of Amplitude Modulator section post **AM OUT** to post **IN** of Analog Buffer on FCL-01. Keep pot **P2** fully clockwise position.
- Connect the output of Analog Buffer post **OUT** to post **TX IN**.
- Slightly unscrew the cap of LED SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the 1-meter fiber into the cap. Now tighten the cap by screwing it back.
- Slightly unscrew the cap of Photo Diode SFH250V. Do not remove the cap from the connector. Once the cap is loosened, insert the other end of fiber into the cap. Now tighten the cap by screwing it back.
- Keep switch **SW2** in **ANALOG OUT** position on FCL-02.
- Connect the output of Photo Diode detector post **OUT** to post **IN** of AM Demodulator Section.
- Observe AM Signal and amplitude demodulated signals at the output of AM Modulator & Demodulator Section respectively in FCL-01 & FCL-02 as show in fig.11.4
- Perform the above procedure again for all the combinations of Transmitter & Receiver.

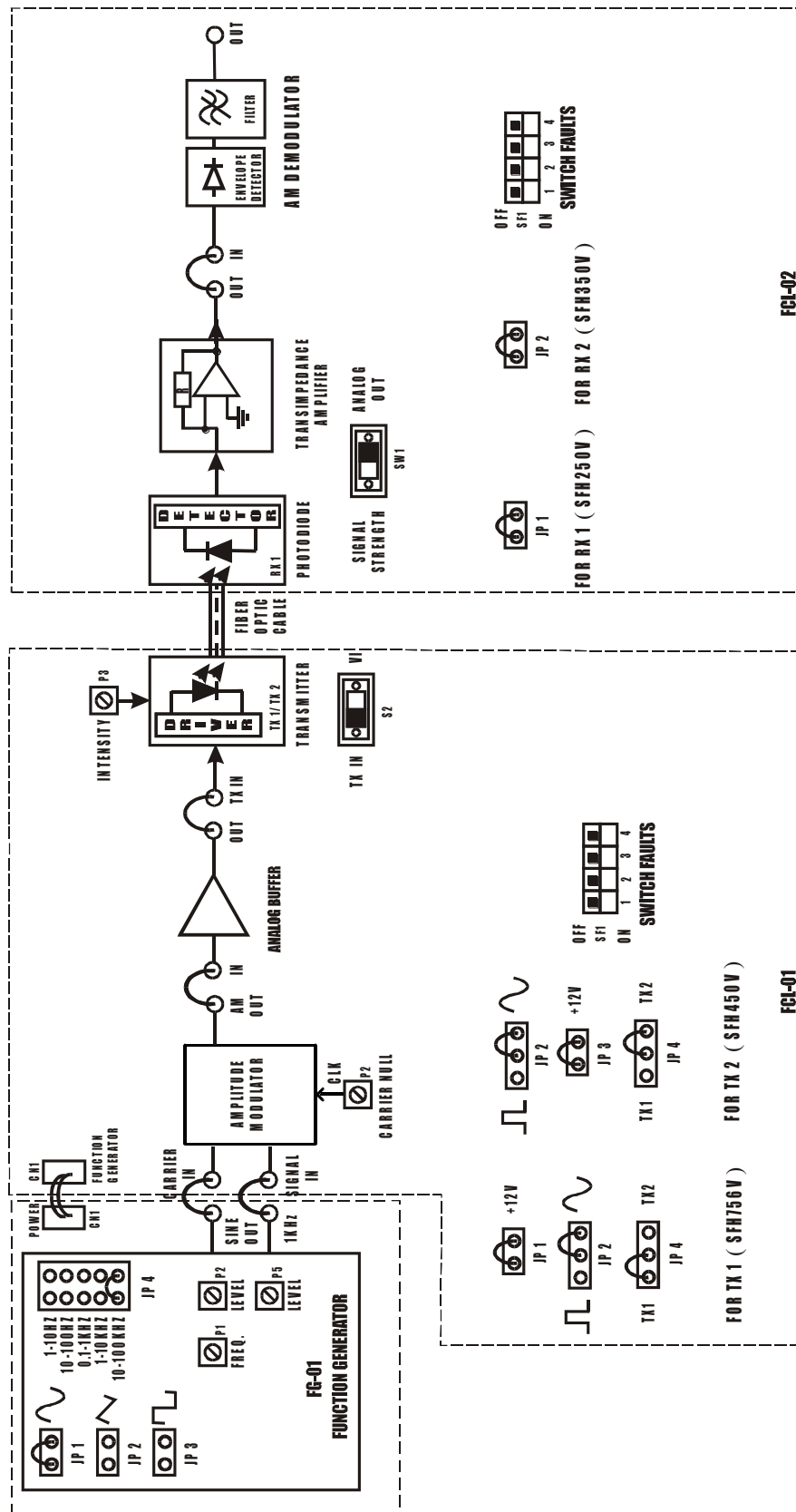
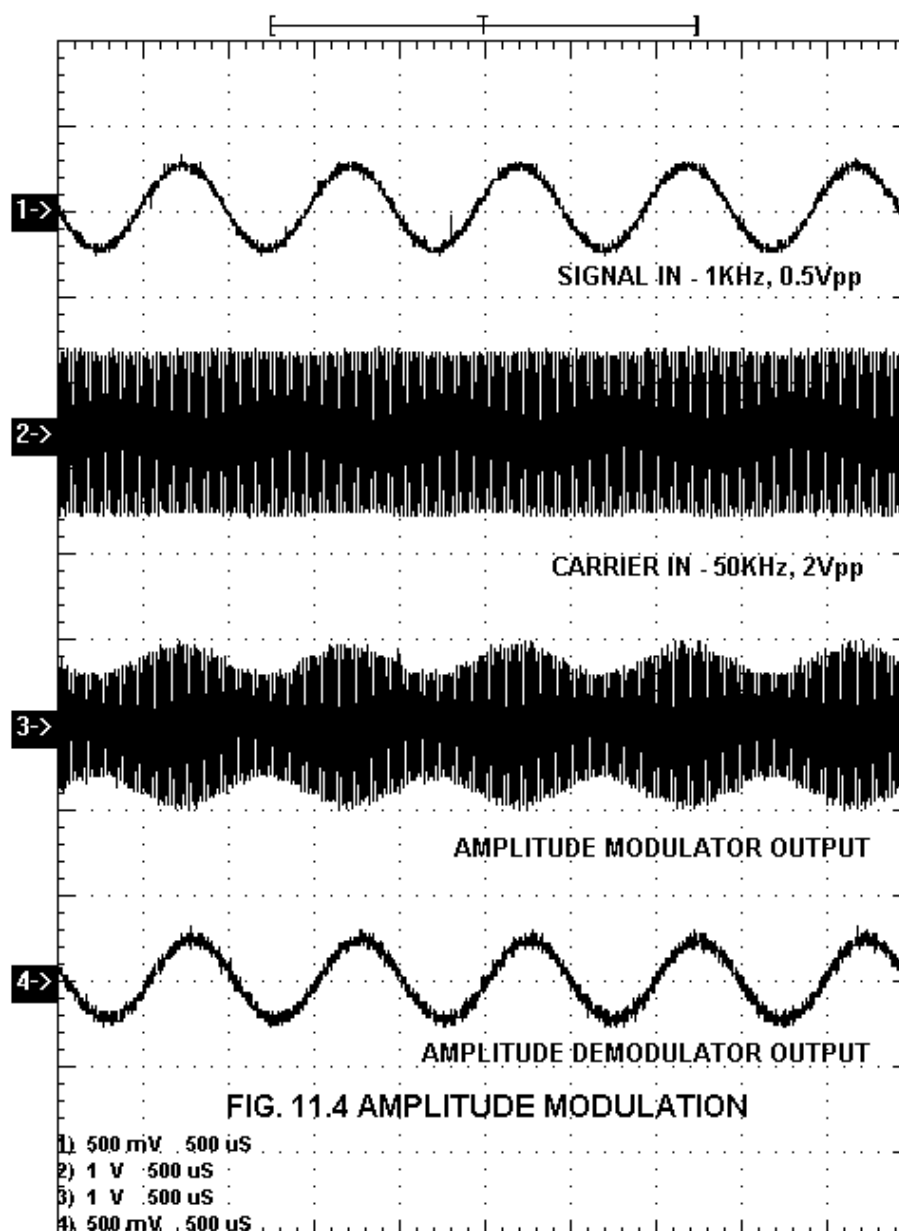


FIG. 11.3 BLOCK DIAGRAM FOR AMPLITUDE MODULATION AND DEMODULATION

SWITCH FAULTS

NOTE: Keep the connections as per the procedure. Now switch **ON** corresponding fault switch button to **ON** position & observe the different effects on the output. The faults are normally used one at a time.

- Put switch **2** of **SF1** in Switch Fault section of FCL-01 to ON position. This will short resistor R168 & R221 at the bias pin of Balanced Modulator from Amplitude Modulator block. Amplitude Modulator output approximates zero.
- Put switch **3** of **SF1** in Switch Fault section of FCL-02 to ON position. This will open capacitors C19 & C20 in Amplitude. Output of Amplitude Demodulator gets distorted.



EXPERIMENT NO.12

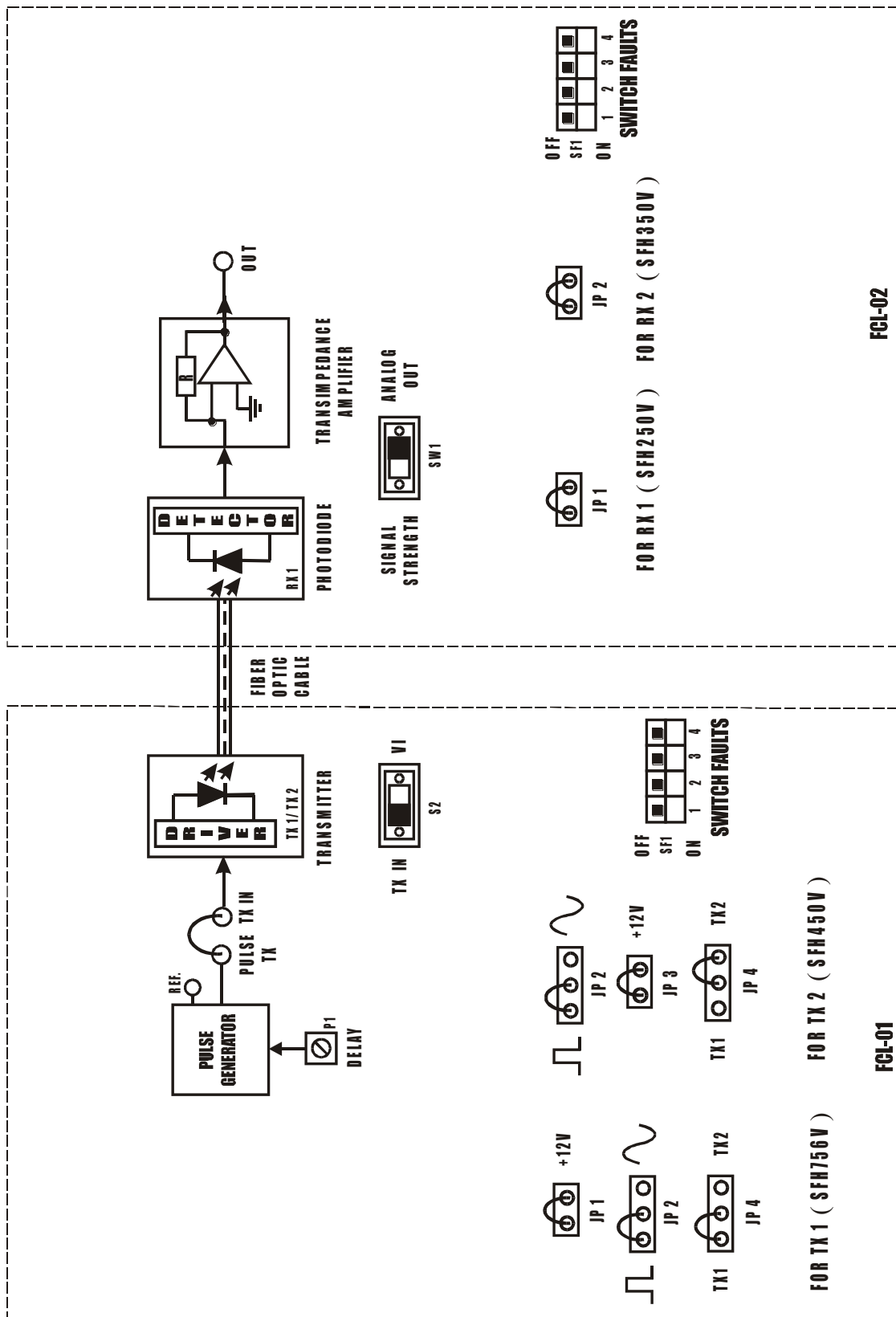


FIG. 12.1 BLOCK DIAGRAM FOR LENGTH MEASUREMENT OF FIBER CABLE

EXPERIMENT NO: 12

NAME

Measurement of Length of Fiber Cable

OBJECTIVE

To study the measurement of length of fiber cable using speed of light method

THEORY

In 1926, Michelson got a grip on light, computed at 299,796 kilometers per second – or 186,284 miles per second. The quest for precision and the continued refinement of the measurement continued till today, really only hard-core mathematicians quibble about the number of decimal places we should assign to the speed of light. The speed of light in a vacuum is now listed as 299,792.4562 kilometers per second. For the purposes of everyday discussion, we generally suffice with a figure of 300 million meters per second or 186,000 miles per second. Up to this point, we've talked primarily about light traveling through a vacuum, such as outer space. When light passes through some medium other than a vacuum (such as a gas, solid or liquid), it slows down, but often not much.

To put these relationships into some form of mathematical perspective, we utilize the term “refractive index” (or “index of refraction”), which refers to the ratio between the speed of light in a vacuum and its speed in some other medium. Here's the equation.

$$\frac{\text{Light speed in a vacuum}}{\text{Light speed in another medium}} = \text{Refractive index}$$

Or messaging things down into more manageable symbols

$$\frac{c}{v} = n$$

Substituting the numerical values of light speed in air, water and glass, we would obtain the following refractive indices:

$$\text{For air: } n = 1.000$$

We now have just about all the information we need to calculate the speed of light. Because the light in our demonstration is traveling through a plastic fiber, which is not a vacuum, we need to know the index of refraction of the plastic fiber to compensate for the slower light speed.

Here, the index of refraction of a particular medium equals the speed of light in a vacuum divided by the speed of light in the medium ($n = c/v$). Today we won't attempt to measure the index of refraction of the plastic fiber. That's a project for another day. For now, we'll rely on the manufacturer's specifications, which say our fiber's refractive index is 1.5.

Use the familiar equation:

$$C = \frac{n \cdot L}{t}$$

Where

C = the speed of light in a vacuum

n = the index of refraction of the plastic fiber

L = the length of the plastic fiber, in meters

t = the travel time of light through the fiber, in nanoseconds

Now take the measurement number you wrote down and substitute it for 't' in the equation. If the results of your measurement resemble 100 nanoseconds, your numbers should look something like this:

$$\begin{aligned} C &= \frac{1.5 \times L}{100 \times 10^{-9}} \\ &= 3 \times 10^8 \text{ meters/second} \end{aligned}$$

$$\text{hence, } L = 20 \text{ meter}$$

Transmitter

In this section one reference pulse is generated with the help of high frequency timer and dual monostable multivibrator. Then it is transmitted through fiber optic driver circuit to fiber optic transmitter SFH756V. The transmitter portion of FCL-01 for Reference pulse generates more than 500,000 light pulses per second, which appear as a cone of red light leaving the red LED. When one end of the fiber is connected to an LED and other end to a Detector or receiver, our red light will make a circular trip through a fiber.

Receiver

This section of kit contains the delay circuitry. The time required for red pulses to travel through 20m long fiber is about 100ns. These received pulses are received at fiber optic detector SFH250V. Delay output can be observed through Signal conditioning and delay circuit.

EQUIPMENTS

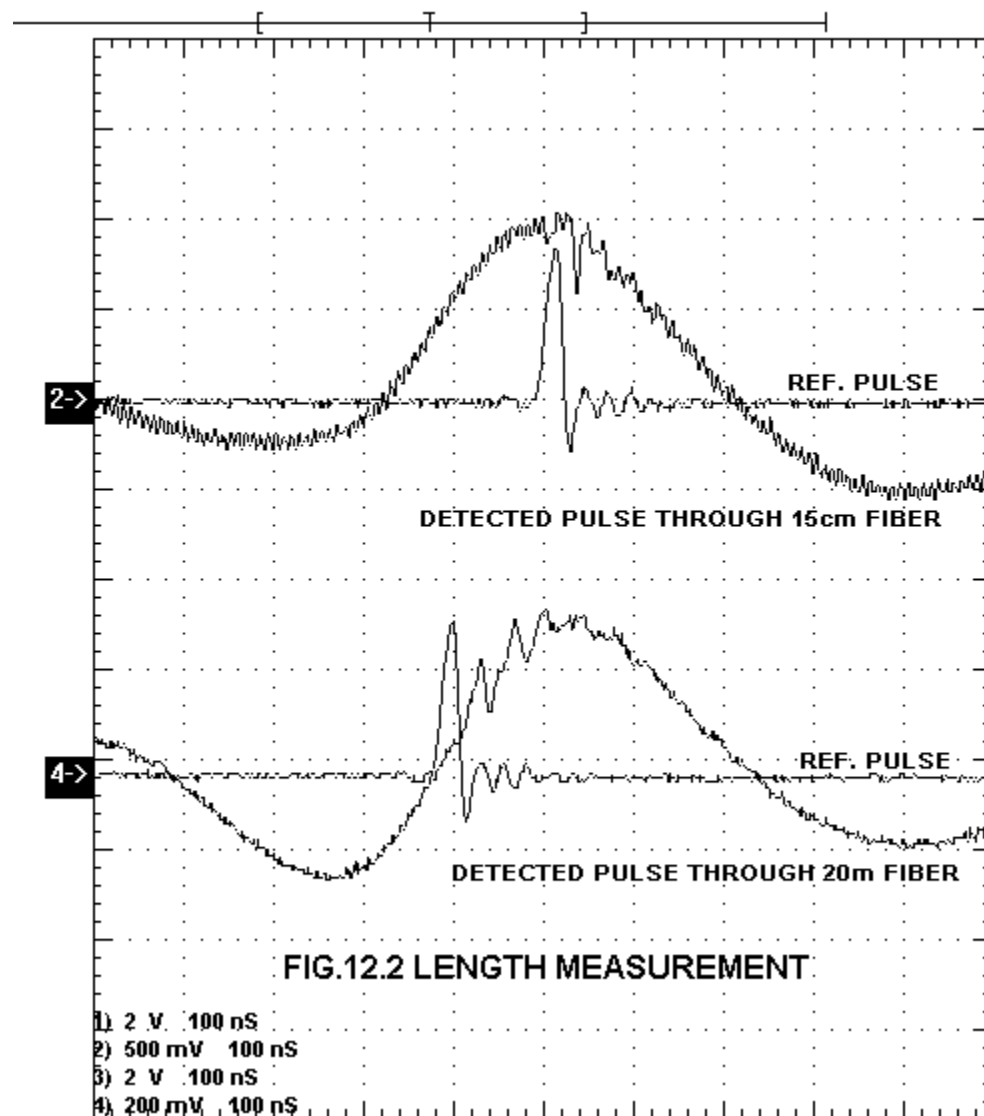
- FCL-01 & FCL-02
- 20MHz Dual Channel Oscilloscope
- 15 cm, 3 m, 20m Fiber cable
- Patch chords
- Power Supply (Use only one provided)

NOTE: Keep All Switch Faults In Off Position.

PROCEDURE

- Refer to fig 12.1 and make the following connections.
- Connect the power supply with proper polarity to FCL-01 and FCL-02. While connecting this, ensure that the power supply is OFF.
- Keep the jumpers **JP1, JP2, JP3 & JP4** on FCL-01 as shown in fig.12.1.
- Keep the jumpers **JP1 & JP2** on FCL-02 as shown in fig.12.1.
- Keep switch **S2** in **TX IN** position on FCL-01.
- Switch on the power supply.
- Connect the post **PULSE TX** to post **TX IN**.
- Connect **REF.** point on FCL-01 to channel one of oscilloscope with the help of probe.
- Observe the Reference pulse, which is nearly 2.2Vpp in amplitude and 150ns in pulse width as shown in Fig. 12.2.
- Slightly unscrew the cap of LED SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the 15-cm fiber into the cap. Now tighten the cap by screwing it back.
- Slightly unscrew the cap of Photo Diode SFH250V. Do not remove the cap from the connector. Once the cap is loosened, insert the other end of fiber into the cap. Now tighten the cap by screwing it back.
- Keep switch **SW2** in **ANALOG OUT** position on FCL-02.
- Connect second channel of Oscilloscope to the output of Photo Diode detector post OUT on FCL-02. To observe signals on Oscilloscope use attenuation (X10) probes only.
- Then observe Delayed output pulse, which is of 2Vpp in amplitude and 300 nanosecond wide. This is the pulse received through the 15cm fiber optic cable.
- Using the vertical and horizontal positioning align the bases of channel 1 and 2 then rotate the “**Delay**” knob **P1** on FCL-01 until the peak of the received pulse coincides with the peak of the **REF.** pulse as shown in Fig. 12.3. To adjust this peak you have to use 10X-magnification knob on the oscilloscope. Fine-tune the Calibration delay adjustment knob **P1** to achieve best overlap/coincidence of the Reference and received pulse.

- Then carefully loosen the fiber cinch nuts and remove the 15cm fiber optic cable and insert 20m-fiber optic cable gently but firmly, into **LED SFH 756** and other end into **DETECTOR SFH 250**.
- Observe the display on Oscilloscope that the received pulse should now have moved to the right of the Reference pulse with reduced amplitude as shown in Fig. 12.4.
- Very carefully measure the time difference between the Reference pulse and the received Delay pulse, in nanosecond (If required in X5 magnification mode) through 20m Optical fiber. Write down the result it should be between 90 and 110 nanoseconds.
- Then calculate the practical length of optical fiber by putting appropriate value into the formula explained above.



EXPERIMENT NO.13

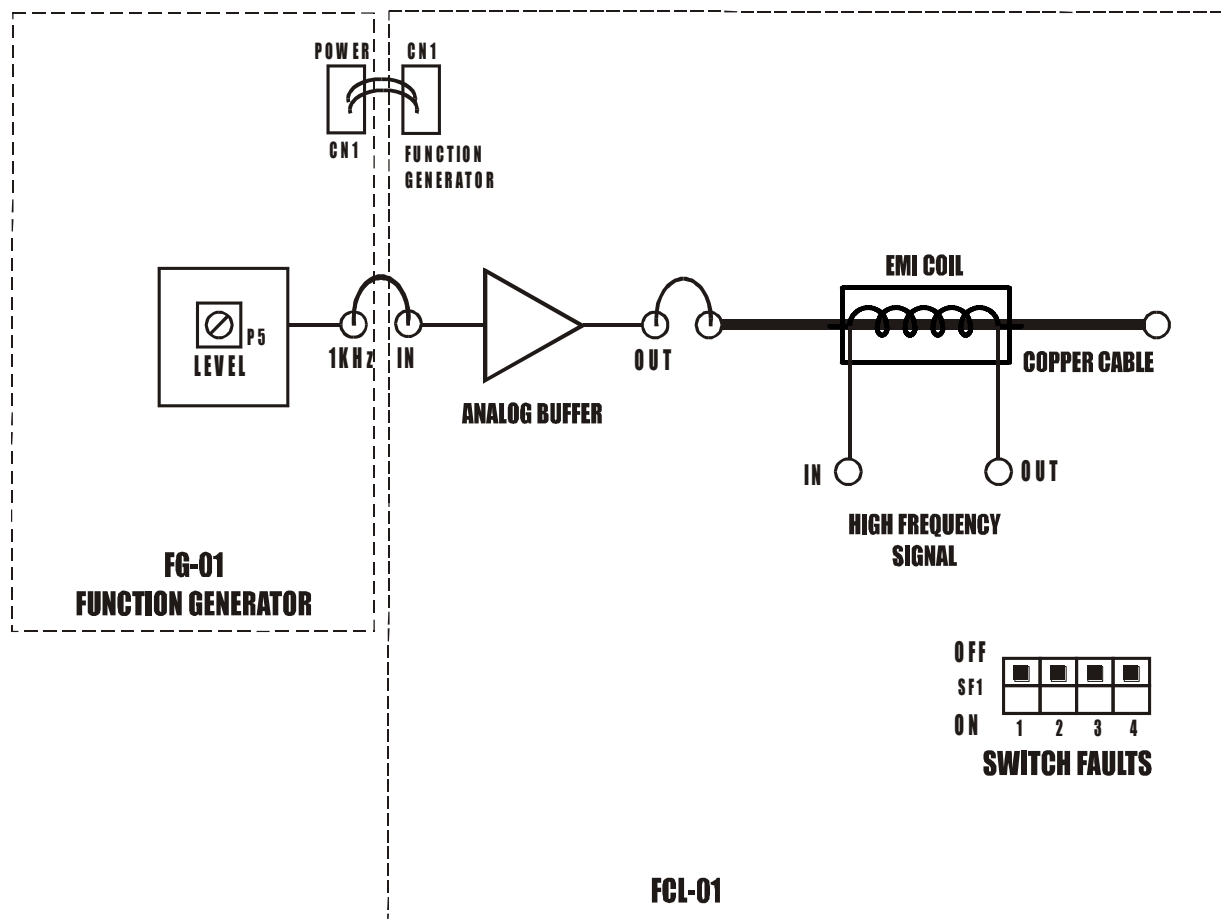


FIG. 13.1 BLOCK DIAGRAM FOR EMI INTERFERENCE ON COPPER CABLE

EXPERIMENT NO: 13

NAME

Comparison of Effect of Emi Interference on Copper Medium And On Optical Fiber Medium

OBJECTIVE

To study comparison of effect of EMI interference on copper medium and on optical fiber medium

THEORY

In many environments today Electromagnetic interference (EMI) and Radio frequency interference (RFI) have become a serious problem affecting even low bit rate data communication over short distances. When optical fiber being totally dielectric, it is immune to EMI/RFI and finds widespread applications in such solutions.

Optical fiber has established itself the medium for long-haul wide bandwidth communications, it has also made a significant impact on many other applications where the link length and bandwidth requirement is large. This is because optical fiber is a dielectric medium, i.e. totally non-metallic.

EQUIPMENTS

- FCL-01 & FCL-02
- FG-01 with power cable
- 1 meter Fiber cable
- EMI Coil Jig and one copper cable
- Patch chords
- Power supply (Use only one provided)
- Function generator 1Hz-10MHz
- 20 MHz Dual Channel Oscilloscope

NOTE: Keep All Switch Faults In Off Position.

PROCEDURE

- Refer to fig 13.1 and make the following connections.
- Connect the power supply with proper polarity to FCL-01 and FCL-02. While connecting this, ensure that the power supply is OFF.
- Connect Function Generator FG-01 to FCL-01 using power cable.
- Keep the jumpers **JP1, JP2, JP3&JP4** on FCL-01 as shown in fig.13.2.
- Keep the jumpers **JP1 & JP2** on FCL-02 as shown in fig.13.2.
- Keep switch **S2** in **TX IN** position on FCL-01.
- Switch on the power supply.
- Connect the **1 KHz, 2Vpp** Signal from FG-01 as a constant signal to the **IN** post of Analog Buffer section on FCL-01.
- Connect the output of Analog Buffer section post **OUT** to post **IN** of EMI section on FCL-01.
- Connect copper cable one end to post **IN** of EMI section on FCL-01.
- Insert the copper cable through EMI Coil Jig.
- Connect other end of copper cable to post **OUT** of EMI section on FCL-01.
- Connect the interference coil of EMI unit to a function generator and excite it with a sinusoidal signal of 1MHz – 5MHz at around 10- 20Vp-p.
- Observe the EMI interference at **EMI OUT** post. It is observed that as the external signals amplitude and frequency increases EMI interference increases on output signal as shown in fig. 13.3.
- Refer to fig 13.2 and make the following connections.
- Connect the **1 KHz, 2Vpp** Signal from FG-01 as a constant signal to the **IN** post of Analog Buffer section on FCL-01.
- Connect the output of Analog Buffer section post **OUT** to post **TX IN** of transmitter section on FCL-01.
- Slightly unscrew the cap of LED SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the 1meter fiber into the cap. Now tighten the cap by screwing it back.
- Insert the fiber cable through EMI Jig.
- Slightly unscrew the cap of Photo Diode SFH250V. Do not remove the cap from the connector. Once the cap is loosened, insert the other end of fiber into the cap. Now tighten the cap by screwing it back.
- Keep switch **SW2** in **ANALOG OUT** position on FCL-02.
- Connect the interference coil of EMI unit to a function generator and excite it with a sinusoidal signal of 1MHz – 5MHz at around 10- 20Vp-p.
- It is observed that as the external signals amplitude and frequency increases there is no effect of EMI interference on the received signal at DETECTOR O/P through fiber medium.

(**NOTE:** Excite the EMI coil when required only.)

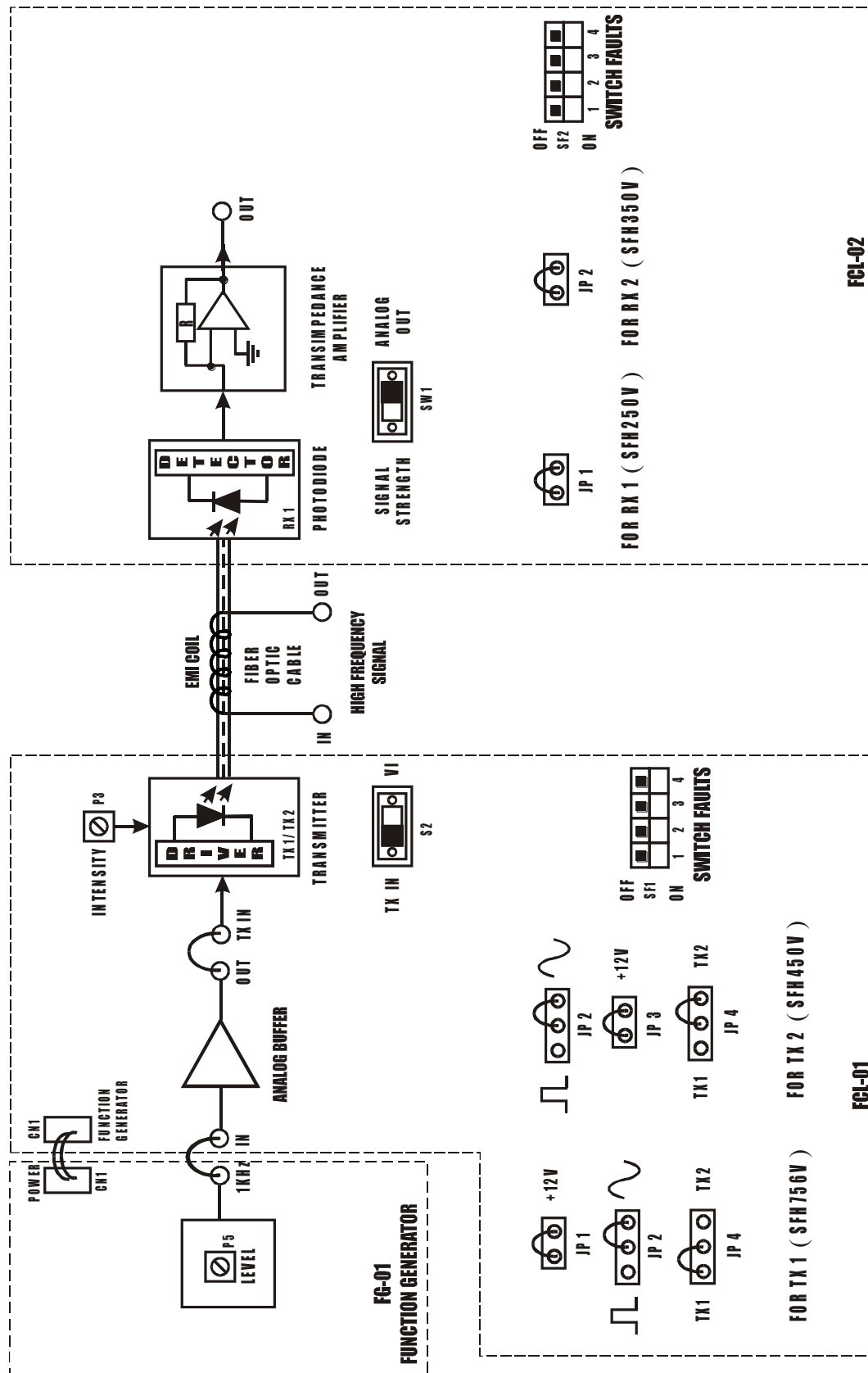
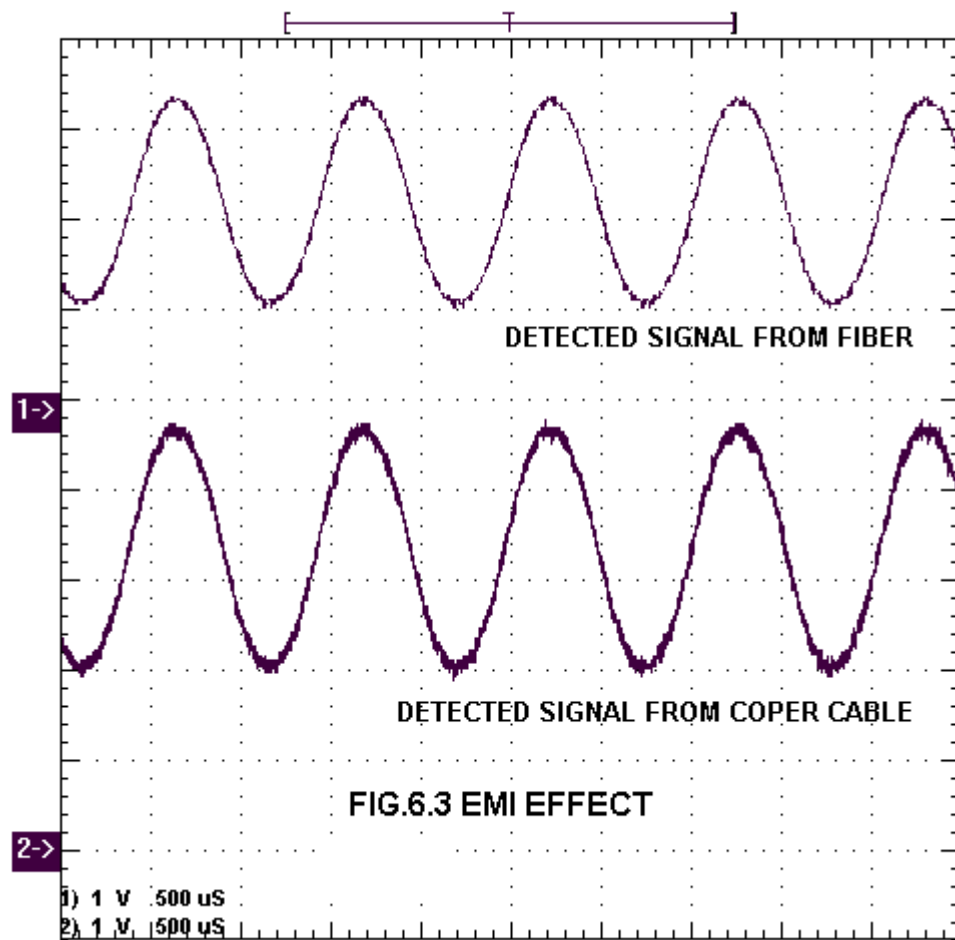


FIG.13.2 BLOCK DIAGRAM FOR EMI INTERFERENCE ON FIBER OPTIC CABLE

CONCLUSION

Data communication on a copper link is adversely affected by high frequency electromagnetic signals in the environment. You observed in the experiment that EMI signals do not effect communication on a fiber.



EXPERIMENT NO.14

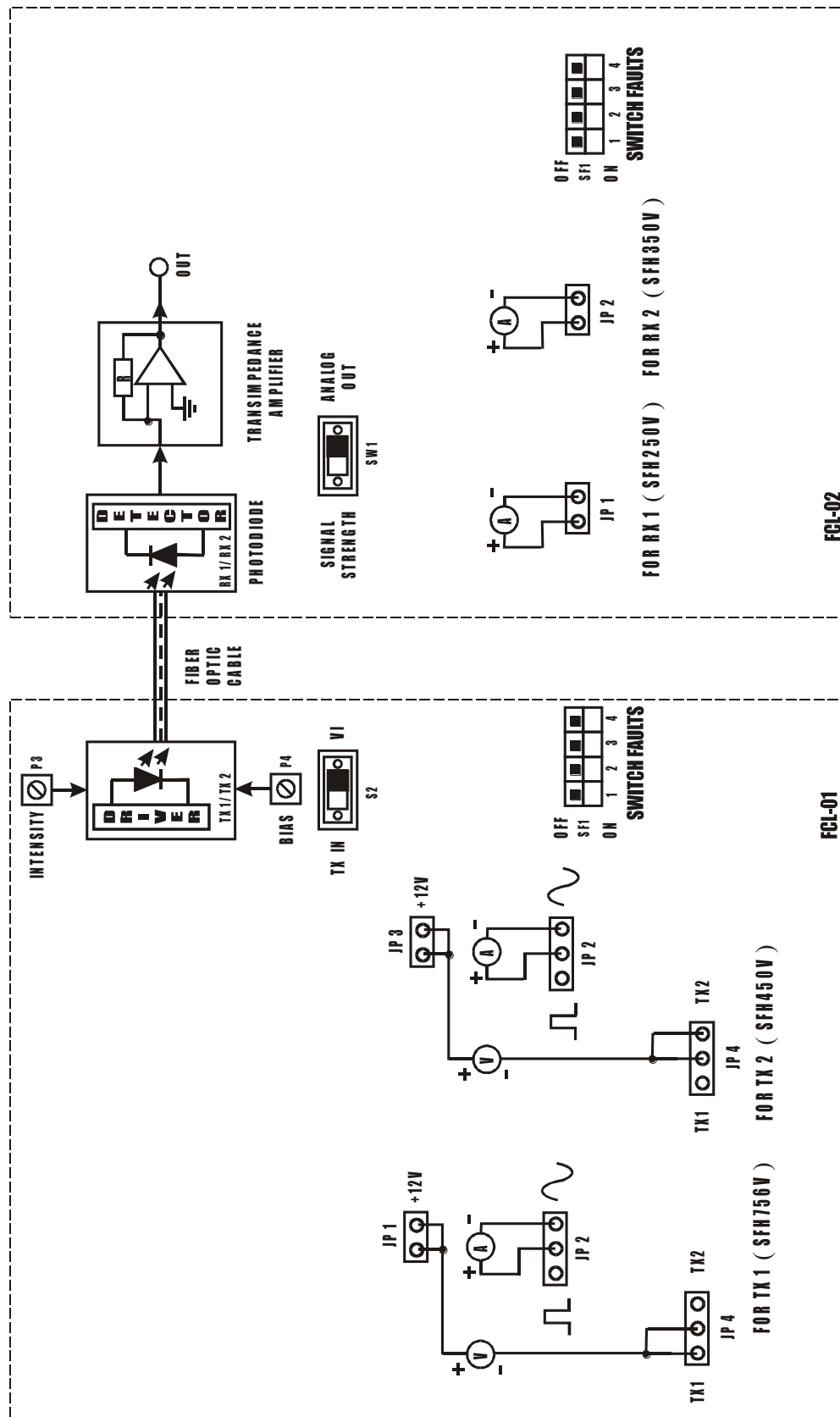


FIG. 14.1 BLOCK DIAGRAM FOR CHARACTERISTICS OF FIBER OPTIC LED'S AND DETECTORS

EXPERIMENT NO: 14

NAME

Study of Characteristics of Fiber Optic LED and Photo-detector

OBJECTIVE

To study the IV characteristics of fiber optic LED and plot the graph of forward current v/s output optical energy and also to study the photo detector response

THEORY

In optical fiber communication system, electrical signal is first converted into optical signal with the help of E / O conversion device as LED. After this optical signal is transmitted through optical fiber, it is retrieved in its original electrical form with the help O / E conversion device as photo detector.

Different technologies employed in chip fabrication lead to significant variation in parameters for the various emitter diodes. All the emitters distinguish themselves in offering high output power coupled into the plastic fiber. Datasheets for LEDs usually specify electrical and optical characteristics, out of which the important ones are peak wavelength of emission, conversion efficiency (usually specified in terms of power launched in optical fiber for specified forward current), optical rise and fall times which put the limitation on operating frequency, maximum forward current through LED and typical forward voltage across LED.

Photo-detectors usually comes in variety of forms like photoconductive, photovoltaic, transistor type output and diode type output. Here also characteristics to be taken into account are response time of the detector, which puts the limitation on the operating frequency, wavelength sensitivity and responsivity.

EQUIPMENTS

- FCL-01 & FCL-02
- 1 meter Fiber cable
- Patch chords
- Jumper to crocodile wires
- Power supply (Use only one provided)

- 20 MHz Dual Channel Oscilloscope
- Volt meter
- Current Meter

NOTE: Keep All Switch Faults In Off Position.

PROCEDURE

- Make connections as shown in fig.14.1. Connect the power supply with proper polarity to FCL-01 & FCL-02 Kits. While connecting this, ensure that the power supply is OFF.
- Slightly unscrew the cap of LED SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the 1-meter fiber into the cap. Now tighten the cap by screwing it back.
- Slightly unscrew the cap of Photo Diode SFH250V. Do not remove the cap from the connector. Once the cap is loosened, insert the other end of fiber into the cap. Now tighten the cap by screwing it back.
- Keep the jumpers **JP1**, **JP2**, **JP3** & **JP4** on FCL-01 as shown in fig.14.1.
- Keep the jumpers **JP1** & **JP2** on FCL-02 as shown in fig.14.1.
- Keep switch **S2** in **VI** position on FCL-01.
- Connect voltmeter and current meter as per the polarities shown in fig.14.1.
- Switch on the power supply.
- Keep the potentiometer **P3** in its maximum position (anti-clockwise rotation). **P3** is used to control current flowing through the LED.
- Keep the potentiometer **P4** in its fully clockwise rotation. **P4** is used to control bias voltage of the LED.
- To get the IV characteristics of LED, rotate **P3** slowly and measure forward current and corresponding forward voltage. Take number of such readings for various current values and plot IV characteristics graph for the LED.
- For each reading taken above, find out the power, which is product of V and I. This is the electrical power supplied to the LED. Data sheets for the LED specify optical power coupled into plastic fiber when forward current was 10 mA as 200 μ W. This means that the electrical power at 10 mA current is converted into 200 μ W of optical energy. Hence the efficiency of the LED comes out to be approx. 1.15%.
- With this efficiency assumed, find out optical power coupled into plastic optical fiber for each of the reading. Plot the graph of forward current v/s output optical power of the LED.
- Similarly measure the current at the detector.
- Plot the graph of receiver current v/s output optical power of the LED.
- Perform the above procedure again for all the combinations of Transmitter & Receiver.
- Calculate the responsivity of the detector

$R = I_p / P_o$ (A/W) = Photocurrent in μA / Optical power in μW

Quantum efficiency (η) = $(I_p / q) / (P_o / hf)$

$q = 1.60218 \times 10^{-19} \text{ C}$

$h = 6.6256 \times 10^{-34} \text{ J.s}$

$f = c / \lambda = (3 \times 10^8 \text{ m/s}) / 850\text{nm}$ or

$\eta = R \cdot hf / q$, R = Responsivity

VI Characteristics Of Fiber Optic Led & Detector

<i>LED CHARACTERISTICS</i>				DETECTOR CHARACTERISTICS		
V _f (V)	I _f (mA)	P _i (mW)	P _o (μW)	V (V)	I _p (μA)	R (mA)
1.559	0.248	0.386	4.4	0.07	1.0	0.22
1.6	0.882	1.4	16.0	0.124	5.3	0.33
1.62	1.461	2.36	27.0	0.240	9.9	0.36
1.64	2.5	4.1	47.0	0.260	17.8	0.37
1.66	3.8	6.3	72.0	0.445	29.7	0.41
1.68	5.8	9.7	111.5	0.750	47.1	0.42
1.70	8.0	13.6	156.4	1.15	68.2	0.43
1.72	10.6	18.2	210.0	1.76	92.3	0.43
1.74	13.6	23.6	271.4	2.81	121.0	0.44
1.76	17.3	30.4	350.0	3.54	152.5	0.43
1.78	21.8	38.8	446.2	4.16	186.1	0.41
1.81	27.5	49.7	571.5	4.58	247.0	0.43

V_f = Forward voltage of LED SFH756

I_f = Forward current of LED SHF 756

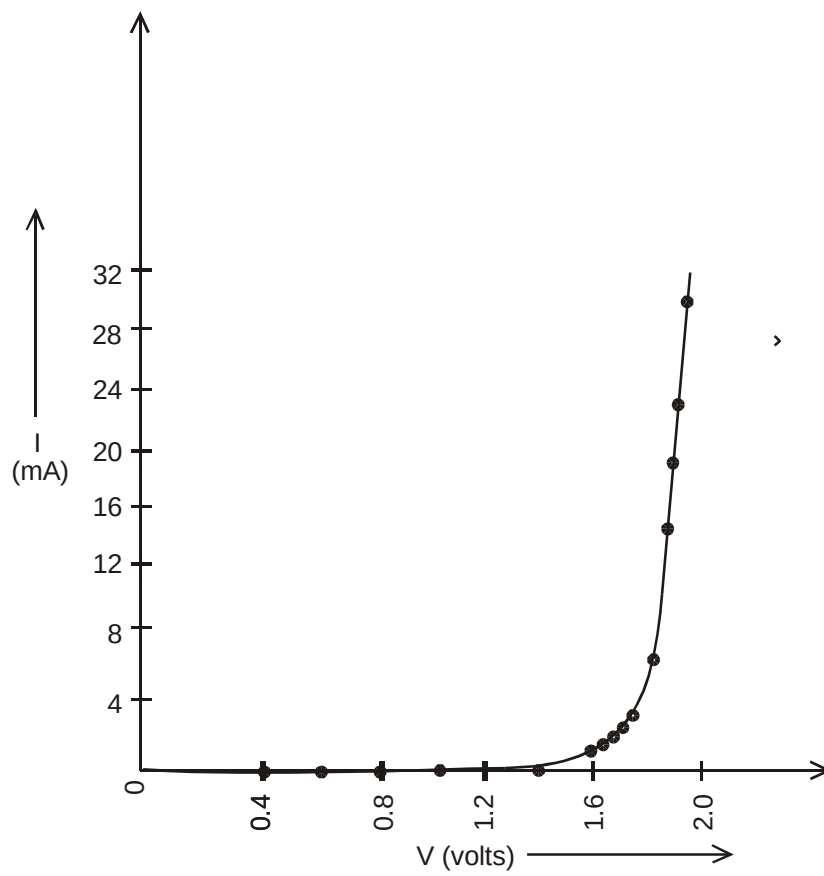
P_i = $V \times I$ (Electrical power)

P_o = $P_i \times 1.15\%$ (Optical Power of LED 756.)

V = Out put Voltage of SFH350

I_p = Out put Current of SFH350

R = I_p / P_o (Responsivity)



VI CHARACTERISTICS

EXPERIMENT NO.15

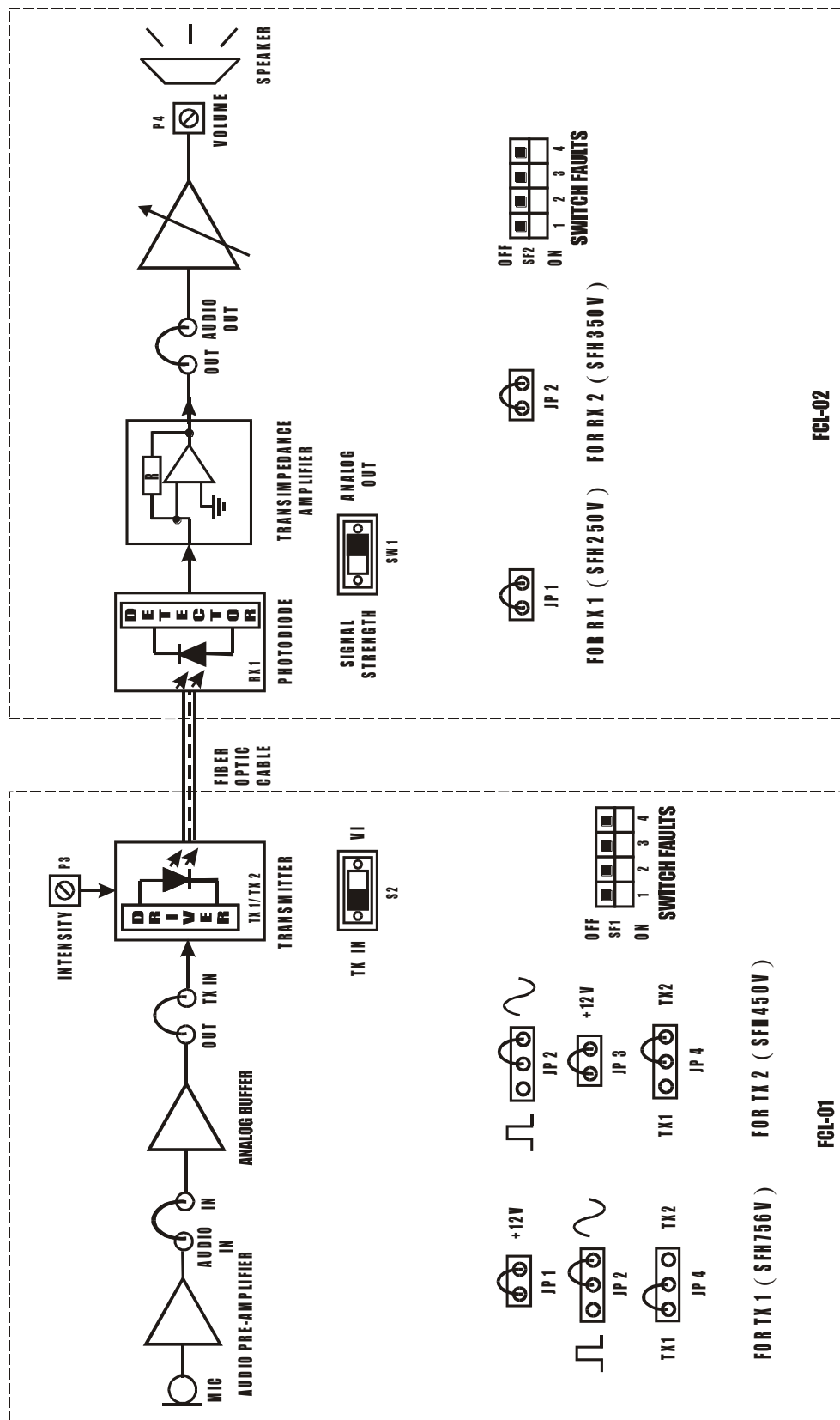


FIG. 15.1 BLOCK DIAGRAM FOR SETTING UP FIBER OPTIC VOICE LINK

EXPERIMENT NO: 15

NAME

Setting Up Simple Fiber Optic Voice Link

OBJECTIVE

Setting up Fiber optic voice link through various modulation techniques

THEORY

On FCL-01 kit, we have provision for Microphone connection. On FCL-02 kit, we have provision for Speaker connection. Audio Pre-Amplifier section on FCL-01 kit will amplify the voice signal from Microphone and it can be applied to any of the modulation sections as an input signal. This signal is converted into light using any of the transmitters. Received signal is converted to analog form first then it is demodulated using respective demodulator section and fed to Audio Amplifier section. Amplified signal is fed to speaker, where it is again converted to voice form.

EQUIPMENTS

- FCL-01 & FCL-02
- 1 meter Fiber cable
- Patch chords
- MIC& Speaker
- Power supply (Use only one provided)

20 MHz Dual Channel Oscilloscope

NOTE: Keep All Switch Faults In Off Position

PROCEDURE

- Make connections as shown in fig.15.1. Connect the power supply cables with proper polarity to FCL-01 & FCL-02 Kits. While connecting this, ensure that the power supply is OFF.

- Keep the jumpers **JP1, JP2, JP3 & JP4** on FCL-01 as shown in fig.15.1.
- Keep the jumpers **JP1 & JP2** on FCL-02 as shown in fig.15.1.
- Keep switch **S2** in **TX IN** position on FCL-01.
- Switch on the power supply.
- Connect MIC to the post **MIC**.
- Connect the **AUDIO IN** post of Audio Pre-Amplifier to the **IN** post of Analog Buffer on FCL-01.
- Connect the output of Analog Buffer post **OUT** to post **TX IN**.
- Slightly unscrew the cap of LED SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the 1 meter fiber into the cap. Now tighten the cap by screwing it back.
- Slightly unscrew the cap of Photo Diode SFH250V. Do not remove the cap from the connector. Once the cap is loosened, insert the other end of fiber into the cap. Now tighten the cap by screwing it back.
- Keep Switch **SW1** to **ANALOG OUT** position in FCL-02.
- Connect the post **OUT** of detector to post **AUDIO OUT**.
- Connect Speaker to the post **SPEAKER**.
- Notice the output signal from speaker, which is exact reproduction of the speech signal from the microphone.
- For clear voice signal vary Optical Power Control Pot **P3** on FCL-01 and Volume pot **P4**. You should get the reproduction of the original transmitted signal.
- Similarly Connect the voice link through Pulse Amplitude Modulation & Demodulation by referring to Expt. No. 9, through Time Division Multiplexing & Demultiplexing by referring to Expt. No. 10, through Amplitude Modulation & Demodulation by referring to Expt. No. 11 modulating signal connect **AUDIOIN** post input.